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Inventor(s)

BAI; Sanmao et al.

VEHICLE CLIMATE CONTROL SYSTEM UTILIZING A FLEXIBLE HEAT PUMP

Abstract

A heat transfer system to alternatively and/or simultaneously provide heating and cooling in a mobile vehicle that includes an electrical power source requiring heating and/or cooling during charging and/or operation and that includes a cabin that requires heat input during low temperature ambient conditions.

Inventors: BAI; Sanmao (Shanghai, CN), VINCENT; Curt (Grand Rapids, MI), PETERSEN; Michael (Buffalo, NY), LIU; Tao (Shanghai, CN), PUROHIT; Nilesh (Charlotte, NC)

Applicant: HONEYWELL INTERNATIONAL INC. (Charlotte, NC)

Family ID: 1000008529151

Assignee: HONEYWELL INTERNATIONAL INC. (Charlotte, NC)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present invention is related to and claims the priority benefit of U.S. Provisional Application No. 63/563,252, filed Mar. 8, 2024 and of U.S. Provisional Application No. 63/635,746, filed Apr. 18, 2024 and of U.S. Provisional Application No. 63/635,747, filed Apr. 18, 2024 and of U.S. Provisional Application No. 63/665,190, filed Jun. 27, 2024 and of U.S. Provisional Application No. 63/666,460, filed Jul. 1, 2024 and of U.S. Provisional Application No. 63/666,464, filed Jul. 1, 2024, each of which is incorporated herein by reference in their entirety.

FIELD

[0002] The present invention relates to thermal management systems for electric vehicles and in particular to a flexible and efficient climate control system arrangement and method utilizing a heat pump for such vehicles.

BACKGROUND

[0003] A vehicle, such as a car or truck, which is propelled solely by one or more electric motors, sometimes referred to as a traction motor, is typically referred to as an electric vehicle or an EV. In a hybrid electric vehicle, or HEV, one or more traction motors are used in conjunction with another power source, such as for example an internal combustion engine, including both gasoline and diesel-powered engines. In both cases, a battery or capacitor bank carried by the vehicle during operation provides an electrical current to the traction motor and other components that are driven by an electric current and which will generally generate heat during operations.

[0004] Because the propulsion systems of EVs do not include an internal combustion engine, a traditional internal combustion engine cooling system is not present, and therefore hot liquid coolant is unavailable for heating the interior of the cabin, cab, or passenger compartment of the vehicle. Although an internal combustion engine is included in HEVs, there are times when it may be desirable to operate the HEV without running the internal combustion engine, in which case heat may be unavailable from circulating hot liquid coolant to heating the interior of the cabin, cab, or passenger compartment. Furthermore, it is frequently required that, in addition to the need to heat the cabin, cab, or passenger compartment for the comfort of the occupants, heat is frequently also required to defrost the vehicle windows.

[0005] The development of a thermal management system to handle the heating and cooling needs of EVs and HEVs is challenging for several reasons. For example, it has been known to provide another source of heat, such as electric heaters, in EVs to provide at least some of the heat needed by the vehicle as described above. Such electric heaters, however, typically draw electric current from the same on-board source of electricity that supplies current to the traction motor that is used to propel the vehicle. It can be a disadvantage to require the use of such a heating source since it can limit the range of the EV or limit the number of miles in which an HEV is propelled by the traction motor.

[0006] Another challenge associated with the development of EVs and HEVs thermal management systems is that such systems also require the ability to cool the cabin, cab, or passenger compartment during warmer weather. In conventional non-electric vehicles, such air conditioning is provided by a compressor that is mechanically driven by the internal combustion engine.

Because an EV lacks an internal combustion engine, and because the internal combustion engine of a hybrid electric vehicle may be turned off for periods of time, it is desirable to provide an alternate source of cooling for the cab, cabin, or passenger compartment for such vehicles when air conditioning is desired.

[0007] Another challenge involves the potential need to manage the temperature of the battery and/or other electrical components of EVs, and potentially for some HEVs, when the vehicle is stationary, and the battery is being charged by an external source of electrical current, such as would occur at a charging station.

[0008] Therefore, heating and cooling of the cab, cabin, or passenger compartment of an EV or an HEV, including defrosting of the vehicle windows, is a challenging task that should provide effective and efficient thermal operation while having the lowest possible impact on the range of the vehicle or on environmental performance of the EV or HEV.

SUMMARY

[0009] The present invention provides heat transfer systems to alternatively and/or simultaneously provide heating and cooling in a mobile vehicle that includes an electrical power source requiring temperature regulation during operation and that includes a cabin that requires heat input during low temperature ambient conditions, said system comprising: [0010] a) a vapor compression refrigeration circuit located in said mobile vehicle comprising: [0011] (i) a heat transfer composition comprising a first refrigerant, [0012] (ii) a compressor for compressing said first refrigerant in the vapor state from a first pressure to a higher second pressure, said compressor optionally but preferably being connected upstream to a refrigerant accumulator, [0013] (iii) an inner condenser for selectively condensing during low temperature ambient conditions at least a portion of said first refrigerant vapor from said compressor by rejecting heat to said cabin, [0014] (iv) an outside heat exchanger located downstream of said inner condenser to selectively either (1) condense during low temperature ambient conditions at least a portion of said higher pressure refrigerant vapor not condensed in said inner condenser by rejecting heat, directly or indirectly, to ambient air and/or to a circulating coolant or (2) evaporate during high temperature ambient conditions low pressure refrigerant liquid from said inner condenser vapor; [0015] (v) a first OCE (as defined hereinafter) connected between said inner condenser and said outside heat exchanger for selectively (1) providing in an expansion mode a flow of reduced pressure liquid refrigerant from said inner condenser to said outside heat exchanger; (2) allowing in an open mode said condensed high pressure refrigerant from said condenser to pass to said outside condenser without pressure drop to said outside heat exchanger; or (3) preventing in a closed mode the flow of refrigerant from said inner condenser to said outside heat exchanger; [0016] (vi) an inside heat exchanger fluidly connectable to said refrigerant downstream of said inner condenser for selectively heating a flow of cabin air; [0017] (vii) a chiller fluidly connectable to said refrigerant downstream of said inner condenser for selectively heating a flow of liquid coolant; [0018] (viii) a bypass channel system connected upstream of said first open/closed/expansion device and downstream of said outside heat exchanger for selectively routing said refrigerant from said inner condenser and/or from said outside heat exchanger: (1) around said first expansion device and to either (A) a second OCE fluidly connected to said inside heat exchanger for selectively (a) providing in an expansion mode a flow of reduced pressure liquid refrigerant from said inner condenser to said inside heat exchanger; (b) allowing in an open mode said condensed high pressure refrigerant from said condenser or from said outside heat exchanger to pass without pressure reduction to said inside heat exchanger; or (c) preventing in a closed mode the flow of refrigerant to said inside heat exchanger; and/or (B) an expansion device fluidly connected to said chiller for selectively (a) providing in an expansion mode a flow of reduced pressure liquid refrigerant to said chiller; or (b) preventing in a closed mode the flow of refrigerant to said chiller; or (2) through said first OCE operating in the expansion mode through said outside heat exchanger to said accumulator; [0019] b) a heat exchange network interconnected with said vapor

compression refrigeration circuit to selectively; (i) deliver, directly or indirectly, at said outside heat exchanger and/or at said chiller evaporative heat from one or more of ambient air and/or heat associated with the generation or use of electrical power within the vehicle and/or at said inside heat exchanger either directly or indirectly from (1) ambient air and/or (2) said electrical power source located in said vehicle.

[0020] For the purposes of convenience, the heat transfer system as described in this paragraph is referred to for convenience as Heat Transfer System 1A.

[0021] The present invention also provides heat transfer systems as described herein, including Heat Transfer System **1**, in which the refrigerant used in the vapor compression refrigeration circuit comprises, or consists essentially of, or consists of, on a weight basis, HFO-1234yf. For the purposes of convenience, the heat transfer system as described in this paragraph is referred to for convenience as Heat Transfer System 1B.

[0022] The present invention also provides heat transfer systems as described herein, including Heat Transfer System **1**, in which the refrigerant used in the vapor compression refrigeration circuit comprises, or consists essentially of, or consists of, on a weight basis, HFO-1234ze(E). For the purposes of convenience, the heat transfer system as described in this paragraph is referred to for convenience as Heat Transfer System 1C.

[0023] The present invention also provides heat transfer systems as described herein, including Heat Transfer System **1**, in which the refrigerant used in the vapor compression refrigeration circuit comprises, or consists essentially of, or consists of, on a weight basis, about 21.5% of R32 and about 78.5% of R1234yf. For the purposes of convenience, the heat transfer system as described in this paragraph is referred to for convenience as Heat Transfer System 2A.

[0024] The present invention also provides heat transfer systems as described herein, including Heat Transfer System **1**, in which the refrigerant used in the vapor compression refrigeration circuit comprises, or consists essentially of, or consists of, on a weight basis, about 3% of carbon dioxide (CO.sub.2), about 21.5% of R32 and about 75.5% of R1234yf. For the purposes of convenience, the heat transfer system as described in this paragraph is referred to for convenience as Heat Transfer System 2B.

[0025] The present invention also provides heat transfer systems as described herein, including Heat Transfer System **1**, in which the refrigerant used in the vapor compression refrigeration circuit comprises, or consists essentially of, or consists of, on a weight basis, about 23% of R1132(E) and about 77% of R1234yf. For the purposes of convenience, the heat transfer system as described in this paragraph is referred to for convenience as Heat Transfer System 2C.

[0026] The present invention also provides heat transfer systems as described herein, including Heat Transfer System **1**, in which the refrigerant used in the vapor compression refrigeration circuit comprises, or consists essentially of, or consists of, on a weight basis, about 31.5% of R1132(E) and about 68.5% of R1234yf. For the purposes of convenience, the heat transfer system as described in this paragraph is referred to for convenience as Heat Transfer System 2D.

[0027] The present invention also provides heat transfer systems as described herein, including Heat Transfer System **1**, in which the refrigerant used in the vapor compression refrigeration circuit comprises, or consists essentially of, or consists of, on a weight basis, about 8.5% of R134a and about 77.5% of R1234yf. For the purposes of convenience, the heat transfer system as described in this paragraph is referred to for convenience as Heat Transfer System 3A.

[0028] The present invention also provides heat transfer systems as described herein, including Heat Transfer System **1**, in which the refrigerant used in the vapor compression refrigeration circuit comprises, or consists essentially of, or consists of, on a weight basis, about 5% of R152a, about 12% of R32 and about 83% of R1234ze(E). For the purposes of convenience, the heat transfer system as described in this paragraph is referred to for convenience as Heat Transfer System 3B.

[0029] The present invention also provides heat transfer systems as described herein, including Heat Transfer System **1**, in which the refrigerant used in the vapor compression refrigeration circuit

comprises, or consists essentially of, or consists of, on a weight basis, about 78% of R1234yf, about 14.5% of R152a, and about 7.5% of R32. For the purposes of convenience, the heat transfer system as described in this paragraph is referred to for convenience as Heat Transfer System 3C.

[0030] The present invention also provides heat transfer systems as described herein, including Heat Transfer System 1, in which the refrigerant used in the vapor compression refrigeration circuit comprises, or consists essentially of, or consists of, on a weight basis, about 82% of R1234yf, about 14% of R152a, and about 4% of R32. For the purposes of convenience, the heat transfer system as described in this paragraph is referred to for convenience as Heat Transfer System 3D.

[0031] The present invention also provides heat transfer systems as described above including Heat Transfer Systems 1-3, in which the heat exchange network comprises a coolant circuit that comprises a coolant that absorbs waste heat from an electrical power source located in said vehicle during low temperature ambient conditions and rejects heat to said refrigerant in said chiller.

[0032] As used herein, reference to a “Heat Transfer System No.” is an explicit reference to all heat transfer systems having that number, including all heat transfer systems with that number, including any suffix. For example, reference to “Heat Transfer System 1” is a reference to each of Heat Transfer System 1A, 1B and 1C

[0033] The systems of the present invention, including each of Heat Transfer Systems 1-3 are particularly well adapted to operate with system pressures that are similar to the pressures that have been present in vapor compression systems that use the refrigerants R-134a, R404A, R22, R507 and R407, but without the substantial environmental disadvantages associated with the use of those refrigerants.

[0034] The systems of the present invention, including particularly each of Heat Transfer Systems 2 are particularly well adapted to operate with system pressures that are similar to the pressures that have been present in vapor compression systems that use the refrigerants R404A, R22, R507 and R407, but without the substantial environmental disadvantages associated with the use of those refrigerants.

[0035] The systems of the present invention, including particularly each of Heat Transfer Systems 3 are particularly well adapted to operate with system pressures that are similar to the pressures that have been present in vapor compression systems that use the refrigerant R-134a, but without the substantial environmental disadvantages associated with the use of those refrigerants.

[0036] As used herein, the term “waste heat from an electrical power source” refers to heat that needs to be and/or can removed from an on-board battery or an electrically powered device or article powered by the on-board battery or an off-board source of electrical power, such as the charging source that is being used to charge the batter. By way of example, such devices include the vehicle battery, motor, inverter and other electrical devices carried by the vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0037] FIG. 1 illustrates a schematic of a thermal management system for an EV or HEV according to one example of the present invention.

[0038] FIGS. 2A and 2B illustrate a schematic of a thermal management system for an EV or HEV according to a second example of the present invention.

[0039] FIG. 3A illustrates a respective schematic of an exemplary thermal management system, generally as illustrated in FIG. 2.

[0040] FIGS. 3B-3C, illustrate preferred generalized alternative modes of operation of the respective schematic generally illustrated in FIG. 3.

[0041] FIGS. 3D and 3D1, 3D2-1, 3D2-2, 3D2-3, 3D3-1, 3D3-2, 3F and 3G illustrate a preferred generalized alternative modes of operation of the respective schematic generally illustrated in FIG.

2.

[0042] FIG. 3E illustrates the COP and Capacity data from Example 1A.

[0043] FIGS. 4-31 illustrate respective schematics of the thermal management system, generally as illustrated in FIG. 2 and FIGS. 3A-3D, for a series of different operating modes as described in Examples 1-30 hereof.

[0044] FIGS. 32A and 32B illustrate that data from Example 1B1.

[0045] FIGS. 33A and 33B illustrate that data from Example 1B2.

[0046] FIGS. 34A and 34B illustrate that data from Example 1B3.

[0047] FIGS. 35A and 35B illustrate that data from Example 1B4.

[0048] Comparative FIG. 1 and Comparative FIG. 2 illustrate schematics of comparative thermal management systems according to Comparative Examples 1 and Comparative Examples 2.

Definitions

[0049] The phrase “coefficient of performance” (herein abbreviated as “COP”) is a universally accepted measure of refrigerant performance, especially useful in representing the relative thermodynamic efficiency of a refrigerant in a specific heating or cooling cycle involving evaporation or condensation of the refrigerant. In refrigeration engineering, this term expresses the ratio of useful refrigeration, cooling or heating capacity to the energy applied by the compressor in compressing the vapor and therefore expresses the capability of a given compressor to pump quantities of heat for a given volumetric flow rate of a heat transfer fluid, such as a refrigerant. In other words, given a specific compressor, a refrigerant with a higher COP will deliver more cooling or heating power. One means for estimating COP of a refrigerant at specific operating conditions is from the thermodynamic properties of the refrigerant using standard refrigeration cycle analysis techniques (see for example, R.C. Downing, FLUOROCARBON REFRIGERANTS HANDBOOK, Chapter 3, Prentice-Hall, 1988 which is incorporated herein by reference in its entirety).

[0050] The phrase “Global Warming Potential” (herein abbreviated as “GWP”) was developed to allow comparisons of the global warming impact of different gases. It compares the amount of heat trapped by a certain mass of a gas to the amount of heat trapped by a similar mass of carbon dioxide over a specific time period of time. Carbon dioxide was chosen by the Intergovernmental Panel on Climate Change (IPCC) as the reference gas and its GWP is taken as 1. The larger GWP, the more that a given gas warms the Earth compared to CO₂ over that time period. As used herein, the term GWP means the value of GWP as measured in accordance with IPCC Fourth Assessment Report, 2014, referred to and abbreviated herein as AR4, except for components that did not have a GWP value measured in AR4 (such as R1234yf), then the values used are according to the Fifth Assessment Report.

[0051] As used herein, the terms “positive temperature coefficient heater,” “PTC heater” and “PTC” mean a heating device that provides heat via electrical current input, including preferably a heating device that comprises a ceramic heating element with a positive temperature coefficient.

[0052] As used herein, the term OCE refers to a device, such as a valve, that can operate in the open position, the closed position and in an expansion mode in which it operates, for example, as an expansion orifice or valve.

[0053] As used herein, the terms 1234yf and R1234yf mean 2,3,3,3-tetrafluoropropene.

[0054] As used herein, the terms 1234ze(E) and R1234ze(E) mean the trans isomer of 1,3,3,3-tetrafluoropropene.

[0055] As used herein, the terms “R-134a” and “HFC-134a” as used herein each mean 1,1,1,2-tetrafluoroethane.

[0056] As used herein, the terms “R-152a” and “HFC-152a” as used herein each mean 1,1-difluoroethane.

[0057] As used herein, the terms “R-1132 (E)” and “HFC-1132 (E)” as used herein each mean the trans isomer of 1,2-difluoroethylene. As used herein, the terms “R-32” and “HFC-32” as used

herein each mean difluoromethane.

[0058] As used herein, the terms “R1132(E)” and “transHFO-1132 (E)” each means the trans isomer of 1,2-difluorethylene.

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[0062] Plattner, M. Tignor, S. K. Allen, J. Boschung, A. Nauels, Y. Xia, V. Bex and P. M. Midgley (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.

[0063] https://www.ipcc.ch/pdf/assessmentreport/ar5/wg1/WG1AR5_Chapter08_FINAL.pdf (p. 73-79)

[0064] As used herein, the term “R454C” means the refrigerant designated by ASHRAE as 454C and which consists of 21.5%+2/-2% of R-32 and 78.5-+2/-2% of HFC-1234yf.

[0065] As used herein, the term “R455A” means the refrigerant designated by ASHRAE as 455A and which consists of 21.5%+2/-1% of R-32, 75.5 of HFC-1234yf+2/-2% and 3%+2/-1% of CO.sub.2.

[0066] As used herein, the term “R474A” means the refrigerant designated by ASHRAE as 474A and which consists of about 23% of R-1132 (E) and about 77% of R-1234yf.

[0067] As used herein, the term “R474B” means the refrigerant designated by ASHRAE as 474A and which consists of about 31.5% of R-1132 (E) and about 68.5% of R-1234yf.

[0068] As used herein, the term “R516A” means the refrigerant designated by ASHRAE as 516A and which consists of about 77.5% of R-1234yf, about 8.5% of R-134a and about 14% R152a.

[0069] As used herein, the term “R444A” means the refrigerant designated by ASHRAE as 444A and which consists of about 83% of R-1234ze(E), about 12% of R-32 and about 5% of R-152a.

[0070] As used herein, the term “R457C” means the refrigerant designated by ASHRAE as 457C and which consists of about 78% of R-1234yf, about 7.5% of R-32 and about 14.5% R152a.

[0071] As used herein, the term “R457D” which consists of about 82% of R-1234ze, about 4% of R-32 and about 14% R152a.

DETAILED DESCRIPTION

[0072] An exemplary thermal management system according to the present invention is illustrated in FIG. 1 hereof. The thermal management system, including the components of the EV of HEV being heated or cooled, are designated generally as 10, and includes an area in which one or more persons would travel, which is referred to generally herein as the “cabin” (not shown) and areas outside the cabin which will generally house the working components of the EV or HEV. Portions of the various thermal management systems of the present invention may be located within the cabin and/or outside the cabin.

[0073] The system 10 of the present invention includes heat pump subsystem, which preferably comprises a vapor compression system, designated generally as 20, thermally interconnected with a coolant circuit, designated generally as 100, a cabin climate control module **200** and potentially also independently fluidly connected with a source of ambient air, designated as 300. It will be understood that since some of the components of the vapor compression circuit interface with some components of the coolant circuit and the climate control module **200**, those portions may be properly designated as components of each of those portions of the system.

[0074] In particular, the heat pump vapor compression system includes a refrigerant, preferably R-1234ze(E), R-1234yf or blends that comprise R-1234yf, or 1234ze(E) or blends that comprise R-1234ze(E) as described herein, that circulates to various components of the present invention, a compressor **21**, optionally but preferably an accumulator **22** on the suction side of the compressor and inner condenser **23** located in the climate control module **200**. Although it is contemplated that

the specific heat exchanger used to for the inner condenser can vary widely according to the particular needs of a particular application, in some embodiments, particularly in which the heating mode of the system is especially important, it is preferred that a four-pass or higher configuration is used, since applicants have found that such a configuration can provide unexpected levels of improvement in COP performance and capacity performance in the heating mode, as well as possible lower compressor discharge temperatures and pressures. The climate control module **200** preferably includes a door **23A** on the inner condenser **23** which can be moved to any position between a fully closed position (as shown) in which no cabin air which enters the control module can flow through the inner condenser to a fully open position in which the door permits air from the cabin to flow fully through the inner condenser and to be heated as it condenses at least a portion of the refrigerant which flows into the condenser from the discharge side of the compressor. [0075] Refrigerant which exits the inner condenser is fluidly connected to an OCE (labeled as OC/EX1). The preferred OC/EX1 is a device that can be configured to take one of three possible actions: (1) change the pressure and temperature of the refrigerant flowing therethrough; (2) open fully so as to allow passage of refrigerant therethrough with minimal change in pressure or temperature; or (3) close so as to prevent the flow of refrigerant therethrough. The OCE devices that are used in the present invention may include an electronic actuator-controlled controller (see FIGS. 2A and 2B), which may cause the actuator to position the expansion device in the wide-open position, in the fully closed position, or a throttled position in which flow is permitted but at a substantially reduced pressure and temperature. The throttled position typically is a partially open position where the controller modulates the size of the valve opening to regulate flow through the device. The controller and OCE devices may be configured to continuously or periodically modulate the throttled position in response to system operating conditions. By throttling the position of the expansion device, the controller can regulate flow, pressure, temperature, and state of the refrigerant as needed.

[0076] By operating the OC/EX1 in the fully opened position, the outside heat exchanger **24** (which is located outside the passenger cabin) can be used during low temperature ambient conditions in a supplemental condensation mode to condense at least a portion of any refrigerant vapor that is not condensed in the inner condenser **23** by rejecting heat to the relatively low temperature ambient air **400** directly, or preferably indirectly after ambient air has passed through the radiator of the circulating coolant system. During periods of high temperature ambient conditions, for example, the OC/EX1 can be operated in the throttled position and the outside heat exchanger can operate as an evaporator or alternatively the outside condenser can be bypassed by operating the OCEX1 in the fully closed position, which will direct the refrigerant flow from the inner condenser through the bypass conduit and to the divert valve **25**.

[0077] The refrigerant which flows through diverter valve **25** can be directed to chiller **26** and/or inner heat exchanger **27** or to bypass each of these and flow through diverter valve **28** directly to accumulator **22**. An open/closed valve OC 1 may be provided downstream of diverter valve **25** and upstream of EXV1, and in the closed position blocks flow towards EXV1, thereby ensuring that refrigerant flows to OC/EX2. As an alternative in some cases, EXV1 may be provided as an OC/EV and operated in a closed position to prevent flow of refrigerant to the chiller, as illustrated in some of the examples below. A second OCE (labeled as OC/EX2) is provided upstream of the inner heat exchanger and can be operated to allow refrigerant to flow to the inner heat exchanger either in the fully open position (i.e., without substantial pressure reduction) or in the throttling mode. The OC/EX2 can also be operated in the fully closed position to prevent the flow of refrigerant to the inner heat exchanger **27**.

[0078] As illustrated particularly in the following examples, the many advantages of the systems of the present include, but are not necessarily limited to: [0079] 1—eliminate or reduce condensing capacity issues in cold weather; [0080] 2—eliminate or reduce icing issues at the outside heat exchanger; [0081] 3—eliminate or reduce the need for high voltage PTC (positive temperature

coefficient) heaters inside the vehicle; [0082] 4—offset heating capacity reduction in cold weather [0083] 5—battery warming with PTC, if present, in very cold weather [0084] 6—using the inside heat exchanger either as an evaporator or a condenser extension and pre warmer [0085] 7—extending the heat pump initial air temperature range for R-1234yf; [0086] 8—using the outside heat exchanger (radiator) to reject heat from components instead of the chiller [0087] 9—warming the motor and inverter for higher efficiency before cold starts; [0088] 10—using the inside condenser for dehumidification reheat, which is more efficient than using PTC; [0089] 11—ability to use all sources for the heat pump (in any combination) at the OHE or chiller; [0090] 12—ability to cool all heat sources at the Radiator [0091] 13—ability to cool all heat sources at the Chiller [0092] 14—ability to self-heat motor and inverter and battery independently. [0093] 15—ability to self-heat motor and inverter and battery in series; and [0094] 16—ability to cool the motor and inverter (at outside heat exchanger (radiator)) and battery (at chiller) concurrently.

[0095] The present invention also provides a heat transfer systems as described herein, including in the Examples below and including each of Heat Transfer System 1-3, in which the heat transfer composition further comprises a lubricant.

[0096] The present invention also provides a heat transfer systems as described herein, including in the Examples below and including each of Heat Transfer Systems 1-3, in which the heat transfer composition further comprises a polyol ester (“POE”) lubricant.

[0097] The present invention also provides a heat transfer systems as described herein, including in the Examples below and including each of Heat Transfer Systems 1-3, in which the heat transfer composition further comprises a poly vinyl ether (“PVE”) lubricant.

EXAMPLES

[0098] The following examples use a thermal management system according to various embodiments according to the present invention as illustrated, inter alia, in FIGS. 2A-2B and 3A-3D.

[0099] The present invention, including embodiments as illustrated in FIGS. 2A and 2B and as referenced in the following examples, is able to provide at least the following advantageous features: [0100] A) the ability to use at least the following four sources of evaporative energy available to use for heating, via heat pump, of an electric vehicle: [0101] 1. Waste (or excess) energy from the motor and inverter [0102] 2. Waste (or excess) energy from the battery [0103] 3. Electrical energy from a heater (PTC) [0104] 4. Free energy from the environment (air) [0105] B) two locations within the system where energy can be absorbed (as the evaporative heat source) and used by the heat pump to warm the vehicle and/or its components: [0106] 1. The outside heat Exchanger (with airflow) [0107] 2. The chiller (with coolant flow)

Comparative Example 1—Heat Pump Mode to Warm Cabin Air

[0108] The operation of a typical prior heat pump system for use in an EV is illustrated in FIG. C1 using the thick solid lines to illustrate the only options available in prior art operation, and the results of the use in this configuration is used as the basis for results of the comparative data reported for this Comparative Example 1 (referred to as “CE1 data”). In this system battery waste heat is carried by a coolant (such as water/glycol for example) away from the battery and the PTC and is used as the evaporative heat source at the chiller of a vapor compression system, as shown in FIG. C1. This configuration may be effective in certain cases, but applicants have come to appreciate that in many circumstances and/or desired modes of operation, including at relatively low ambient temperature conditions, full condensing is frequently not achieved at the inner condenser 1, which detracts from the capacity and effectiveness of such systems in such situations. This Comparative Example 1 and the embodiment of the present invention described in connection with Example 1A which follows, is based upon the use of R-1234yf as the refrigerant.

Example 1A—Heat Pump Mode with Inner Evaporator/Condenser in the Loop to Warm Cabin Air

[0109] Applicants have come to appreciate that when ambient temperatures are relatively low, EVs as previously configured, including as described in Comparative Example 1 and illustrated by the

thick solid lines in Comparative FIG. 1, can have a problem with insufficient condenser surface area at the inner condenser 1 to provide complete condensation, which can result in problems with system capacity and efficiency (COP). Applicants have found that a system of the present invention as described and illustrated herein can dramatically improve performance with relatively simple and low-cost modifications that provide not only unexpectedly superior performance but also high levels of operability over a wide variety of ambient conditions and of modes of cooling and heating to be carried out by the system. The system of the present invention in accordance with this Example 1A is configured for operation to heat cabin air during periods of low ambient temperatures and is illustrated in FIG. 3A.

[0110] In this system, and in the remaining systems illustrated in the Examples, the label “Inner Cond” designates the same heat exchanger referenced in FIG. 1 as the internal condenser 23 or “IC” 23 and the heat exchanger designated as Evap/Cond designates the same heat exchanger designated as “internal heat exchanger” or “IHE” 27, as described and shown in FIG. 1 located in essentially the same relative positions and arrangement, including with presence of a door and cabin air as illustrated and explained in connection with FIG. 1. In addition, each of the Figures according to the present invention will have as needed an open/close valve to prevent the flow of refrigerant to the EXV leading to the chiller, even though such valve is not always illustrated in these figures for convenience. It will be understood that these relative positions and features are present but not always illustrated strictly for the purposes of convenience in this figure and the remaining figures to facilitate easier illustration of the system.

[0111] As illustrated in FIG. 3A, the present system allows the ability to selectively alter the flow of refrigerant from the inner condenser 1 to the inner heat exchanger 2 through an open OCE, that is, entering the heat exchanger 2 at substantially the same pressure and temperature at the exit of the inner condenser. In this way, the inner heat exchanger 2 provides additional condensing surface and at the same time serves as a preheater (with door fully open, thereby allowing the preheated cabin air to enter the inner condenser) for the cabin air entering the inner condenser.

[0112] The conditions tested and the relative capacity and effectiveness of the two systems operating in this manner are reported in the Tables 1 and 2 and illustrated for convenience as FIG. 3E, with the results from this Example 1A (using 1234yf as the refrigerant) reported as EWG-HP and the results from Comparative Example 1 (using 1234yf as the refrigerant) reported as WG-HP.

TABLE-US-00001

TABLE 1 Example operating conditions		Outdoor conditions		Indoor conditions	
Water-Glycol Condition	Ambient/Temp	Temp	Air Flow	Target Temp	Temp WG In Test Air In Air
In Rate at Outlet	Tamb + 5° C.	Flow Rate	Name [° C.] [° C.] [kg/min]	[° C.] [° C.]	[L/min]
-30	-30	4	50/Max	-25	8
-20a	-20	-20	6	50	-15
8	-20b	0	4	-10a	-10
-10	-10	6	50	-5	8
-10b	-10	5	4		
0a	0	0	6	50	5
8	0b	10	4	5a	5
5	5	4	50	10	8
15a	15	15	4	40	20
8					

TABLE-US-00002

TABLE 2 Temperature and pressure data		WG-HP (configuration as per solid EWG-HP (configuration as per thick lines in Comparative FIG. 1) thick solid lines in FIG. 3A)	
T4	T5	T6	T7
p1	p2	p3	p4
T4	T5	T6	T7
p1	p2	p3	p4
Units	Condition	° C.	° C.
° C.	° C.	° C.	° C.
bar	bar	bar	bar
-30a	27.52	NA	-34.47
27.52	9.56	NA	0.81
9.56	18.00	5.34	-35.89
-27.82	23.49	6.54	-20a
36.2	NA	-25.88	36.2
11.88	NA	1.18	11.88
23.49	6.54	-20b	60.59
NA	-23.07	60.59	20.71
NA	1.33	20.71	48.35
12.52	-24.87	48.35	-10a
55.27	NA	-15.76	55.27
18.84	NA	1.78	18.84
44.99	11.53	-18.22	-10b
58.94	NA	-13.89	58.94
19.99	NA	1.92	19.99
50.44	13.16	-15.30	50.44
13.16	13.16	0a	60.37
NA	-6.43	60.37	20.61
NA	2.52	20.61	51.42
13.47	-8.29	51.42	13.47
13.47	0b	57.33	NA
-4.45	57.33	19.3	NA
2.71	19.3	50.73	13.25
-5.32	50.73	13.25	50.73
13.25	13.25	5a	58.95
NA	-0.02	58.95	19.99
NA	3.15	19.99	50.64
13.22	-1.04	50.64	13.22
50.64	13.22	15a	42.57
NA	10.78	42.57	13.84
NA	4.48	13.84	40.57
10.33	10.49	40.57	10.33
40.57	10.33	10.33	

In the table above, the temperature and pressure conditions correspond to those indicated in FIGS. 2A and 2B hereof, where applicable.

[0113] From the results reported above and in FIG. 3E, it can be seen that the present thermal management system produces in this operating mode a COP on average 34.1% (22.3%-43.1%) higher than the prior heat pump systems and a heating capacity that is on average 7.0% (5.4%-9.2%) higher than the prior systems, for conditions -30a, -20a and -10a conditions.

Examples 1B5-1B8

[0114] Example 1A is repeated, except that the refrigerant blends identified in Table E1B' below as Refrigerants RE1B5-RE1B8 are used instead of 1234yf under the same set of conditions identified in Table 1 above. Applicants note that the use of Refrigerants RE1B5 RE1B8 in this configuration of the present invention results in system operation at pressures about the same as when using 1234yf (and also using R134a), that is, the refrigeration system operates at pressures in the ranges typically experience with prior stationary air conditioning that use the refrigerants R404A and R22. The present invention is thus shown to be capable of operating, particularly with Refrigerants RE1B1-RE1B4, within these higher pressure ranges, to achieve advantageous capacity and efficiency at such pressures while at the same time without the substantial environmental disadvantages associated with the use of those high GWP refrigerants.

TABLE-US-00003 TABLE E1B REFRIGERANTS ASHRAE COMPONENTS, WT % GWP

Example Refrigerant Name CO.sub.2 R1234yf R1132(E) R32 (AR4) Ex1B1 RE1B1 R454C 0 78.5 0 21.5 148 Ex1B2 RE1B2 R455A 3.0 75.5 0 21.5 148 Ex1B3 RE1B3 R474A 0.0 77.0 23 0 3 Ex1B4 RE1B4 R474B 0.0 68.5 31.5 0 3

[0115] The temperatures and pressures in the respective systems in accordance with this example operating in this manner are reported in the Tables E1B1 through E1B4, with the results using the refrigerant blends of this Example 1B reported as EWG-HP and the results from the use of the system of Comparative Example 1 (but using R1234yf of the present invention) reported as WG-HP.

TABLE-US-00004 TABLE E1B1 Temperature and pressure data - RE1B1 Refrigerant (R454C)

WG-HP HP (configuration as per solid thick EWG-HP (configuration as per thick lines in Comparative FIG. 1 using 1234yf) solid lines in FIG. 3A using R454C) T4 T5 T6 T7 p1 p2 p3 p4 T4 T5 T6 T7 p1 p2 p3 p4 Units Condition ° C. ° C. ° C. ° C. bar bar bar bar ° C. ° C. ° C. ° C. bar bar bar bar -30a 27.5 NA -34.4 27.5 9.81 NA 0.81 9.81 17.9 17.9 -35.8 17.9 9.88 9.9 1.11 9.88 -20a 36.2 NA -25.88 36.2 12.22 NA 1.19 12.22 23.4 23.4 -27.7 23.4 11.44 11.4 1.58 11.44 -20b 60.6 NA -23.0 60.6 21.38 NA 1.34 21.38 48.3 48.3 -24.8 48.3 20.87 20.9 1.79 20.87 -10a 55.3 NA -15.7 55.3 19.44 NA 1.80 19.44 44.9 44.9 -18.1 44.9 19.34 19.3 2.34 19.34 -10b 58.9 NA -13.8 58.9 20.63 NA 1.94 20.63 50.4 50.4 -15.2 50.4 21.85 21.8 2.62 21.85 0a 60.4 NA -6.3 60.4 21.27 NA 2.56 21.27 51.4 51.4 -8.2 51.4 22.32 22.3 3.40 22.32 0b 57.3 NA -4.3 57.3 19.91 NA 2.75 19.91 50.7 50.7 -5.2 50.7 21.99 22.0 3.78 21.99 5a 58.9 NA 0.0 58.9 20.63 NA 3.20 20.63 50.6 50.6 -0.9 50.6 21.94 21.9 4.38 21.94 15a 42.6 NA 10.8 42.6 14.25 NA 4.57 14.25 40.5 40.5 10.6 40.5 17.47 17.5 6.37 17.47

[0116] It is expected that the present thermal management system produces for each of the conditions using RE1B1 in this operating mode a COP that is about the same as or higher than the prior heat pump systems and a heating capacity that is about the same as or higher than the prior systems.

TABLE-US-00005 TABLE E1B2 Temperature and pressure data - RE1B2 Refrigerant (R455A)

WG-HP (configuration as per solid thick EWG-HP (configuration as per thick lines in Comparative FIG. 1 using 1234yf) solid lines in FIG. 3A using R455A) T4 T5 T6 T7 p1 p2 p3 p4 T4 T5 T6 T7 p1 p2 p3 p4 Units Condition ° C. ° C. ° C. ° C. bar bar bar bar ° C. ° C. ° C. ° C. bar bar bar bar -30a 27.5 NA -34.4 27.5 9.81 NA 0.81 9.81 17.9 17.9 -35.8 17.9 11.60 11.6 1.18 11.60 -20a 36.2 NA -25.88 36.2 12.22 NA 1.19 12.22 23.4 23.4 -27.7 23.4 13.34 13.3 1.69 13.34 -20b 60.6 NA -23.0 60.6 21.38 NA 1.34 21.38 48.3 48.3 -24.8 48.3 23.60 23.6 1.91 23.60 -10a 55.3 NA -15.7 55.3 19.44 NA 1.80 19.44 44.9 44.9 -18.1 44.9 21.96 22.0 2.50 21.96 -10b 58.9 NA -13.8 58.9 20.63 NA 1.94 20.63 50.4 50.4 -15.2 50.4 24.66 24.7 2.80 24.66 0a 60.4 NA -6.3 60.4 21.27

NA 2.56 21.27 51.4 51.4 -8.2 51.4 25.17 25.2 3.63 25.17 0b 57.3 NA -4.3 57.3 19.91 NA 2.75
19.91 50.7 50.7 -5.2 50.7 24.81 24.8 4.03 24.81 5a 58.9 NA 0.0 58.9 20.63 NA 3.20 20.63 50.6
50.6 -0.9 50.6 24.76 24.8 4.68 24.76 15a 42.6 NA 10.8 42.6 14.25 NA 4.57 14.25 40.5 40.5 10.6
40.5 19.93 19.9 6.79 19.93

[0117] It is expected that the present thermal management system produces for each of the conditions using RE1B2 in this operating mode a COP that is about the same as or higher than the prior heat pump systems and a heating capacity that is about the same as or higher than the prior systems.

TABLE-US-00006 TABLE E1B3 Temperature and pressure data - RE1B3 Refrigerant (R474A)

WG-HP (configuration as per solid thick EWG-HP (configuration as per thick solid lines in Comparative FIG. 1 using 1234yf) lines in FIG. 3A using R474A) Condition T4 T5 T6 T7 p1 p2 p3 p4 T4 T5 T6 T7 p1 p2 p3 p4 Units ° C. ° C. ° C. ° C. bar bar bar bar ° C. ° C. ° C. ° C. Bar bar bar bar
bar -30a 27.5 NA -34.4 27.5 9.81 NA 0.81 9.81 17.9 17.9 -35.8 17.9 5.70 5.7 0.76 5.70 -20a 36.2
NA -25.88 36.2 12.22 NA 1.19 12.22 23.4 23.4 -27.7 23.4 6.69 6.7 1.19 6.69 -20b 60.6 NA -23.0
60.6 21.38 NA 1.34 21.38 48.3 48.3 -24.8 48.3 12.88 12.9 1.24 12.88 -10a 55.3 NA -15.7 55.3
19.44 NA 1.80 19.44 44.9 44.9 -18.1 44.9 11.85 11.9 1.64 11.85 -10b 58.9 NA -13.8 58.9 20.63
NA 1.94 20.63 50.4 50.4 -15.2 50.4 13.54 13.5 1.84 13.54 0a 60.4 NA -6.3 60.4 21.27 NA 2.56
21.27 51.4 51.4 -8.2 51.4 13.86 13.9 2.39 13.86 0b 57.3 NA -4.3 57.3 19.91 NA 2.75 19.91 50.7
50.7 -5.2 50.7 13.64 13.6 2.67 13.64 5a 58.9 NA 0.0 58.9 20.63 NA 3.20 20.63 50.6 50.6 -0.9
50.6 13.60 13.6 3.10 13.60 15a 42.6 NA 10.8 42.6 14.25 NA 4.57 14.25 40.5 40.5 10.6 40.5 10.61
10.6 4.53 10.61

[0118] It is expected that the present thermal management system produces for each of the conditions using RE1B3 in this operating mode a COP that is about the same as or higher than the prior heat pump systems and a heating capacity that is about the same as or higher than the prior systems.

TABLE-US-00007 TABLE E1B4 Temperature and pressure data - RE1B4 Refrigerant (R474B)

WG-HP (configuration as per solid thick EWG-HP (configuration as per thick lines in Comparative FIG. 1 using 1234yf) solid lines in FIG. 3A using R474B) Condition T4 T5 T6 T7 p1 p2 p3 p4 T4 T5 T6 T7 p1 p2 p3 p4 Units ° C. ° C. ° C. ° C. bar bar bar bar ° C. ° C. ° C. ° C. Bar bar bar bar
bar -30a 27.5 NA -34.4 27.5 9.81 NA 0.81 9.81 17.9 17.9 -35.8 17.9 9.64 9.6 1.17 9.64 -20a 36.2 NA
-25.88 36.2 12.22 NA 1.19 12.22 23.4 23.4 -27.7 23.4 11.16 11.2 1.66 11.16 -20b 60.6 NA -23.0
60.6 21.38 NA 1.34 21.38 48.3 48.3 -24.8 48.3 20.32 20.3 1.88 20.32 -10a 55.3 NA -15.7 55.3
19.44 NA 1.80 19.44 44.9 44.9 -18.1 44.9 18.83 18.8 2.45 18.83 -10b 58.9 NA -13.8 58.9 20.63
NA 1.94 20.63 50.4 50.4 -15.2 50.4 21.28 21.3 2.73 21.28 0a 60.4 NA -6.3 60.4 21.27 NA 2.56
21.27 51.4 51.4 -8.2 51.4 21.74 21.7 3.53 21.74 0b 57.3 NA -4.3 57.3 19.91 NA 2.75 19.91 50.7
50.7 -5.2 50.7 21.41 21.4 3.92 21.41 5a 58.9 NA 0.0 58.9 20.63 NA 3.20 20.63 50.6 50.6 -0.9
50.6 21.36 21.4 4.54 21.36 15a 42.6 NA 10.8 42.6 14.25 NA 4.57 14.25 40.5 40.5 10.6 40.5 17.01
17.0 6.56 17.01

[0119] It is expected that the present thermal management system produces for each of the conditions using RE1B4 in this operating mode a COP that is about the same as or higher than the prior heat pump systems and a heating capacity that is about the same as or higher than the prior systems.

Examples 1B1-1B4

[0120] Example 1A is repeated, except that the refrigerant blends identified in Table E1B below as Refrigerants RE1B5-RE1B8 are used instead of 1234yf under the same set of conditions identified in Table 1 above.

TABLE-US-00008 TABLE E1B REFRIGERANTS ASHRAE COMPONENTS, WT % GWP

Example Refrigerant Name R134a R1234yf R152a R32 R1234ze (AR4) Ex1B5 RE1B5 R516A
8.5 77.5 14.0 0.0 0.0 142 Ex1B6 RE1B6 R444A 0.0 0.0 5.0 12.0 83.0 92 Ex1B7 RE1B7 R457C 0.0
78.0 14.5 7.5 0.0 72 Ex1B8 RE1B8 R457D 0.0 82.0 14.0 4.0 0.0 48

[0121] The temperatures and pressures in the respective systems in accordance with this example operating in this manner are reported in the Table E1B5 below with the results using the refrigerant blends of this Examples 1B5 reported as EWG-HP and the results from the use of the system of Comparative Example 1 (but using R1234yf of the present invention) reported as WG-HP.

TABLE-US-00009 TABLE E1B5 Temperature and pressure data - RE1B5 Refrigerant (R516A)

WG-HP	EWG-HP	Condition	T4	T5	T6	T7	p1	p2	p3	p4	T4	T5	T6	T7	p1	p2	p3	p4	Units
° C.	° C.	° C.	° C.	° C.	° C.	° C.	bar	bar	bar	bar	° C.	° C.	° C.	° C.	bar	bar	bar	bar	° C.
-30a	27.5	NA	-34.4	27.5	9.81	NA	0.81	9.81	17.9	17.9	-35.8	17.9	5.70	5.7	0.76	5.70	-20a	36.2	NA
-25.88	36.2	12.22	NA	1.19	12.22	23.4	23.4	-27.7	23.4	6.69	6.7	1.19	6.69	-20b	60.6	NA	-23.0	60.6	21.38
NA	1.34	21.38	48.3	48.3	-24.8	48.3	12.88	12.9	1.24	12.88	-10a	55.3	NA	-15.7	55.3	19.44	NA	1.80	19.44
44.9	44.9	-18.1	44.9	11.85	11.9	1.64	11.85	-10b	58.9	NA	-13.8	58.9	20.63	NA	1.94	20.63	50.4	50.4	-15.2
50.4	13.54	13.5	1.84	13.54	0a	60.4	NA	-6.3	60.4	21.27	NA	2.56	21.27	51.4	51.4	-8.2	51.4	13.86	13.9
2.39	13.86	0b	57.3	NA	-4.3	57.3	19.91	NA	2.75	19.91	50.7	50.7	-5.2	50.7	13.64	13.6	2.67	13.64	5a
58.9	NA	0.0	58.9	20.63	NA	3.20	20.63	50.6	50.6	-0.9	50.6	13.60	13.6	3.10	13.60	15a	42.6	NA	10.8
42.6	14.25	NA	4.57	14.25	40.5	40.5	10.6	40.5	10.61	10.6	4.53	10.61							

[0122] The relative capacity and effectiveness (COP) of the two systems operating in this manner are reported and illustrated in FIGS. 32A and 32B, with the results from this Example 1B reported as EWG-HP and the results from the use of the system of Comparative Example 1 reported as WG-HP.

[0123] From the results reported above and in FIGS. 32A and 32B, it can be seen that the present thermal management system produces for each of the conditions using RE1B5 in this operating mode a COP that is about the same as or higher than the prior heat pump systems and a heating capacity that is about the same as or higher than the prior systems.

[0124] The temperatures and pressures in the respective systems in accordance with this example operating in this manner with R444A (RE1B6) are reported in the Table E1B6 below with the results using the refrigerant blends of this Example 1B6 reported as EWG-HP and the results from the use of the system of Comparative Example 1 (but using R1234yf of the present invention) reported as WG-HP.

TABLE-US-00010 TABLE E1B6 Temperature and pressure data - RE1B6 Refrigerant (R444A)

WG-HP	EWG-HP	Condition	T4	T5	T6	T7	p1	p2	p3	p4	T4	T5	T6	T7	p1	p2	p3	p4	Units
° C.	° C.	° C.	° C.	° C.	° C.	° C.	bar	bar	bar	bar	° C.	° C.	° C.	° C.	bar	bar	bar	bar	° C.
-30a	27.5	NA	-34.4	27.5	11.29	NA	0.63	11.29	17.9	17.9	-35.8	17.9	6.73	6.7	0.59	6.73	-20a	36.2	NA
-25.8	36.2	13.90	NA	0.95	13.90	23.4	23.4	-27.7	23.4	7.84	7.8	0.87	7.84	-20b	60.6	NA	-23.0	60.6	23.61
NA	1.08	23.61	48.3	48.3	-24.8	48.3	14.62	14.6	0.99	14.62	-10a	55.3	NA	-15.6	55.3	21.59	NA	1.48	21.59
44.9	44.9	-18.1	44.9	13.51	13.5	1.33	13.51	-10b	58.9	NA	-13.8	58.9	22.83	NA	1.60	22.83	50.4	50.4	-15.2
50.4	15.3	1.51	15.33	0a	60.4	NA	-6.3	60.4	23.50	NA	2.15	23.50	51.4	51.4	-8.2	51.4	15.68	15.7	2.00
15.68	0b	57.3	NA	-4.3	57.3	22.09	NA	2.32	22.09	50.7	50.7	-5.2	50.7	15.43	15.4	2.25	15.43	5a	58.9
NA	0.0	58.9	22.83	NA	2.73	22.83	50.6	50.6	-0.9	50.6	15.40	15.4	2.64	15.40	15a	42.6	NA	10.8	42.6
16.09	NA	3.99	16.09	40.5	40.5	10.6	40.5	12.16	12.2	3.95	12.16								

[0125] The relative capacity and effectiveness (COP) of the two systems operating in this manner are reported and illustrated in FIGS. 33A and 33B, with the results from this Example 1B6 reported as EWG-HP and the results from the use of the system of Comparative Example 1 reported as WG-HP.

[0126] From the results reported above and from FIGS. 33A and 33B, it can be seen that the present thermal management system produces for each of the conditions using RE1B6 in this operating mode a COP that is higher than the prior heat pump systems and a heating capacity that is about the same as or higher than the prior systems.

[0127] The temperatures and pressures in the respective systems in accordance with this example operating in this manner with R457C (RE1B7) are reported in the Table E1B7 below with the results using the refrigerant blends of this Example 1B7 reported as EWG-HP and the results from

the use of the system of Comparative Example 1 (but using R1234yf of the present invention) reported as WG-HP.

TABLE-US-00011 TABLE E1B7 Temperature and pressure data - RE1B7 Refrigerant (R457C)																																																						
WG-HP		EWG-HP		Condition		T4	T5	T6	T7	p1	p2	p3	p4	T4	T5	T6	T7	p1	p2	p3	p4	Units																																
° C.	° C.	bar	bar	bar	bar	° C.	° C.	° C.	° C.	bar	bar	bar	bar	° C.	° C.	° C.	° C.	bar	bar	bar	bar	° C.																																
12.18	17.9	17.9	-35.8	17.9	7.27	7.3	0.86	7.27	-20a	36.2	NA	-25.8	36.2	15.00	NA	1.34	15.00	23.4	23.4	-27.7	23.4	8.47	8.5	1.24	8.47	-20b	60.6	NA	-23.0	60.6	25.52	NA	1.51	25.52	48.3	48.3	-24.8																	
48.3	15.77	15.8	1.40	15.77	-10a	55.3	NA	-15.6	55.3	23.32	NA	2.02	23.32	44.9	44.9	-18.1	44.9	14.58	14.6	1.84	14.58	-10b	58.9	NA	-13.8	58.9	24.68	NA	2.18	24.68	50.4	50.4	-15.2	50.4	16.54	16.5	2.06	16.54	0a	60.4	NA	-6.3	60.4	25.41	NA	2.86	25.41	51.4	51.4	-8.2	51.4	16.92	16.9	2.68
16.92	0b	57.3	NA	-4.3	57.3	23.86	NA	3.08	23.86	50.7	50.7	-5.2	50.7	16.65	16.7	2.98	16.65	5a	58.9	NA	0.0	58.9	24.68	NA	3.58	24.68	50.6	50.6	-0.9	50.6	16.62	16.6	3.47	16.62	15a	42.6	NA	10.8	42.6	17.36	NA	5.10	17.36	40.5	40.5	10.6	40.5	13.12	13.1	5.06	13.12			

[0128] From the results reported above and in FIGS. 35A and 35B, it can be seen that the present thermal management system produces for each of the conditions using RE1B8 in this operating mode a COP that is higher than the prior heat pump systems and a heating capacity that is about the same as or higher than the prior systems.

[0129] The temperatures and pressures in the respective systems in accordance with this example operating in this manner are reported in the Table E1B8 below with the results using the refrigerant blends of this Example 1B8 reported as EWG-HP and the results from the use of the system of Comparative Example 1 (but using R1234yf of the present invention) reported as WG-HP.

TABLE-US-00012 TABLE E1B8 Temperature and pressure data - RE1B4 Refrigerant (R457D)																						
WG-HP		EWG-HP		Condition		T4	T5	T6	T7	p1	p2	p3	p4	T4	T5	T6	T7	p1	p2	p3	p4	Units
° C.	° C.	bar	bar	bar	bar	° C.	° C.	° C.	° C.	bar	bar	bar	bar	° C.	° C.	° C.	° C.	bar	bar	bar	bar	° C.
12.18	17.9	17.9	-35.8	17.9	7.27	7.3	0.86	7.27	-20a	36.2	NA	-25.8	36.2	15.00	NA	1.34	15.00	23.4	23.4	-27.7	23.4	8.47
8.5	1.24	8.47	-20b	60.6	NA	-23.0	60.6	25.52	NA	1.51	25.52	48.3	48.3	-24.8	48.3	15.77	15.8	1.40	15.77	-10a	55.3	NA
-15.6	55.3	23.32	NA	2.02	23.32	44.9	44.9	-18.1	44.9	14.58	14.6	1.84	14.58	-10b	58.9	NA	-13.8	58.9	24.68	NA	2.18	24.68
50.4	50.4	-15.2	50.4	16.54	16.5	2.06	16.54	0a	60.4	NA	-6.3	60.4	25.41	NA	2.86	25.41	51.4	51.4	-8.2	51.4	16.92	16.9
2.68	16.92	0b	57.3	NA	-4.3	57.3	23.86	NA	3.08	23.86	50.7	50.7	-5.2	50.7	16.65	16.7	2.98	16.65	5a	58.9	NA	
0.0	58.9	24.68	NA	3.58	24.68	50.6	50.6	-0.9	50.6	16.62	16.6	3.47	16.62	15a	42.6	NA	10.8	42.6	17.36	NA	5.10	17.36
40.5	40.5	10.6	40.5	13.12	13.1	5.06	13.12															

[0130] From the results reported above and from FIGS. 35A and 35B, it can be seen that the present thermal management system produces for each of the conditions using RE1B8 in this operating mode a COP that is higher than the prior heat pump systems and a heating capacity that is about the same as or higher than the prior systems.

Example 1C—Heat Pump Mode with Inner Evaporator/Condenser in the Loop to Warm Cabin Air and PTC Upstream of the Chiller

[0131] As with Example 1A, applicants have come to appreciate that when ambient temperatures are relatively low, EVs as previously configured, including as described in Comparative Example 1 and illustrated by the thick solid lines in Comparative FIG. 1, can have a problem with insufficient condenser surface area at the inner condenser 1 to provide complete condensation, which can result in problems with system capacity and efficiency (COP). In addition, Applicants have found that systems of the present invention as described and illustrated herein, particularly in connection with FIG. 3A, can be even further dramatically improved in terms of overall performance with relatively simple and low-cost further modifications involving primarily the relative placement of the chiller, the PTC and the coolant pump. In particular, applicant has come to appreciate that in relatively low ambient temperatures the relative power consumption associated with the operation of the coolant pump located downstream of the PTC, which itself is located downstream of the chiller, as illustrated in FIG. 3A can be undesirably high. This undesirable feature can occur due to the

relatively low viscosity at the suction side of the coolant pump, which causes a potentially dramatic and undesirable increase in power consumption in the circuit. In one desirable alternative to this configuration, as illustrated in FIG. 3B, the PTC is moved to a point upstream of the chiller, which results in a reduction in the power consumption associated with the operation of the system. An illustration of using the configuration of FIG. 3B in the system as otherwise configured in FIG. 3A is illustrated in FIG. 3F.

Example 1D—Heat Pump Mode with Inner Evaporator/Condenser in the Loop to Warm Cabin Air and PTC and Coolant Pump Upstream of the Chiller

[0132] In this Example 1D, the configuration of Example 1C is repeated, except that a further modification includes locating the coolant pump upstream of the chiller and downstream of the PTC, as illustrated in FIG. 3C. This configuration is a specially preferred, as with the configuration in Example 1C, in relatively low ambient temperatures conditions in order to minimize or at least reduce the relative power consumption associated with the operation of the coolant pump.

Applicants have come to appreciate that locating both the PTC and the coolant pump upstream of the chiller, as illustrated in FIG. 3C, is even more preferred from the standpoint of minimizing the power required for the coolant pump to operate this portion of the systems of the present invention.

Example 1C—Heat Pump Mode with Inner Evaporator/Condenser in the Loop to Warm Cabin Air and PTC Upstream of the Chiller

[0133] As with Example 1A, applicants have come to appreciate that when ambient temperatures are relatively low, EVs as previously configured, including as described in Comparative Example 1 and illustrated by the thick solid lines in Comparative FIG. 1, can have a problem with insufficient condenser surface area at the inner condenser 1 to provide complete condensation, which can result in problems with system capacity and efficiency (COP). In addition, Applicants have found that systems of the present invention as described and illustrated herein, particularly in connection with FIG. 3A, can be even further dramatically improved in terms of overall performance with relatively simple and low-cost further modifications involving primarily the relative placement of the chiller, the PTC and the coolant pump. In particular, applicant has come to appreciate that in relatively low ambient temperatures the relative power consumption associated with the operation of the coolant pump located downstream of the PTC, which itself is located downstream of the chiller, as illustrated in FIG. 3A can be undesirably high. This undesirable feature can occur due to the relatively low viscosity at the suction side of the coolant pump, which causes a potentially dramatic and undesirable increase in power consumption in the circuit. In one desirable alternative to this configuration, as illustrated in FIG. 3B, the PTC is moved to a point upstream of the chiller, which results in a reduction in the power consumption associated with the operation of the system. An illustration of a system using the configuration of FIG. 3B in the system as otherwise configured in FIG. 3A is illustrated in FIG. 3G.

Examples 2-30: Heating and/or Cooling in Ambient Conditions from -35C to 45C and For Several Combinations of Heating and Cooling Needs in an EV

[0134] For the purposes of understanding the advantages and features of the present inventive system compared to prior heat pump systems over the relevant range of ambient temperature conditions and over exemplary alternative scenarios relating to heating and/or cooling needs in an operating and/or charging EV, it is noted that typical prior systems would operate in warm temperatures to primarily provide cabin air conditioning to an EV using the typical configuration illustrated by the solid thick lines in Comparative FIG. 2. In this typical prior air conditioning cycle configuration in which cabin air is to be cooled, the OHE 3 is the predominate source for condensing the refrigerant and the evaporator 2 is the cooling unit for cabin air.

[0135] Applicants have come to appreciate that while heat pump systems need an evaporative heat (energy) source, EVs also have cooling needs beyond the cooling of cabin air, that is, heat sources that represent waste heat for use as the evaporative source (highly efficient) and that the present improved system has been developed so as to take advantage of these features when possible and to

achieve results not previously achievable. In particular, applicants have noted that there are two main areas that may need cooling, depending on conditions, on all EVs: [0136] 1. The battery needs cooling during charging and may, at times, need cooling during discharging (vehicle operation). The battery may also need warming initially in very cold weather. [0137] 2. The vehicle drive motor(s)/inverter require cooling during vehicle operation. These devices also benefit in efficiency from the possibility of receiving warming in cold weather.

Unfortunately for systems of prior designs, the cooling needs and/or heating needs, and the ideal temperatures of the battery and motor/inverter, vary greatly depending on the vehicle ambient conditions, on the driving conditions, on stationary battery charging, on how long the vehicle has been off before running or on how long the vehicle has been driving. In some circumstances, one or both of these (battery or motor/inverter) may need cooling while the other does not or may need warming.

[0138] Applicants' highly flexible system is able to achieve highly efficient evaporative heat conditions using available heat while not compromising the other sources. The present system provides highly beneficial performance by unique combinations of components, including the possibility to use three evaporative heat source locations (chiller, Outside Heat Exchanger [OHE] or an Inner Heat Exchanger, while at the same time waste heat from the battery during charging and/or discharging can be used at the chiller or the OHE and ambient air can be used at the OHE or the Inner Heat Exchanger as the heat source. In addition, one or both of the above heat sources can be warming up while another is used as the evaporative heat source for the heat pump. In addition, an electrically operated Positive Temperature Coefficient (PTC) heater can also be used alone or in series with the evaporative heat sources at the chiller and/or the OHE. The available evaporative heat sources usable according to the systems of the present invention include those listed below: [0139] 1. Air only (at the OHE) [0140] 2. Motor and inverter (at the OHE) [0141] 3. Motor and inverter (at the chiller) [0142] 4. Motor and Inverter and PTC (at the chiller) [0143] 5. Battery (at the chiller) [0144] 6. Battery and PTC (at the chiller) [0145] 7. Battery (at the OHE) [0146] 8. Motor, inverter and battery (at the chiller) [0147] 9. Motor, inverter, battery and PTC (at the chiller) [0148] 10. Motor, inverter, battery (at the OHE) [0149] 11. PTC (at the chiller) [0150] 12. PTC (at the OHE) [0151] 13. Motor, Inverter and PTC (at the OHE) [0152] 14. Motor, inverter, battery and PTC (at the OHE) [0153] 15. Battery and PTC (at the OHE) [0154] 16. Air only (at the chiller) [0155] 17. Battery, motor and inverter (at the chiller) [0156] 18. Dehumidification (at the evaporator)

[0157] Several of many possible applicable system operating modes of the present invention for the evaporative heat sources and their use location are shown in the following Examples (with associated figures), with the relevant ambient temperatures (ambient) as well as the condition of the cabin being hot, cool, cold or acceptable to the passenger (OK). Furthermore, the condition of the battery, motor and inverter are specified as being hot, warm, cool or cold or acceptable (OK) are defined. The driving condition between start and comfort as well as the battery charge being active (Yes) or not active (No) are described. The applicable operating conditions for the different heat sources are indicated. In the Figures, thick lines in the vapor compression system indicate refrigerant flow at relatively high pressure (only line pressure drop from compressor discharge), dashed thick lines in the vapor compression system indicate refrigerant flow at a reduced pressure (after throttling in an expansion valve), and thin lines indicate refrigerant conduits (and corresponding units) that have been bypassed. Similarly, thick lines in the coolant section indicate active coolant flow and thin lines indicate coolant conduits that have been bypassed, while dashed thick lines indicate the coolant could optionally be flowing but is not for the results reported in the example.

Example 2A—Vehicle Heating Between 0° C. and 15° C.—with Evaporative Heat Source 1—Air only at OHE and under Applicable Conditions of 15° C., 5° C. and -5° C. Using RE1B1

[0158] In this example, a system of the present invention is configured for operation to heat cabin

air during periods of low ambient temperatures is illustrated in FIG. 4. The pressure data obtained by operating the system of this example with R1234yf and with RE1B1 is reported in the following Table E2A1. The use of Refrigerant RE1B1 (R545C) is reported in Table E2A2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using identified evaporative heat source(s) being used.

TABLE-US-00013 TABLE E2A1 Temperature and pressure data For Ambient Temperature of 15° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 55.7 NA 10.6 55.7 10.6 55.7 18.6 NA RE1B1 55.7 NA 10.6 55.7 10.6 55.7 30.1 NA (R545C)

TABLE-US-00014 TABLE E2A2 Using R545C with Evaporative Heat Source 1 - Air only at OHE and under applicable conditions of 15° C., 5° C. and -5° C. Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

This is an efficient mode for vehicle heating between 0° C. and 15° C. It can and will likely be used in conjunction with self-heating of the battery and the motor and inverter either in series or parallel. Example 2B—Vehicle Heating Between 0° C. and 15° C.—with Evaporative Heat Source 1—Air only at OHE and under Applicable Conditions of 15° C., 5° C. and -5° C. Using RE1B2

[0159] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 4. The temperature and pressure data obtained by operating the system of this example with R1234yf and with RE1B2 is reported in the following Table E2B1. The use of Refrigerant RE1B2 (R455A) is reported in Table E2B2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using identified evaporative heat source(s) being used. TABLE-US-00015 TABLE E2B1 Temperature and pressure data For Ambient Temperature of 15° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 55.7 NA 10.6 55.7 10.6 55.7 18.6 NA RE1B2 55.7 NA 10.6 55.7 10.6 55.7 33.4 NA (R455A)

TABLE-US-00016 TABLE E2B2 Using R455A with Evaporative Heat Source 1 - Air only at OHE and under applicable conditions of 15° C., 5° C. and -5° C. Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

This is an efficient mode for vehicle heating between 0° C. and 15° C. It can and will likely be used in conjunction with self-heating of the battery and the motor and inverter either in series or parallel. Example 2C—Vehicle Heating Between 0° C. And 15° C.—with Evaporative Heat Source 1-Air Only at OHE and Under Applicable Conditions of 15° C., 5° C. And -5° C. Using RE1B3

[0160] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 4. The pressure data obtained by operating the system of this example with R1234yf and with RE1B3 is reported in the following Table E2C1. The use of Refrigerant RE1B3 (R474A) is reported in Table E2C2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using identified evaporative heat source(s) being used.

TABLE-US-00017 TABLE E2C1 Temperature and pressure data For Ambient Temperature of 15° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 55.7 NA 10.6 55.7 10.6 55.7 18.6 NA RE1B3 55.7 NA 10.6 55.7 10.6 55.7 27.0 NA (R474A)

TABLE-US-00018 TABLE E2C2 Using R474A with Evaporative Heat Source 1 - Air only at OHE and under applicable conditions of 15° C., 5° C. and -5° C. Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

This is an efficient mode for vehicle heating between 0° C. and 15° C. It can and will likely be used in conjunction with self-heating of the battery and the motor and inverter either in series or parallel. Example 2D—Vehicle Heating Between 0° C. and 15° C.—with Evaporative Heat Source 1-Air only at OHE and under Applicable Conditions of 15° C., 5° C. and -5° C. Using RE1B4

[0161] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 4. The pressure data obtained by operating the system of this example with R1234yf and with RE1B4 is reported in the following Table E2D1. The use of Refrigerant RE1B4 (R474B) is reported in Table E2D2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using identified evaporative heat source(s) being used.

TABLE-US-00019 TABLE E2D1 Temperature and pressure data For Ambient Temperature of 15° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 55.7 NA 10.6 55.7 10.6 55.7 18.6 NA RE1B4 55.7 NA 10.6 55.7 10.6 55.7 29.3 NA (R474B)

TABLE-US-00020 TABLE E2D2 Using R474B with Evaporative Heat Source 1 - Air only at OHE and under applicable conditions of 15° C., 5° C. and -5° C. Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

This is an efficient mode for vehicle heating between 0° C. and 15° C. It can and will likely be used in conjunction with self-heating of the battery and the motor and inverter either in series or parallel. Example 2E—Vehicle Heating Between 0° C. and 15° C.—with Evaporative Heat Source 1 —Air only at OHE and under Applicable Conditions of 15° C., 5° C. and -5° C. Using RE1B5

[0162] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 4. The pressure data obtained by operating the system of this example with R1234yf and with RE1B5 is reported in the following Table E2E1.

[0163] The use of Refrigerant RE1B5 (R516A) is reported in Table E2E2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using identified evaporative heat source(s) being used.

TABLE-US-00021 TABLE E2E1 Temperature and pressure data For Ambient Temperature of 15° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 55.7 NA 10.6 55.7 18.6 NA 4.5 18.6 RE1B6 55.7 NA 10.6 55.7 19.2 NA 4.5 19.2 (R516A)

TABLE-US-00022 TABLE E2E2 Using R516A with Evaporative Heat Source 1 - Air only at OHE and under applicable conditions of 15° C., 5° C. and -5° C. Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35**

Cool Cool Cool Start Yes Cold Cold Cold Comfort No

This is an efficient mode for vehicle heating between 0° C. and 15° C. It can and will likely be used in conjunction with self-heating of the battery and the motor and inverter either in series or parallel. Example 2F—Vehicle Heating Between 0° C. and 15° C.—with Evaporative Heat Source 1-Air only at OHE and under Applicable Conditions of 15° C., 5° C. and -5° C. Using RE1B2

[0164] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 4. The pressure data obtained by operating the system of this example with R1234yf and with RE1B6 is reported in the following Table E2F1. The use of Refrigerant RE1B6 (R444A) is reported in Table E2F2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using identified evaporative heat source(s) being used.

TABLE-US-00023 TABLE E2F1 Temperature and pressure data For Ambient Temperature of 15° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 3.4 59.0 20.0 NA 2.2 20.0 RE1B6 59.0 NA 3.4 59.0 22.8 NA 1.8 22.8 (R444A)

TABLE-US-00024 TABLE E2F2 Using R444A with Evaporative Heat Source 1 - Air only at OHE and under applicable conditions of 15° C., 5° C. and -5° C. Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

This is an efficient mode for vehicle heating between 0° C. and 15° C. It can and will likely be used in conjunction with self-heating of the battery and the motor and inverter either in series or parallel. Example 2G—Vehicle Heating Between 0° C. And 15° C.—with Evaporative Heat Source 1-Air Only at OHE and Under Applicable Conditions of 15° C., 5° C. And -5° C. Using RE1B7

[0165] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 4. The pressure data obtained by operating the system of this example with R1234yf and with RE1B7 is reported in the following Table E2G1. The use of Refrigerant RE1B7 (R457C) is reported in Table E2G2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using identified evaporative heat source(s) being used.

TABLE-US-00025 TABLE E2G1 Temperature and pressure data For Ambient Temperature of 15° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 3.4 59.0 20.0 NA 2.2 20.0 RE1B7 59.0 NA 3.4 59.0 24.7 NA 2.5 24.7 (R457C)

TABLE-US-00026 TABLE E2G2 Using R457C with Evaporative Heat Source 1 - Air only at OHE and under applicable conditions of 15° C., 5° C. and -5° C. Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

This is an efficient mode for vehicle heating between 0° C. and 15° C. It can and will likely be used in conjunction with self-heating of the battery and the motor and inverter either in series or parallel. Example 2H—Vehicle Heating Between 0° C. And 15° C.—with Evaporative Heat Source 1-Air Only at OHE and Under Applicable Conditions of 15° C., 5° C. And -5° C. Using RE1B8

[0166] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 4. The pressure data obtained by operating the system of this example with R1234yf and with RE1B8 is reported in the following

Table E2H1. The use of Refrigerant RE1B8 (R457D) is reported in Table E2H2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using identified evaporative heat source(s) being used.

TABLE-US-00027 TABLE E2H1 Temperature and pressure data For Ambient Temperature of 15° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 3.4 59.0 20.0 NA 2.2 20.0 RE1B4 59.0 NA 3.4 59.0 22.9 NA 2.3 22.9 (R457D)

TABLE-US-00028 TABLE E2H2 Using R457D with Evaporative Heat Source 1 - Air only at OHE and under applicable conditions of 15° C., 5° C. and -5° C. Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

This is an efficient mode for vehicle heating between 0° C. and 15° C. It can and will likely be used in conjunction with self-heating of the battery and the motor and inverter either in series or parallel. Example 3A—Vehicle Heating Between -10° C. and 15° C.—Using R454C with Evaporative Heat Source 2: Motor and Inverter at OHE

[0167] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 5. The data obtained by operating the system of this example with R1234yf and with RE1B1 is reported in the following Table E3A1. The use of Refrigerant RE1B1 (R454C) is reported in Table E3A2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00029 TABLE E3A1 Temperature and pressure data For Ambient Temperature of 5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 3.4 59.0 20.0 NA 2.2 20.0 RE1B1 59.0 NA 3.4 59.0 32.1 NA 3.1 32.1 (R454C)

TABLE-US-00030 TABLE E3A2 Using R454C with Evaporative Heat Source 2: Motor and Inverter at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0168] This is an efficient mode for vehicle heating between -10° C. and 15° C. after the motor and inverter are warmed up and need (or can tolerate) some cooling. It can and will likely be used in conjunction with self-heating of the battery or cooling of the battery at the chiller. This can also be used to de-ice the OHE.

Example 3B—Vehicle Heating Between -10° C. and 15° C.—Using R455A with Evaporative Heat Source 2: Motor and Inverter at OHE

[0169] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 5. The data obtained by operating the system of this example with R1234yf and with RE1B2 is reported in the following Table E3B1. The use of Refrigerant RE1B2 (R455A) is reported in Table E3B2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00031 TABLE E3B1 Temperature and pressure data For Ambient Temperature of 5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 3.4 59.0 20.0 NA 2.2 20.0 RE1B2 59.0 NA 3.4 59.0 35.5 NA 3.3 35.5 (R455A)

TABLE-US-00032 TABLE E3B2 Using R455A with Evaporative Heat Source 2: Motor and Inverter at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25** Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0170] This is an efficient mode for vehicle heating between -10° C. and 15° C. after the motor and inverter are warmed up and need (or can tolerate) some cooling. It can and will likely be used in conjunction with self-heating of the battery or cooling of the battery at the chiller. This can also be used to de-ice the OHE.

Example 3C—Vehicle Heating Between -10° C. and 15° C.—Using R474A with Evaporative Heat Source 2: Motor and Inverter at OHE

[0171] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 5. The data obtained by operating the system of this example with R1234yf and with RE1B3 is reported in the following Table E3C1. The use of Refrigerant RE1B3 (R474A) is reported in Table E3C2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00033 TABLE E3C1 Temperature and pressure data For Ambient Temperature of 5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 3.4 59.0 20.0 NA 2.2 20.0 RE1B3 59.0 NA 3.4 59.0 28.9 NA 2.9 28.9 (R474A)

TABLE-US-00034 TABLE E3B2 Using R474C with Evaporative Heat Source 2: Motor and Inverter at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25** Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0172] This is an efficient mode for vehicle heating between -10° C. and 15° C. after the motor and inverter are warmed up and need (or can tolerate) some cooling. It can and will likely be used in conjunction with self-heating of the battery or cooling of the battery at the chiller. This can also be used to de-ice the OHE.

Example 3D—Vehicle Heating Between -10° C. and 15° C.—Using R474B with Evaporative Heat Source 2: Motor and Inverter at OHE

[0173] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 5. The data obtained by operating the system of this example with R1234yf and with RE1B4 is reported in the following Table E3D1. The use of Refrigerant RE1B4 (R474B) is reported in Table E3D2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00035 TABLE E3D1 Temperature and pressure data For Ambient Temperature of 5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 3.4 59.0 20.0 NA 2.2 20.0 RE1B4 59.0 NA 3.4 59.0 31.3 NA 3.3 31.3 (R474B)

TABLE-US-00036 TABLE E3D2 Using R474D with Evaporative Heat Source 2: Motor and Inverter at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool**

Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cold Comfort No -5 OK OK OK Start
Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25
Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No

[0174] This is an efficient mode for vehicle heating between -10° C. and 15° C. after the motor and inverter are warmed up and need (or can tolerate) some cooling. It can and will likely be used in conjunction with self-heating of the battery or cooling of the battery at the chiller. This can also be used to de-ice the OHE.

Example 3E—Vehicle Heating Between -10° C. and 15° C.—Using R516A with Evaporative Heat Source 2: Motor and Inverter at OHE

[0175] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 5. The data obtained by operating the system of this example with R1234yf and with RE1B5 is reported in the following Table E3E1. The use of Refrigerant RE1B5 (R516A) is reported in Table E3E2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00037 TABLE E3E1 Temperature and pressure data For Ambient Temperature of 5° C.

T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA
3.4 59.0 20.0 NA 2.2 20.0 RE1B5 59.0 NA 3.4 59.0 20.6 NA 2.2 20.6 (R516A)

TABLE-US-00038 TABLE E3E2 Using R516A with Evaporative Heat Source 2: Motor and Inverter at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25
Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool
Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start
Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25**
Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No

[0176] This is an efficient mode for vehicle heating between -10° C. and 15° C. after the motor and inverter are warmed up and need (or can tolerate) some cooling. It can and will likely be used in conjunction with self-heating of the battery or cooling of the battery at the chiller. This can also be used to de-ice the OHE.

Example 3F—Vehicle Heating Between -10° C. and 15° C.—Using R444A with Evaporative Heat Source 2: Motor and Inverter at OHE Using R444A

[0177] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 5. The data obtained by operating the system of this example with R1234yf and with RE1B6 is reported in the following Table E3F1. The use of Refrigerant RE1B6 (R444A) is reported in Table E3B6 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00039 TABLE E3F1 Temperature and pressure data For Ambient Temperature of 5° C.

T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA
3.4 59.0 20.0 NA 2.2 20.0 RE1F2 59.0 NA 3.4 59.0 22.8 NA 1.8 22.8 (R444A)

TABLE-US-00040 TABLE E3F2 Using R444A with Evaporative Heat Source 2: Motor and Inverter at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25
Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool
Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start
Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25**
Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No

[0178] This is an efficient mode for vehicle heating between -10°C . and 15°C . after the motor and inverter are warmed up and need (or can tolerate) some cooling. It can and will likely be used in conjunction with self-heating of the battery or cooling of the battery at the chiller. This can also be used to de-ice the OHE.

Example 3G—Vehicle Heating Between -10°C . and 15°C .—Using R457C with Evaporative Heat Source 2: Motor and Inverter at OHE Using R457C

[0179] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 5. The data obtained by operating the system of this example with R1234yf and with RE1B7 is reported in the following Table E3G1. The use of Refrigerant RE1B7 (R457C) is reported in Table E3G2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00041 TABLE E3G1 Temperature and pressure data For Ambient Temperature of 5°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 3.4 59.0 20.0 NA 2.2 20.0 RE1B7 59.0 NA 3.4 59.0 24.7 NA 2.5 24.7 (R457C)

TABLE-US-00042 TABLE E3B7 Using R457C with Evaporative Heat Source 2: Motor and Inverter at OHE Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0180] This is an efficient mode for vehicle heating between -10°C . and 15°C . after the motor and inverter are warmed up and need (or can tolerate) some cooling. It can and will likely be used in conjunction with self-heating of the battery or cooling of the battery at the chiller. This can also be used to de-ice the OHE.

Example 3H—Vehicle Heating Between -10°C . and 15°C .—Using R457D with Evaporative Heat Source 2: Motor and Inverter at OHE Using R457D

[0181] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 5. The data obtained by operating the system of this example with R1234yf and with RE1B4 is reported in the following Table E3H1. The use of Refrigerant RE1B8 (R457D) is reported in Table E3H2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00043 TABLE E3H1 Temperature and pressure data For Ambient Temperature of 5°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 3.4 59.0 20.0 NA 2.2 20.0 RE1B8 59.0 NA 3.4 59.0 22.9 NA 2.3 22.9 (R457D)

TABLE-US-00044 TABLE E3H2 Using R457D with Evaporative Heat Source 2: Motor and Inverter at OHE Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0182] This is an efficient mode for vehicle heating between -10°C . and 15°C . after the motor and inverter are warmed up and need (or can tolerate) some cooling. It can and will likely be used in conjunction with self-heating of the battery or cooling of the battery at the chiller. This can also be used to de-ice the OHE.

Example 4A—Vehicle Heating Between -15°C . and 15°C .—Evaporative Heat Source 2: Motor and Inverter at Chiller Using R454C

[0183] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 6. The data obtained by operating the system of this example with R1234yf and with RE1B1 produces the results as reported in the following Table E4A1. The use of Refrigerant RE1B1 (R454C) is reported in Table E4A2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00045 TABLE E4A1 Temperature and pressure data For Ambient Temperature of -5°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B1 62.3 NA -13.05 62.3 34.2 NA 2.9 34.2 (R454C)

TABLE-US-00046 TABLE E4A2 Evaporative heat source 2: Motor and Inverter at Chiller Motor & Ambient, $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0184] This is also an efficient mode for vehicle heating between -15°C . and 15°C . after the motor and inverter are warmed up and need or can tolerate some cooling. It can also be used to cool the motor and inverter in warm weather. It will likely be used when the battery is at an appropriate or acceptable temperature.

Example 4B—Vehicle Heating Between -15°C . and 15°C .—Evaporative Heat Source 2: Motor and Inverter at Chiller Using R455A

[0185] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 6. The data obtained by operating the system of this example with R1234yf and with RE1B2 produces the results as reported in the following Table E4B1. The use of Refrigerant RE1B2 (R455A) is reported in Table E4B2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00047 TABLE E4B1 Temperature and pressure data For Ambient Temperature of -5°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B2 62.3 NA -13.05 62.3 37.7 NA 3.0 37.7 (R455A)

TABLE-US-00048 TABLE E4B2 Evaporative heat source 2: Motor and Inverter at Chiller Motor & Ambient, $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0186] This is also an efficient mode for vehicle heating between -15°C . and 15°C . after the motor and inverter are warmed up and need or can tolerate some cooling. It can also be used to cool the motor and inverter in warm weather. It will likely be used when the battery is at an appropriate or acceptable temperature.

Example 4C—Vehicle Heating Between -15°C . and 15°C .—Evaporative Heat Source 2: Motor and Inverter at Chiller Using R474A

[0187] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 6. The data obtained by operating the system of this example with R1234yf and with RE1B3 produces the results as reported in the following Table E4C1. The use of Refrigerant RE1B3 (R474A) is reported in Table

E4C2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00049 TABLE E4C1 Temperature and pressure data For Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B3 62.3 NA -13.05 62.3 30.8 NA 2.7 30.8 (R474A)

TABLE-US-00050 TABLE E4C2 Evaporative heat source 2: Motor and Inverter at Chiller Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cold Cold Cold Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No** [0188] This is also an efficient mode for vehicle heating between -15° C. and 15° C. after the motor and inverter are warmed up and need or can tolerate some cooling. It can also be used to cool the motor and inverter in warm weather. It will likely be used when the battery is at an appropriate or acceptable temperature.

Example 4D—Vehicle Heating Between -15° C. and 15° C.—Evaporative Heat Source 2: Motor and Inverter at Chiller Using R474B

[0189] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 6. The data obtained by operating the system of this example with R1234yf and with RE1B4 produces the results as reported in the following Table E4D1. The use of Refrigerant RE1B4 (R474B) is reported in Table E4D2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00051 TABLE E4D1 Temperature and pressure data For Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B3 62.3 NA -13.05 62.3 30.8 NA 2.7 30.8 (R474B)

TABLE-US-00052 TABLE E4D2 Evaporative heat source 2: Motor and Inverter at Chiller Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cold Cold Cold Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No** [0190] This is also an efficient mode for vehicle heating between -15° C. and 15° C. after the motor and inverter are warmed up and need or can tolerate some cooling. It can also be used to cool the motor and inverter in warm weather. It will likely be used when the battery is at an appropriate or acceptable temperature.

Example 4E—Vehicle Heating Between -15° C. and 15° C.—Evaporative Heat Source 2: Motor and Inverter at Chiller Using R516A

[0191] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 6. The data obtained by operating the system of this example with R1234yf and with RE1B5 produces the results as reported in the following Table E4E1. The use of Refrigerant RE1B5 (R516A) is reported in Table E4E2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00053 TABLE E4E1 Temperature and pressure data For Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B5 62.3 NA -13.05 62.3 22.1 NA 2.0 22.1 (R516A)

TABLE-US-00054 TABLE E4E2 Evaporative heat source 2: Motor and Inverter at OHE Motor &

Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0192] This is also an efficient mode for vehicle heating between -15° C. and 15° C. after the motor and inverter are warmed up and need or can tolerate some cooling. It can also be used to cool the motor and inverter in warm weather. It will likely be used when the battery is at an appropriate or acceptable temperature.

Example 4F—Vehicle Heating Between -15° C. and 15° C.—Evaporative Heat Source 2: Motor and Inverter at Chiller Using R444A

[0193] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 6. The data obtained by operating the system of this example with R1234yf and with RE1B6 produces the results as reported in the following Table E4F1. The use of Refrigerant RE1B6 (R444A) is reported in Table E4F2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00055 TABLE E4F1 Temperature and pressure data For Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B6 62.3 NA -13.05 62.3 24.4 NA 1.7 24.4 (R444A)

TABLE-US-00056 TABLE E4F2 Evaporative heat source 2: Motor and Inverter at Chiller Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0194] This is also an efficient mode for vehicle heating between -15° C. and 15° C. after the motor and inverter are warmed up and need or can tolerate some cooling. It can also be used to cool the motor and inverter in warm weather. It will likely be used when the battery is at an appropriate or acceptable temperature.

Example 4G—Vehicle Heating Between -15° C. and 15° C.—Evaporative Heat Source 2: Motor and Inverter at Chiller Using 457C

[0195] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 6. The data obtained by operating the system of this example with R1234yf and with RE1B7 produces the results as reported in the following Table E4G1. The use of Refrigerant RE1B7 (R457C) is reported in Table E4G2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00057 TABLE E4G1 Temperature and pressure data For Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B7 62.3 NA -13.05 62.3 26.48 NA 2.2 26.4 (R457C)

TABLE-US-00058 TABLE E4G2 Evaporative heat source 2: Motor and Inverter at Chiller Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0196] This is also an efficient mode for vehicle heating between -15°C . and 15°C . after the motor and inverter are warmed up and need or can tolerate some cooling. It can also be used to cool the motor and inverter in warm weather. It will likely be used when the battery is at an appropriate or acceptable temperature.

Example 4H—Vehicle Heating Between -15°C . and 15°C .—Evaporative Heat Source 2: Motor and Inverter at Chiller Using R457D

[0197] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 6. The data obtained by operating the system of this example with R1234yf and with RE1B8 produces the results as reported in the following Table E4H1. The use of Refrigerant RE1B8 (R457D) is reported in Table E4D2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00059 TABLE E4H1 Temperature and pressure data For Ambient Temperature of -5°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B8 62.3 NA -13.05 62.3 24.5 NA 2.1 24.5 (R457D)

TABLE-US-00060 TABLE E4H2 Evaporative heat source 2: Motor and Inverter at Chiller Motor & Ambient, $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cold Cold Cold Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0198] This is also an efficient mode for vehicle heating between -15°C . and 15°C . after the motor and inverter are warmed up and need or can tolerate some cooling. It can also be used to cool the motor and inverter in warm weather. It will likely be used when the battery is at an appropriate or acceptable temperature.

Example 5A—Motor and Inverter Temperature Control While Heating the EV—Evaporative Heat Source 4: Motor, Inverter and PTC at Chiller Using R454C

[0199] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 7. The data obtained by operating the system with R1234yf and with RE1B1 is reported in the following Table E5A1. The use of Refrigerant RE1B1 (R454C) is reported in Table E5A2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00061 TABLE E5A1 Temperature and pressure data For Ambient Temperature of -5°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B1 62.3 NA -13.05 62.3 34.2 NA 2.9 34.2 (R454C)

TABLE-US-00062 TABLE E5A2 Evaporative Heat Source 4: Motor, Inverter and PTC at chiller Motor & Ambient, $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0200] In this mode the motor and inverter temperature can be maintained while heating the vehicle. In this case, the battery is assumed to be warming up while charging or at an appropriate temperature.

Example 5B—Motor and Inverter Temperature Control While Heating the EV—Evaporative Heat Source 4: Motor, Inverter and PTC at Chiller Using R455A

[0201] In this example, a system of the present invention configured for operation to heat cabin air

during periods of low ambient temperatures is illustrated in FIG. 7. The data obtained by operating the system with R1234yf and with RE1B2 is reported in the following Table E5B1. The use of Refrigerant RE1B2 (R455A) is reported in Table E5B2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00063 TABLE E5B1 Temperature and pressure data For Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B2 62.3 NA -13.05 62.3 37.7 NA 3.0 37.7 (R455A)

[0202] In this mode the motor and inverter temperature can be maintained while heating the vehicle. In this case, the battery is assumed to be warming up while charging or at an appropriate temperature.

Example 5C—Motor and Inverter Temperature Control While Heating the EV—Evaporative Heat Source 4: Motor, Inverter and PTC at Chiller R474A

[0203] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 7. The data obtained by operating the system with R1234yf and with RE1B3 is reported in the following Table E5C1. The use of Refrigerant RE1C2 (R474A) is reported in Table E5C2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00064 TABLE E5C1 Temperature and pressure data For Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B3 62.3 NA -13.05 62.3 30.8 NA 2.7 30.8 (R474A)

[0204] In this mode the motor and inverter temperature can be maintained while heating the vehicle. In this case, the battery is assumed to be warming up while charging or at an appropriate temperature.

Example 5D—Motor and Inverter Temperature Control While Heating the EV—Evaporative Heat Source 4: Motor, Inverter and PTC at Chiller Using R474B

[0205] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 7. The data obtained by operating the system with R1234yf and with RE1B4 is reported in the following Table E5D1. The use of Refrigerant REID2 (R474B) is reported in Table E5D2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00065 TABLE E5D1 Temperature and pressure data For Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B4 62.3 NA -13.05 62.3 33.4 NA 3.0 33.4 (R474B)

[0206] In this mode the motor and inverter temperature can be maintained while heating the vehicle. In this case, the battery is assumed to be warming up while charging or at an appropriate temperature.

Example 5E—Motor and Inverter Temperature Control While Heating the EV—Evaporative Heat Source 4: Motor, Inverter and PTC at Chiller Using R516A

[0207] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 7. The data obtained by operating the system with R1234yf and with RE1B5 is reported in the following Table E5E1. The use of Refrigerant RE1B5 (R516A) is reported in Table E5E2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00066 TABLE E5E1 Temperature and pressure data For Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B5 62.3 NA -13.05 62.3 22.1 NA 2.0 22.1 (R516A)

TABLE-US-00067 TABLE E5E2 Evaporative Heat Source 4: Motor, Inverter and PTC at chiller Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No** [0208] In this mode the motor and Inverter temperature can be maintained while heating the vehicle. In this case, the battery is assumed to be warming up while charging or at an appropriate temperature.

Example 5F—Motor and Inverter Temperature Control While Heating the EV—Evaporative Heat Source 4: Motor, Inverter and PTC at Chiller Using R444A

[0209] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 7. The data obtained by operating the system with R1234yf and with RE1B6 is reported in the following Table E5F1. The use of Refrigerant RE1B6 (R444A) is reported in Table E5F2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00068 TABLE E5F1 Temperature and pressure data For Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B6 62.3 NA -13.05 62.3 24.4 NA 1.7 24.4 (R444A)

[0210] In this mode the motor and Inverter temperature can be maintained while heating the vehicle. In this case, the battery is assumed to be warming up while charging or at an appropriate temperature.

Example 5G—Motor and Inverter Temperature Control While Heating the EV—Evaporative Heat Source 4: Motor, Inverter and PTC at Chiller R457C

[0211] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 7. The data obtained by operating the system with R1234yf and with RE1B7 is reported in the following Table E5G1. The use of Refrigerant RE1B7 (R457C) is reported in Table E5G2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00069 TABLE E5G1 Temperature and pressure data For Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B7 62.3 NA -13.05 62.3 26.4 NA 2.2 36.4 (R457C)

[0212] In this mode the motor and inverter temperature can be maintained while heating the vehicle. In this case, the battery is assumed to be warming up while charging or at an appropriate temperature.

Example 5H—Motor and Inverter Temperature Control While Heating the EV—Evaporative Heat Source 4: Motor, Inverter and PTC at Chiller Using R457D

[0213] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 7. The data obtained by operating the system with R1234yf and with RE1B8 is reported in the following Table E5H1. The use of Refrigerant RE1B8 (R457D) is reported in Table E5H2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00070 TABLE E5H1 Temperature and pressure data For Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.05 62.3 21.5 NA 2.0 21.5 RE1B8 62.3 NA -13.05 62.3 24.5 NA 2.1 24.5 (R457D)

[0214] In this mode the motor and inverter temperature can be maintained while heating the

vehicle. In this case, the battery is assumed to be warming up while charging or at an appropriate temperature.

Example 6—Vehicle Heating Between -25°C . and 5°C . With Battery Charging

[0215] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 8 and in FIGS. 3A and 3B. The data obtained by operating the system of this example with R1234yf and with each of RE1B1, RE1B2, RE1B3, RE1B4, RE1B5, RE1B6, RE1B7 and RE1B8 is reported in the following Table E6.

TABLE-US-00071 TABLE E6 Temperature and pressure data for Ambient Temperature of -15°C .
T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 60.6 NA
-8.94 60.6 20.7 NA 2.3 20.7 RE1B1 60.6 NA -8.94 60.6 33.1 NA 3.3 33.1 (R454C) RE1B2 60.6
NA -8.94 60.6 36.6 NA 3.5 36.6 (R455A) RE1B3 60.6 NA -8.94 60.6 29.8 NA 3.1 29.8 (R474A)
RE1B4 60.6 NA -8.94 60.6 32.3 NA 3.5 32.3 (R474B) RE1B5 60.6 NA -8.94 60.6 21.4 NA 2.3
21.4 (R516A) RE1B6 60.6 NA -8.94 60.6 23.6 NA 2.0 23.6 (R444A) RE1B7 60.6 NA -8.94 60.6
25.5 NA 2.6 25.5 (R547C) RE1B8 60.6 NA -8.94 60.6 23.6 NA 2.5 23.6 (R547D)

[0216] This is an efficient mode for vehicle heating between -25°C . and 5°C . while the battery is charging. Excess heat from charging or from the charging source can be used to heat the vehicle.

Example 7A—Vehicle Heating with Ambient Between -35°C . and 5°C . With Battery Charging-Evaporative Heat Source 5: Motor, Inverter and PTC at Chiller

[0217] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 9. This system is operated with R1234yf and with each of RE1B1, RE1B2, RE1B3, RE1B4, RE1B5, RE1B6, RE1B7 and RE1B8 is reported in the following Table E7A1 and produces the results as reported in the following Table E7A2 by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00072 TABLE E7A1 Temperature and pressure data for Ambient Temperature of -25°C .
T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 60.7 NA
5.6 60.7 20.8 NA 3.8 20.8 RE1B1 (R454C) 60.7 NA 5.6 60.7 33.2 NA 5.5 33.2 RE1B2 (R455A)
60.7 NA 5.6 60.7 36.7 NA 5.8 36.7 RE1B3 (R474A) 60.7 NA 5.6 60.7 29.9 NA 5.1 29.9 RE1B4
(R474B) 60.7 NA 5.6 60.7 32.4 NA 5.6 32.4 RE1B5 (R516A) 60.7 NA 5.6 60.7 21.4 NA 3.9 21.4
RE1B6 (R444A) 60.7 NA 5.6 60.7 23.7 NA 3.3 23.7 RE1B7 (R457C) 60.7 NA 5.6 60.7 25.6 NA
4.3 25.6 RE1B4 (R474D) 60.7 NA 5.6 60.7 23.7 NA 4.1 23.7

TABLE-US-00073 TABLE E7A2 Evaporative Heat Source 5: Motor, Inverter and PTC at Chiller
Motor & Ambient, $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort**
No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold**
Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort
No

[0218] This is an efficient mode for vehicle heating while maintaining the battery temperature between ambient conditions of -35°C . and 5°C . This mode would likely be used at the start of a drive after charging.

Example 8A—Vehicle Heating with Ambient Between -15°C . and 5°C . With Battery Charging-Evaporative Heat Source 7: Battery, PTC and Air at OHE

[0219] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 10. This system is operated with R1234yf and with RE1B1 and produces the results as reported in the following Table E8A1. The use of Refrigerant RE1B1 (R454C) is reported in Table E8A2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions

when using the identified evaporative heat source(s).

TABLE-US-00074 TABLE E8A1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA -6.69 59.0 20.0 NA 2.5 20.0 RE1B1 59.0 NA -6.69 59.0 32.1 NA 3.6 32.1 (R454C)

TABLE-US-00075 TABLE E8A2 Evaporative Heat Source 7: Battery, PTC and Air at OHE Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0220] This is an efficient mode for vehicle heating between -15° C. and 5° C. while the battery is charging. Excess heat from charging can be used to heat the vehicle at the OHE.

Example 8B—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-Evaporative Heat Source 7: Battery, PTC and Air at OHE

[0221] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 10. This system is operated with R1234yf and with RE1B2 and produces the results as reported in the following Table E8B1. The use of Refrigerant RE1B2 (R455A) is reported in Table E8B2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00076 TABLE E8B1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA -6.69 59.0 20.0 NA 2.5 20.0 RE1B2 59.0 NA -6.69 59.0 35.5 NA 3.8 35.5 (R455A)

TABLE-US-00077 TABLE E8B2 Evaporative Heat Source 7: Battery, PTC and Air at OHE Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0222] This is an efficient mode for vehicle heating between -15° C. and 5° C. while the battery is charging. Excess heat from charging can be used to heat the vehicle at the OHE.

Example 8C—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-Evaporative Heat Source 7: Battery, PTC and Air at OHE

[0223] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 10. This system is operated with R1234yf and with RE1B3 and produces the results as reported in the following Table E8C1. The use of Refrigerant RE1B3 (R474A) is reported in Table E8C2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00078 TABLE E8C1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA -6.69 59.0 20.0 NA 2.5 20.0 RE1B3 59.0 NA -6.69 59.0 28.9 NA 3.4 28.9 (R474A)

TABLE-US-00079 TABLE E8C2 Evaporative Heat Source 7: Battery, PTC and Air at OHE Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes**

Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Comfort No
[0224] This is an efficient mode for vehicle heating between -15° C. and 5° C. while the battery is charging. Excess heat from charging can be used to heat the vehicle at the OHE.

Example 8D—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-Evaporative Heat Source 7: Battery, PTC and Air at OHE

[0225] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 10. This system is operated with R1234yf and with RE1B4 and produces the results as reported in the following Table E8D1. The use of Refrigerant RE1B4 (R474B) is reported in Table E8D2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00080 TABLE E8D1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA -6.69 59.0 20.0 NA 2.5 20.0 RE1B4 59.0 NA -6.69 59.0 31.3 NA 3.7 31.3 (R474B)

TABLE-US-00081 TABLE E8D2 Evaporative Heat Source 7: Battery, PTC and Air at OHE Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0226] This is an efficient mode for vehicle heating between -15° C. and 5° C. while the battery is charging. Excess heat from charging can be used to heat the vehicle at the OHE.

Example 8E—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-Evaporative Heat Source 7: Battery, PTC and Air at OHE Using R516A

[0227] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 10. This system is operated with R1234yf and with RE1B5 and produces the results as reported in the following Table E8E1. The use of Refrigerant RE1B5 (R516A) is reported in Table E8E2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00082 TABLE E8E1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA -6.69 59.0 20.0 NA 2.5 20.0 RE1B5 59.0 NA -6.69 59.0 20.6 NA 2.5 20.6 (R516A)

TABLE-US-00083 TABLE E8E2 Evaporative Heat Source 7: Battery, PTC and Air at OHE Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0228] This is an efficient mode for vehicle heating between -15° C. and 5° C. while the battery is charging. Excess heat from charging can be used to heat the vehicle at the OHE.

Example 8F—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-Evaporative Heat Source 7: Battery, PTC and Air at OHE Using R444A

[0229] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 10. This system is operated with R1234yf and with RE1B6 and produces the results as reported in the following Table E8F1. The use of Refrigerant RE1B6 (R444A) is reported in Table E8F2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions

when using the identified evaporative heat source(s).

TABLE-US-00084 TABLE E8F1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA -6.69 59.0 20.0 NA 2.5 20.0 RE1B6 59.0 NA -6.69 59.0 22.8 NA 2.1 22.8 (R444A)

TABLE-US-00085 TABLE E8F2 Evaporative Heat Source 7: Battery, PTC and Air at OHE Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0230] This is an efficient mode for vehicle heating between -15° C. and 5° C. while the battery is charging. Excess heat from charging can be used to heat the vehicle at the OHE.

Example 8G—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-Evaporative Heat Source 7: Battery, PTC and Air at OHE Using R457C

[0231] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 10. This system is operated with R1234yf and with RE1B7 and produces the results as reported in the following Table E8G1. The use of Refrigerant RE1B7 (R457C) is reported in Table E8G2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00086 TABLE E8G1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA -6.69 59.0 20.0 NA 2.5 20.0 RE1B7 59.0 NA -6.69 59.0 24.7 NA 2.8 24.7 (R457C)

TABLE-US-00087 TABLE E8G2 Evaporative Heat Source 7: Battery, PTC and Air at OHE Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0232] This is an efficient mode for vehicle heating between -15° C. and 5° C. while the battery is charging. Excess heat from charging can be used to heat the vehicle at the OHE.

Example 8H—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-Evaporative Heat Source 7: Battery, PTC and Air at OHE

[0233] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 10. This system is operated with R1234yf and with RE1B8 and produces the results as reported in the following Table E8H1. The use of Refrigerant RE1B8 (R457D) is reported in Table E8H2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00088 TABLE E8H1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA -6.69 59.0 20.0 NA 2.5 20.0 RE1B8 59.0 NA -6.69 59.0 22.9 NA 2.7 22.9 (R457D)

TABLE-US-00089 TABLE E8H2 Evaporative Heat Source 7: Battery, PTC and Air at OHE Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes**

Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Comfort No

[0234] This is an efficient mode for vehicle heating between -15° C. and 5° C. while the battery is charging. Excess heat from charging can be used to heat the vehicle at the OHE.

Example 9A—Vehicle Heating with Ambient Between -15° C. and 15° C. While Driving—

Evaporative Heat Source 8: Inverter and Battery at Chiller Using R454C

[0235] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 11. This system is operated with R1234yf and with RE1B1 and produces the results as reported in the following Table E9A1. The use of Refrigerant RE1B1 (R454C) is reported in Table E9A2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00090 TABLE E9A1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.27 62.3 21.5 NA 2.0 21.5 RE1B1 62.3 NA -13.27 62.3 34.2 NA 2.8 34.2 (R454C)

TABLE-US-00091 TABLE E9A2 Evaporative Heat Source 8: Inverter and Battery at Chiller Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm

Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0236] Excess heat from the battery and motor and inverter can be used at the chiller. This can also be used with the enhanced heat pump configuration (dotted line).

Example 9B—Vehicle Heating with Ambient Between -15° C. and 15° C. While Driving—

Evaporative Heat Source 8: Inverter and Battery at Chiller Using R454C

[0237] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 11. This system is operated with R1234yf and with RE1B2 and produces the results as reported in the following Table E9B1. The use of Refrigerant RE1B2 (R454C) is reported in Table E9B2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00092 TABLE E9B1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.27 62.3 21.5 NA 2.0 21.5 RE1B2 62.3 NA -13.27 62.3 37.7 NA 3.0 37.7 (R455A)

TABLE-US-00093 TABLE E9B2 Evaporative Heat Source 8: Inverter and Battery at Chiller Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm

Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0238] Excess heat from the battery and motor and inverter can be used at the chiller. This can also be used with the enhanced heat pump configuration (dotted line).

Example 9C—Vehicle Heating with Ambient Between -15° C. and 15° C. While Driving—

Evaporative Heat Source 8: Inverter and Battery at Chiller Using R474A

[0239] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 11. This system is operated with R1234yf and with RE1B3 and produces the results as reported in the following Table E9C1. The use of Refrigerant RE1B3 (R474A) is reported in Table E9C2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions

when using the identified evaporative heat source(s).

TABLE-US-00094 TABLE E9C1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.27 62.3 21.5 NA 2.0 21.5 RE1B3 62.3 NA -13.27 62.3 30.8 NA 2.7 30.8 (R474A)

TABLE-US-00095 TABLE E9C2 Evaporative Heat Source 8: Inverter and Battery at Chiller Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0240] This is an efficient mode for vehicle heating between -15° C. and 15° C. while driving.

Excess heat from the battery and motor and inverter can be used at the chiller. This can also be used with the enhanced heat pump configuration (dotted line).

Example 9D—Vehicle Heating with Ambient Between -15° C. and 15° C. While Driving—Evaporative Heat Source 8: Inverter and Battery at Chiller Using R474B

[0241] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 11. This system is operated with R1234yf and with RE1B4 and produces the results as reported in the following Table E9D1. The use of Refrigerant RE1B4 (R474B) is reported in Table E9D2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00096 TABLE E9D1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.27 62.3 21.5 NA 2.0 21.5 RE1B4 62.3 NA -13.27 62.3 33.4 NA 3.0 33.4 (R474B)

TABLE-US-00097 TABLE E9D2 Evaporative Heat Source 8: Inverter and Battery at Chiller Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0242] Excess heat from the battery and motor and inverter can be used at the chiller. This can also be used with the enhanced heat pump configuration (dotted line).

Example 9E—Vehicle Heating with Ambient Between -15° C. and 15° C. While Driving—Evaporative Heat Source 8: Inverter and Battery at Chiller Using R516A

[0243] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 11. This system is operated with R1234yf and with RE1B5 and produces the results as reported in the following Table E9E1. The use of Refrigerant RE1B5 (R516A) is reported in Table E9E2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00098 TABLE E9E1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA -6.69 59.0 20.0 NA 2.5 20.0 RE1B5 59.0 NA -6.69 59.0 20.9 NA 6.6 30.9 (R516A)

TABLE-US-00099 TABLE E9E2 Evaporative Heat Source 8: Inverter and Battery at Chiller Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold**

Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start
Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No
[0244] Excess heat from the battery and motor and inverter can be used at the chiller. This can also be used with the enhanced heat pump configuration (dotted line).

Example 9F—Vehicle Heating with Ambient Between -15° C. and 15° C. While Driving—
Evaporative Heat Source 8: Inverter and Battery at Chiller R444A

[0245] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 11. This system is operated with R1234yf and with RE1B6 and produces the results as reported in the following Table E9F1. The use of Refrigerant RE1B6 (R444A) is reported in Table E9F2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00100 TABLE E9F1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA -6.69 59.0 20.0 NA 2.5 20.0 RE1B6 59.0 NA -6.69 59.0 23.1 NA 5.9 23.1 (R444A)

TABLE-US-00101 TABLE E9F2 Evaporative Heat Source 8: Inverter and Battery at Chiller Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start**
Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No
[0246] This is an efficient mode for vehicle heating between -15° C. and 15° C. while driving. Excess heat from the battery and motor and inverter can be used at the chiller. This can also be used with the enhanced heat pump configuration (dotted line).

Example 9G—Vehicle Heating with Ambient Between -15° C. and 15° C. While Driving—
Evaporative Heat Source 8: Inverter and Battery at Chiller Using R457C

[0247] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 11. This system is operated with R1234yf and with RE1B7 and produces the results as reported in the following Table E9G1. The use of Refrigerant RE1B7 (R457C) is reported in Table E9G2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00102 TABLE E9G1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA -6.69 59.0 20.0 NA 2.5 20.0 RE1B7 59.0 NA -6.69 59.0 25.0 NA 7.3 23.2 (R457C)

TABLE-US-00103 TABLE E9G2 Evaporative Heat Source 8: Inverter and Battery at Chiller Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start**
Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No
[0248] Excess heat from the battery and motor and inverter can be used at the chiller. This can also be used with the enhanced heat pump configuration (dotted line).

Example 9H—Vehicle Heating with Ambient Between -15° C. and 15° C. While Driving—
Evaporative Heat Source 8: Inverter and Battery at Chiller Using R457D

[0249] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 11. This system is operated with R1234yf and with RE1B8 and produces the results as reported in the following Table E9H1. The

use of Refrigerant RE1B8 (R457D) is reported in Table E9H2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00104 TABLE E9H1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA -6.69 59.0 20.0 NA 2.5 20.0 RE1B8 59.0 NA -6.69 59.0 22.9 NA 2.7 22.9 (R457D)

TABLE-US-00105 TABLE E9H2 Evaporative Heat Source 8: Inverter and Battery at Chiller Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No** [0250] Excess heat from the battery and motor and inverter can be used at the chiller. This can also be used with the enhanced heat pump configuration (dotted line).

Example 10A—Vehicle Heating with Ambient Between -25° C. and 5° C. While Driving—Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Using R 454C

[0251] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 12. This system is operated with R1234yf and with RE1B1 and produced the results as reported in the following Table E10A1. The use of Refrigerant RE1B1 (R454C) is reported in Table E10A2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00106 TABLE E10A1 Temperature and pressure data for Ambient Temperature of -15° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.6 NA 22.77 59.6 20.3 NA 6.4 20.3 RE1B1 59.6 NA 22.77 59.6 32.5 NA 9.2 32.5 (R454C)

TABLE-US-00107 TABLE E10A2 Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0252] This is an efficient mode for vehicle heating between -25° C. and 5° C. while driving. PTC heat can be used to heat the vehicle while maintaining the battery and motor and inverter temperatures. This can also be used with the enhanced heat pump configuration (dotted line).

Example 10B—Vehicle Heating with Ambient Between -25° C. and 5° C. While Driving—Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Using R455A

[0253] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 12. This system is operated with R1234yf and with RE1B2 and produced the results as reported in the following Table E10B1. The use of Refrigerant RE1B2 (R455A) is reported in Table E10B2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00108 TABLE E10B1 Temperature and pressure data for Ambient Temperature of -15° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 59.6 NA 22.77 59.6 20.3 NA 6.4 20.3 RE1B2 59.6 NA 22.77 59.6 35.9 NA 9.8 35.9 (R455A)

TABLE-US-00109 TABLE E10B2 Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Motor & Ambient, $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes

OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot
Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool**
Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes
Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool
Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold
Comfort No

[0254] This is an efficient mode for vehicle heating between -25° C. and 5° C. while driving. PTC heat can be used to heat the vehicle while maintaining the battery and motor and inverter temperatures. This can also be used with the enhanced heat pump configuration (dotted line).

Example 10C—Vehicle Heating with Ambient Between -25° C. and 5° C. While Driving—
Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Using R474A

[0255] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 12. This system is operated with R1234yf and with RE1B3 and produced the results as reported in the following Table E10C1. The use of Refrigerant RE1B3 (R474A) is reported in Table E10C2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00110 TABLE E10C1 Temperature and pressure data for Ambient Temperature of -15° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf
59.6 NA 22.77 59.6 20.3 NA 6.4 20.3 RE1B3 59.6 NA 22.77 59.6 29.2 NA 8.6 29.2 (R474A)

TABLE-US-00111 TABLE E10C2 Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes
OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot
Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool**
Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes
Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool
Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold
Comfort No

[0256] This is an efficient mode for vehicle heating between -25° C. and 5° C. while driving. PTC heat can be used to heat the vehicle while maintaining the battery and motor and inverter temperatures. This can also be used with the enhanced heat pump configuration (dotted line).

Example 10D—Vehicle Heating with Ambient Between -25° C. and 5° C. While Driving—
Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Using R474B

[0257] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 12. This system is operated with R1234yf and with RE1B4 and produced the results as reported in the following Table E10D1. The use of Refrigerant RE1B4 (R474B) is reported in Table E10D2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00112 TABLE E10D1 Temperature and pressure data for Ambient Temperature of -15° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.6
NA 22.77 59.6 20.3 NA 6.4 20.3 RE1B4 (R474B) 59.6 NA 22.77 59.6 31.7 NA 9.4 31.7

TABLE-US-00113 TABLE E10D2 Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes
OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot
Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool**
Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes
Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool
Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold
Comfort No

[0258] This is an efficient mode for vehicle heating between -25°C . and 5°C . while driving. PTC heat can be used to heat the vehicle while maintaining the battery and motor and inverter temperatures. This can also be used with the enhanced heat pump configuration (dotted line).

Example 10E—Vehicle Heating with Ambient Between -25°C . and 5°C . While Driving—
Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Using R516A

[0259] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 12. This system is operated with R1234yf and with RE1B5 and produced the results as reported in the following Table E10E1. The use of Refrigerant RE1B5 (R516A) is reported in Table E10E2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00114 TABLE E10E1 Temperature and pressure data for Ambient Temperature of -15°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.27 62.3 21.5 NA 2.0 21.5 RE1B5 (R454C) 62.3 NA -13.27 62.3 22.1 NA 2.0 22.1

TABLE-US-00115 TABLE E10E2 Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Motor & Ambient, $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0260] This is an efficient mode for vehicle heating between -25°C . and 5°C . while driving. PTC heat can be used to heat the vehicle while maintaining the battery and motor and inverter temperatures. This can also be used with the enhanced heat pump configuration (dotted line).

Example 10F—Vehicle Heating with Ambient Between -25°C . and 5°C . While Driving—
Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Using R444A

[0261] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 12. This system is operated with R1234yf and with RE1B6 and produced the results as reported in the following Table E10F1. The use of Refrigerant RE1B6 (R444A) is reported in Table E10F2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00116 TABLE E10F1 Temperature and pressure data for Ambient Temperature of -15°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -13.27 62.3 21.5 NA 2.0 21.5 RE1B6 (R444A) 62.3 NA -13.27 62.3 24.4 NA 1.6 24.4

TABLE-US-00117 TABLE E10F2 Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Motor & Ambient, $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0262] This is an efficient mode for vehicle heating between -25°C . and 5°C . while driving. PTC heat can be used to heat the vehicle while maintaining the battery and motor and inverter temperatures. This can also be used with the enhanced heat pump configuration (dotted line).

Example 10G—Vehicle Heating with Ambient Between -25°C . and 5°C . While Driving—
Evaporative Heat Source 9: Motor, Inverter, Battery and PTC at Chiller Using R474A

[0263] In this example, a system of the present invention configured for operation to heat cabin air

C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 2.55 59.0 20.0 NA 3.4 20.0 RE1B1 (R454C) 59.0 NA 2.55 59.0 32.1 NA 4.9 32.1

TABLE-US-00123 TABLE E11A2 Evaporative Heat Source 10: Motor, inverter, battery at OHE
Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort**
No 5 OK OK OK Start Yes **Cool Cool Cool Comfort No -5 OK OK OK** Start Yes **Cold Cold**
Cold **Comfort No -15 OK OK OK** Start Yes **Cold Cold Cold Comfort No -25 Cool Cool Cool**
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No
[0268] Excess heat from the battery and motor and inverter can be used at the OHE. This can also
be used to defrost the OHE in the event of freezing.

Example 11B—Vehicle Heating with Ambient Between -15° C. and 15° C. While Driving—
Evaporative Heat Source 10: Motor, Inverter, Battery at OHE

[0269] In this example, a system of the present invention configured for operation to heat cabin air
during periods of low ambient temperatures is illustrated in FIG. 13. This system is operated with
R1234yf and with RE1B2 and produces the results as reported in the following Table E11B1. The
use of Refrigerant RE1B2 (R455A) is reported in Table E11B2 below by using bold type as an
indicator for acceptable operation at the indicated locations and for indicated ambient conditions
when using the identified evaporative heat source(s).

TABLE-US-00124 TABLE E11B1 Temperature and pressure data for Ambient Temperature of 5°
C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA
2.55 59.0 20.0 NA 3.4 20.0 RE1B2 (R455A) 59.0 NA 2.55 59.0 35.5 NA 5.3 35.5

TABLE-US-00125 TABLE E11B2 Evaporative Heat Source 10: Motor, inverter, battery at OHE
Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort**
No 5 OK OK OK Start Yes **Cool Cool Cool Comfort No -5 OK OK OK** Start Yes **Cold Cold**
Cold **Comfort No -15 OK OK OK** Start Yes **Cold Cold Cold Comfort No -25 Cool Cool Cool**
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No
[0270] Excess heat from the battery and motor and inverter can be used at the OHE. This can also
be used to defrost the OHE in the event of freezing.

Example 11C—Vehicle Heating with Ambient Between -15° C. and 15° C. While Driving—
Evaporative Heat Source 10: Motor, Inverter, Battery at OHE

[0271] In this example, a system of the present invention configured for operation to heat cabin air
during periods of low ambient temperatures is illustrated in FIG. 13. This system is operated with
R1234yf and with RE1B3 and produces the results as reported in the following Table E11C1. The
use of Refrigerant RE1B3 (R474A) is reported in Table E11C2 below by using bold type as an
indicator for acceptable operation at the indicated locations and for indicated ambient conditions
when using the identified evaporative heat source(s).

TABLE-US-00126 TABLE E11C1 Temperature and pressure data for Ambient Temperature of 5°
C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA
2.55 59.0 20.0 NA 3.4 20.0 RE1B3 (R474A) 59.0 NA 2.55 59.0 28.9 NA 4.6 28.9

TABLE-US-00127 TABLE E11C2 Evaporative Heat Source 10: Motor, inverter, battery at OHE
Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort**
No 5 OK OK OK Start Yes **Cool Cool Cool Comfort No -5 OK OK OK** Start Yes **Cold Cold**
Cold **Comfort No -15 OK OK OK** Start Yes **Cold Cold Cold Comfort No -25 Cool Cool Cool**
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No
[0272] Excess heat from the battery and motor and inverter can be used at the OHE. This can also

be used to defrost the OHE in the event of freezing.

Example 11D—Vehicle Heating with Ambient Between -15°C . and 15°C . While Driving—
Evaporative Heat Source 10: Motor, Inverter, Battery at OHE

[0273] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 13. This system is operated with R1234yf and with RE1B4 and produces the results as reported in the following Table E11D1. The use of Refrigerant RE1B4 (R474BA) is reported in Table E11D2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00128 TABLE E11D1 Temperature and pressure data for Ambient Temperature of 5°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 2.55 59.0 20.0 NA 3.4 20.0 RE1B4 (R474B) 59.0 NA 2.55 59.0 31.3 NA 5.1 31.3

TABLE-US-00129 TABLE E11D2 Evaporative Heat Source 10: Motor, inverter, battery at OHE
Motor & Ambient, $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort**
No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold
Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0274] This is an efficient mode for vehicle heating between -15°C . and 15°C . while driving. Excess heat from the battery and motor and inverter can be used at the OHE. This can also be used to defrost the OHE in the event of freezing.

Example 11E—Vehicle Heating with Ambient Between -15°C . and 15°C . While Driving—
Evaporative Heat Source 10: Motor, Inverter, Battery at OHE Using R454C

[0275] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 13. This system is operated with R1234yf and with RE1B5 and produces the results as reported in the following Table E11E1. The use of Refrigerant RE1B5 (R454C) is reported in Table E11E2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00130 TABLE E11E1 Temperature and pressure data for Ambient Temperature of 5°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 2.55 59.0 20.0 NA 3.4 20.0 RE1B5 (R516A) 59.0 NA 2.55 59.0 20.6 NA 3.5 20.6

TABLE-US-00131 TABLE E11E2 Evaporative Heat Source 10: Motor, inverter, battery at OHE
Motor & Ambient, $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort**
No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold
Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0276] Excess heat from the battery and motor and inverter can be used at the OHE. This can also be used to defrost the OHE in the event of freezing.

Example 11F—Vehicle Heating with Ambient Between -15°C . and 15°C . While Driving—
Evaporative Heat Source 10: Motor, Inverter, Battery at OHE Using R444B

[0277] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 13. This system is operated with R1234yf and with RE1B6 and produces the results as reported in the following Table E11F1. The use of Refrigerant RE1B6 (R444A) is reported in Table E11F2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00132 TABLE E11F1 Temperature and pressure data for Ambient Temperature of 5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 2.55 59.0 20.0 NA 3.4 20.0 RE1B6 (R455A) 59.0 NA 2.55 59.0 22.8 NA 3.0 22.8

TABLE-US-00133 TABLE E11F2 Evaporative Heat Source 10: Motor, inverter, battery at OHE Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No** 5 **OK OK OK Start Yes Cool Cool Cool Comfort No** -5 **OK OK OK Start Yes Cold Cold Cold Comfort No** -15 **OK OK OK Start Yes Cold Cold Cold Comfort No** -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No [0278] Excess heat from the battery and motor and inverter can be used at the OHE. This can also be used to defrost the OHE in the event of freezing.

Example 11G—Vehicle Heating with Ambient Between -15° C. and 15° C. While Driving—Evaporative Heat Source 10: Motor, Inverter, Battery at OHE Using R457C [0279] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 13. This system is operated with R1234yf and with REIB3 and produces the results as reported in the following Table E11G1. The use of Refrigerant RE1B7 (R457C) is reported in Table E11G2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00134 TABLE E11G1 Temperature and pressure data for Ambient Temperature of 5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 2.55 59.0 20.0 NA 3.4 20.0 RE1B7 (R457C) 59.0 NA 2.55 59.0 24.7 NA 3.9 24.7

TABLE-US-00135 TABLE E11G2 Evaporative Heat Source 10: Motor, inverter, battery at OHE Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No** 5 **OK OK OK Start Yes Cool Cool Cool Comfort No** -5 **OK OK OK Start Yes Cold Cold Cold Comfort No** -15 **OK OK OK Start Yes Cold Cold Cold Comfort No** -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No [0280] Excess heat from the battery and motor and inverter can be used at the OHE. This can also be used to defrost the OHE in the event of freezing.

Example 11H—Vehicle Heating with Ambient Between -15° C. and 15° C. While Driving—Evaporative Heat Source 10: Motor, Inverter, Battery at OHE Using R457D [0281] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 13. This system is operated with R1234yf and with RE1B8 and produces the results as reported in the following Table E11H1. The use of Refrigerant RE1B8 (R457D) is reported in Table E11H2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00136 TABLE E11H1 Temperature and pressure data for Ambient Temperature of 5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 59.0 NA 2.55 59.0 20.0 NA 3.4 20.0 RE1B8 (R457D) 59.0 NA 2.55 59.0 22.9 NA 3.7 22.9

TABLE-US-00137 TABLE E11H2 Evaporative Heat Source 10: Motor, inverter, battery at OHE Motor & Ambient, ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No** 5 **OK OK OK Start Yes Cool Cool Cool Comfort No** -5 **OK OK OK Start Yes Cold Cold Cold Comfort No** -15 **OK OK OK Start Yes Cold Cold Cold Comfort No** -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0282] Excess heat from the battery and motor and inverter can be used at the OHE. This can also be used to defrost the OHE in the event of freezing.

Example 12A—Vehicle Heating with Ambient Between -35°C . and 5°C . With Battery Charging-Evaporative Heat Source 11: PTC at Chiller Using R454C

[0283] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 14. This system is operated with R1234yf and with REIB1 and produces the results as reported in the following Table E12A1. The use of Refrigerant REIB1 (R454C) is reported in Table E12A2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00138 TABLE E12A1 Temperature and pressure data for Ambient Temperature of -25°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 60.9 NA -7.04 60.9 20.8 NA 2.5 20.8 RE1B1 (R454C) 60.9 NA -7.04 60.9 33.3 NA 3.6 33.3

TABLE-US-00139 TABLE E12A2 Evaporative Heat Source 11: PTC at Chiller Motor & Ambient, $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0284] This is an efficient mode for vehicle heating between -35°C . and -5°C . after charging to prep the vehicle cabin before driving. Heat from the PTC is used by the heat pump to warm the cabin. This can also be used with the enhanced heat pump configuration (dotted line).

Example 12B—Vehicle Heating with Ambient Between -35°C . and 5°C . With Battery Charging-Evaporative Heat Source 11: PTC at Chiller Using R455A

[0285] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 14. This system is operated with R1234yf and with RE1B2 and produces the results as reported in the following Table E12B1. The use of Refrigerant RE1B2 (R455A) is reported in Table E12B2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00140 TABLE E12B1 Temperature and pressure data for Ambient Temperature of -25°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 60.9 NA -7.04 60.9 20.8 NA 2.5 20.8 RE1B2 (R455A) 60.9 NA -7.04 60.9 36.8 NA 3.8 36.8

TABLE-US-00141 TABLE E12B2 Evaporative Heat Source 11: PTC at Chiller Motor & Ambient, $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0286] This is an efficient mode for vehicle heating between -35°C . and -5°C . after charging to prep the vehicle cabin before driving. Heat from the PTC is used by the heat pump to warm the cabin. This can also be used with the enhanced heat pump configuration (dotted line).

Example 12C—Vehicle Heating with Ambient Between -35°C . and 5°C . With Battery Charging-Evaporative Heat Source 11: PTC at Chiller Using R474A

[0287] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 14. This system is operated with R1234yf and with RE1B3 and produces the results as reported in the following Table E12C1. The use of Refrigerant RE1B3 (R474A) is reported in Table E12C2 below by using bold type as an

indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00142 TABLE E12C1 Temperature and pressure data for Ambient Temperature of -25°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 60.9 NA -7.04 60.9 20.8 NA 2.5 20.8 RE1B3 (R474A) 60.9 NA -7.04 60.9 30.0 NA 3.3 30.0

TABLE-US-00143 TABLE E12C2 Evaporative Heat Source 11: PTC at Chiller Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0288] This is an efficient mode for vehicle heating between -35°C . and -5°C . after charging to prep the vehicle cabin before driving. Heat from the PTC is used by the heat pump to warm the cabin. This can also be used with the enhanced heat pump configuration (dotted line).

Example 12D—Vehicle Heating with Ambient Between -35°C . and 5°C . With Battery Charging-Evaporative Heat Source 11: PTC at Chiller Using R474B

[0289] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 14. This system is operated with R1234yf and with RE1B4 and produces the results as reported in the following Table E12D1. The use of Refrigerant RE1B4 (R474B) is reported in Table E12D2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00144 TABLE E12D1 Temperature and pressure data for Ambient Temperature of -25°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 60.9 NA -7.04 60.9 20.8 NA 2.5 20.8 RE1B4 (R474B) 60.9 NA -7.04 60.9 32.5 NA 3.7 32.5

TABLE-US-00145 TABLE E12D2 Evaporative Heat Source 11: PTC at Chiller Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0290] This is an efficient mode for vehicle heating between -35°C . and -5°C . after charging to prep the vehicle cabin before driving. Heat from the PTC is used by the heat pump to warm the cabin. This can also be used with the enhanced heat pump configuration (dotted line).

Example 12E—Vehicle Heating with Ambient Between -35°C . and 5°C . With Battery Charging-Evaporative Heat Source 11: PTC at Chiller Using R516A

[0291] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 14. This system is operated with R1234yf and with RE1B5 and produces the results as reported in the following Table E12E1. The use of Refrigerant RE1B5 (R516A) is reported in Table E12E2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00146 TABLE E12E1 Temperature and pressure data for Ambient Temperature of -25°C . T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 60.9 NA -7.04 60.9 20.8 NA 2.5 20.8 RE1B5 (R454C) 60.9 NA -7.04 60.9 21.5 NA 2.5 21.5

TABLE-US-00147 TABLE E12E2 Evaporative Heat Source 11: PTC at Chiller Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

Warm Comfort No 15 OK OK Warm Start Yes Cool Cool Comfort No 5 OK OK OK
Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No
-15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold
Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0292] This is an efficient mode for vehicle heating between -35° C. and -5° C. after charging to prep the vehicle cabin before driving. Heat from the PTC is used by the heat pump to warm the cabin. This can also be used with the enhanced heat pump configuration (dotted line).

Example 12F—Vehicle Heating with Ambient Between -35° C. and 5° C. With Battery Charging-Evaporative Heat Source 11: PTC at Chiller Using R444B

[0293] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 14. This system is operated with R1234yf and with RE1B6 and produces the results as reported in the following Table E12F1. The use of Refrigerant RE1B6 (R444A) is reported in Table E12F2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00148 TABLE E12F1 Temperature and pressure data for Ambient Temperature of -25° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 60.9 NA -7.04 60.9 20.8 NA 2.5 20.8 RE1B6 (R444A) 60.9 NA -7.04 60.9 23.7 NA 2.1 23.7

TABLE-US-00149 TABLE E12F2 Evaporative Heat Source 11: PTC at Chiller Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0294] This is an efficient mode for vehicle heating between -35° C. and -5° C. after charging to prep the vehicle cabin before driving. Heat from the PTC is used by the heat pump to warm the cabin. This can also be used with the enhanced heat pump configuration (dotted line).

Example 12G—Vehicle Heating with Ambient Between -35° C. and 5° C. With Battery Charging-Evaporative Heat Source 11: PTC at Chiller Using R 457C

[0295] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 14. This system is operated with R1234yf and with RE1B7 and produces the results as reported in the following Table E12G1. The use of Refrigerant RE1B7 (R457C) is reported in Table E12G2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00150 TABLE E12G1 Temperature and pressure data for Ambient Temperature of -25° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 60.9 NA -7.04 60.9 20.8 NA 2.5 20.8 RE1B7 (R457C) 60.9 NA -7.04 60.9 25.7 NA 2.8 25.7

TABLE-US-00151 TABLE E12G2 Evaporative Heat Source 11: PTC at Chiller Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0296] This is an efficient mode for vehicle heating between -35° C. and -5° C. after charging to prep the vehicle cabin before driving. Heat from the PTC is used by the heat pump to warm the cabin. This can also be used with the enhanced heat pump configuration (dotted line).

Example 12H—Vehicle Heating with Ambient Between -35° C. and 5° C. With Battery Charging-

Evaporative Heat Source 11: PTC at Chiller

[0297] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 14. This system is operated with R1234yf and with RE1B8 and produces the results as reported in the following Table E12H1. The use of Refrigerant RE1B8 (R457D) is reported in Table E12H2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00152 TABLE E12H1 Temperature and pressure data for Ambient Temperature of −25° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 60.9 NA −7.04 60.9 20.8 NA 2.5 20.8 RE1B8 (R457D) 60.9 NA −7.04 23.8 NA 2.7 23.8 23.8

TABLE-US-00153 TABLE E12H2 Evaporative Heat Source 11: PTC at Chiller Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No −5 OK OK OK Start Yes Cold Cold Cold Comfort No −15 OK OK OK Start Yes Cold Cold Cold Comfort No −25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No −35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0298] This is an efficient mode for vehicle heating between −35° C. and −5° C. after charging to prep the vehicle cabin before driving. Heat from the PTC is used by the heat pump to warm the cabin. This can also be used with the enhanced heat pump configuration (dotted line).

Example 13A—Vehicle Heating with Ambient Between −15° C. and 5° C. With Battery Charging or Driving-Evaporative Heat Source 12: PTC at OHE

[0299] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 15. This system is operated with R1234yf and with RE1B1 and produces the results as reported in the following Table E13A1. The use of Refrigerant RE1B1 (R454C) is reported in Table E13A2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00154 TABLE E13A1 Temperature and pressure data for Ambient Temperature of −5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C] [C] [C] [C] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA −7.27 62.3 21.5 NA 2.4 21.5 RE1B1 (R454C) 62.3 NA −7.27 62.3 34.2 NA 3.5 34.2

TABLE-US-00155 TABLE E13A2 Evaporative Heat Source 12: PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No −5 OK OK OK Start Yes Cold Cold Cold Comfort No −15 OK OK OK Start Yes Cold Cold Cold Comfort No −25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No −35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0300] This is an efficient mode for vehicle heating between −15° C. and 5° C. while driving or charging when the battery and motor and inverter are at appropriate temperatures. Heat from the PTC can be used at the OHE.

Example 13B—Vehicle Heating with Ambient Between −15° C. and 5° C. With Battery Charging or Driving-Evaporative Heat Source 12: PTC at OHE

[0301] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 15. This system is operated with R1234yf and with RE1B2 and produces the results as reported in the following Table E13B1. The use of Refrigerant RE1B2 (R455A) is reported in Table E13B2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00156 TABLE E13B1 Temperature and pressure data for Ambient Temperature of −5°

C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -7.27 62.3 21.5 NA 2.4 21.5 RE1B2 62.3 NA -7.27 62.3 37.7 NA 3.8 37.7 (R455A)

TABLE-US-00157 TABLE E13B2 Evaporative Heat Source 12: PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25** Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0302] This is an efficient mode for vehicle heating between -15° C. and 5° C. while driving or charging when the battery and motor and inverter are at appropriate temperatures. Heat from the PTC can be used at the OHE.

Example 13C—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging or Driving-Evaporative Heat Source 12: PTC at OHE

[0303] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 15. This system is operated with R1234yf and with RE1B3 and produces the results as reported in the following Table E13C1. The use of Refrigerant RE1B3 (R474A) is reported in Table E13C2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00158 TABLE E13C1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -7.27 62.3 21.5 NA 2.4 21.5 RE1B3 62.3 NA -7.27 62.3 30.8 NA 3.3 30.8 (R474A)

TABLE-US-00159 TABLE E13C2 Evaporative Heat Source 12: PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25** Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0304] This is an efficient mode for vehicle heating between -15° C. and 5° C. while driving or charging when the battery and motor and inverter are at appropriate temperatures. Heat from the PTC can be used at the OHE.

Example 13D—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging or Driving-Evaporative Heat Source 12: PTC at OHE

[0305] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 15. This system is operated with R1234yf and with RE1B4 and produces the results as reported in the following Table E13D1. The use of Refrigerant RE1B4 (R474B) is reported in Table E13D2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00160 TABLE E13D1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -7.27 62.3 21.5 NA 2.4 21.5 RE1B4 62.3 NA -7.27 62.3 33.4 NA 3.7 33.4 (R474B)

TABLE-US-00161 TABLE E13D2 Evaporative Heat Source 12: PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25** Cool Cool Cool Start Yes Cold Cold Cold

Comfort No -35 Cool Cool Start Yes Cold Cold Comfort No

[0306] This is an efficient mode for vehicle heating between -15° C. and 5° C. while driving or charging when the battery and motor and inverter are at appropriate temperatures. Heat from the PTC can be used at the OHE.

Example 13E—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging or Driving-Evaporative Heat Source 12: PTC at OHE Using R516A

[0307] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 15. This system is operated with R1234yf and with RE1B5 and produces the results as reported in the following Table E13E1. The use of Refrigerant RE1B5 (R516A) is reported in Table E13E2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00162 TABLE E13E1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -7.27 62.3 21.5 NA 2.4 21.5 RE1B5 62.3 NA -7.27 62.3 22.2 NA 2.5 22.2 (R454C)

TABLE-US-00163 TABLE E13E2 Evaporative Heat Source 12: PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0308] This is an efficient mode for vehicle heating between -15° C. and 5° C. while driving or charging when the battery and motor and inverter are at appropriate temperatures. Heat from the PTC can be used at the OHE.

Example 13F—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging or Driving-Evaporative Heat Source 12: PTC at OHE Using R444B

[0309] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 15. This system is operated with R1234yf and with RE1B2 and produces the results as reported in the following Table E13F1. The use of Refrigerant RE1B6 (R444A) is reported in Table E13F2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00164 TABLE E13F1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -7.27 62.3 21.5 NA 2.4 21.5 RE1B6 62.3 NA -7.27 62.3 24.4 NA 2.1 24.4 (R444A)

TABLE-US-00165 TABLE E13F2 Evaporative Heat Source 12: PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0310] This is an efficient mode for vehicle heating between -15° C. and 5° C. while driving or charging when the battery and motor and inverter are at appropriate temperatures. Heat from the PTC can be used at the OHE.

Example 13G—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging or Driving-Evaporative Heat Source 12: PTC at OHE Using R444B

[0311] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 15. This system is operated with

R1234yf and with RE1B7 and produces the results as reported in the following Table E13G1. The use of Refrigerant RE1B7 (R457C) is reported in Table E13G2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00166 TABLE E13G1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -7.27 62.3 21.5 NA 2.4 21.5 RE1B7 62.3 NA -7.27 62.3 26.4 NA 2.8 26.4 (R457C)

TABLE-US-00167 TABLE E13G2 Evaporative Heat Source 12: PTC at OHE Ambient, Motor & $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0312] This is an efficient mode for vehicle heating between -15° C. and 5° C. while driving or charging when the battery and motor and inverter are at appropriate temperatures. Heat from the PTC can be used at the OHE.

Example 13H—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging or Driving-Evaporative Heat Source 12: PTC at OHE Using R457D

[0313] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 15. This system is operated with R1234yf and with RE1B8 and produces the results as reported in the following Table E13H1. The use of Refrigerant RE1B8 (R457D) is reported in Table E13H2 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00168 TABLE E13H1 Temperature and pressure data for Ambient Temperature of -5° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 62.3 NA -7.27 62.3 21.5 NA 2.4 21.5 RE1B8 62.3 NA -7.27 62.3 24.5 NA 2.6 24.5 (R457D)

TABLE-US-00169 TABLE E13H2 Evaporative Heat Source 12: PTC at OHE Ambient, Motor & $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0314] This is an efficient mode for vehicle heating between -15° C. and 5° C. while driving or charging when the battery and motor and inverter are at appropriate temperatures. Heat from the PTC can be used at the OHE.

Example 14A—Vehicle Heating with Ambient Between -15° C. and 5° C.—Evaporative Heat Source 13: Motor, Inverter and PTC at OHE Using R454C

[0315] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 16 and is operated with R1234yf and with REIB1. The use of Refrigerant REIB1 (R454C) is reported in Table E14A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00170 TABLE E14A Evaporative Heat Source 13: Motor, Inverter and PTC at OHE Ambient, Motor & $^{\circ}$ C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold**

Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No
[0316] This is a less efficient mode for vehicle heating between -15° C. and 5° C. while the motor and inverter are warming up (the PTC energy can be used in this mode). This mode of operation can also be used to de-ice the OHE.

Example 14B—Vehicle Heating with Ambient Between -15° C. and 5° C.—Evaporative Heat Source 13: Motor, Inverter and PTC at OHE

[0317] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 16 and is operated with R1234yf and with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E14B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00171 TABLE E14B Evaporative Heat Source 13: Motor, Inverter and PTC at OHE
Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No
5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold**
Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0318] This is a less efficient mode for vehicle heating between -15° C. and 5° C. while the motor and inverter are warming up (the PTC energy can be used in this mode). This mode of operation can also be used to de-ice the OHE.

Example 14C—Vehicle Heating with Ambient Between -15° C. and 5° C.—Evaporative Heat Source 13: Motor, Inverter and PTC at OHE Using R474A

[0319] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 16 and is operated with R1234yf and with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E14C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00172 TABLE E14C Evaporative Heat Source 13: Motor, Inverter and PTC at OHE
Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No
5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold**
Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0320] This is a less efficient mode for vehicle heating between -15° C. and 5° C. while the motor and inverter are warming up (the PTC energy can be used in this mode). This mode of operation can also be used to de-ice the OHE.

Example 14D—Vehicle Heating with Ambient Between -15° C. and 5° C.—Evaporative Heat Source 13: Motor, Inverter and PTC at OHE Using R474B

[0321] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 16 and is operated with R1234yf and with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E14D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00173 TABLE E14D Evaporative Heat Source 13: Motor, Inverter and PTC at OHE
Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No

5 OK OK OK Start Yes Cold Cold Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No
[0322] This is a less efficient mode for vehicle heating between -15° C. and 5° C. while the motor and inverter are warming up (the PTC energy can be used in this mode). This mode of operation can also be used to de-ice the OHE.

Example 14E—Vehicle Heating with Ambient Between -15° C. and 5° C.—Evaporative Heat Source 13: Motor, Inverter and PTC at OHE Using R516A

[0323] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 16 and is operated with R1234yf and with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E14E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00174 TABLE E14E Evaporative Heat Source 13: Motor, Inverter and PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**
[0324] This is a less efficient mode for vehicle heating between -15° C. and 5° C. while the motor and inverter are warming up (the PTC energy can be used in this mode). This mode of operation can also be used to de-ice the OHE.

Example 14F—Vehicle Heating with Ambient Between -15° C. and 5° C.—Evaporative Heat Source 13: Motor, Inverter and PTC at OHE Using R444A

[0325] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 16 and is operated with R1234yf and with RE1B6. The use of Refrigerant RE1B6 (R444A) is reported in Table E14F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00175 TABLE E14F Evaporative Heat Source 13: Motor, Inverter and PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**
[0326] This is a less efficient mode for vehicle heating between -15° C. and 5° C. while the motor and inverter are warming up (the PTC energy can be used in this mode). This mode of operation can also be used to de-ice the OHE.

Example 14G—Vehicle Heating with Ambient Between -15° C. and 5° C.—Evaporative Heat Source 13: Motor, Inverter and PTC at OHE Using R457C

[0327] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 16 and is operated with R1234yf and with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E14G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00176 TABLE E14G Evaporative Heat Source 13: Motor, Inverter and PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot

Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No
5 **OK OK OK Start Yes Cool Cool Cool Comfort No** -5 **OK OK OK Start Yes Cold Cold
Cold Comfort No** -15 **OK OK OK Start Yes Cold Cold Cold Comfort No** -25 Cool Cool Cool
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No
[0328] This is a less efficient mode for vehicle heating between -15° C. and 5° C. while the motor
and inverter are warming up (the PTC energy can be used in this mode). This mode of operation
can also be used to de-ice the OHE.

Example 14H—Vehicle Heating with Ambient Between -15° C. and 5° C.—Evaporative Heat
Source 13: Motor, Inverter and PTC at OHE Using R457D

[0329] In this example, a system of the present invention configured for operation to heat cabin air
during periods of low ambient temperatures is illustrated in FIG. 16 and is operated with R1234yf
and with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E14H below by
using bold type as an indicator for acceptable operation at the indicated locations and for indicated
ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00177 TABLE E14H Evaporative Heat Source 13: Motor, Inverter and PTC at OHE
Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No**
5 **OK OK OK Start Yes Cool Cool Cool Comfort No** -5 **OK OK OK Start Yes Cold Cold
Cold Comfort No** -15 **OK OK OK Start Yes Cold Cold Cold Comfort No** -25 Cool Cool Cool
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No
[0330] This is a less efficient mode for vehicle heating between -15° C. and 5° C. while the motor
and inverter are warming up (the PTC energy can be used in this mode). This mode of operation
can also be used to de-ice the OHE.

Example 15A—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-
Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE

[0331] In this example, a system of the present invention is configured for operation to heat cabin
air during periods of low ambient temperatures is illustrated in FIG. 17 and is operated with
R1234yf and with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E15A below
by using bold type as an indicator for acceptable operation at the indicated locations and for
indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00178 TABLE E15A Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at
OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes
OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot
Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool
Comfort No** 5 **OK OK OK Start Yes Cool Cool Cool Comfort No** -5 **OK OK OK Start Yes
Cold Cold Cold Comfort No** -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool
Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold
Comfort No

[0332] This is a less efficient mode for vehicle heating between -15° C. and 15° C. while the motor
and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-
ice the OHE.

Example 15B—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-
Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE

[0333] In this example, a system of the present invention is configured for operation to heat cabin
air during periods of low ambient temperatures is illustrated in FIG. 17 and is operated with
R1234yf and with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E15B
below by using bold type as an indicator for acceptable operation at the indicated locations and for
indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00179 TABLE E15B Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at

OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes
 OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot
 Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool**
Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes
Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool
 Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold
 Comfort No

[0334] This is a less efficient mode for vehicle heating between -15° C. and 15° C. while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 15C—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE

[0335] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 17 and is operated with R1234yf and with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E15C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00180 TABLE E15C Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes
 OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot
 Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool**
Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes
Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool
 Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold
 Comfort No

[0336] This is a less efficient mode for vehicle heating between -15° C. and 15° C. while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 15D—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE

[0337] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 17 and is operated with R1234yf and with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E15D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00181 TABLE E15D Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes
 OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot
 Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool**
Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes
Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool
 Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold
 Comfort No

[0338] This is a less efficient mode for vehicle heating between -15° C. and 15° C. while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 15E—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE

[0339] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 17 and is operated with

R1234yf and with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E15E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00182 TABLE E15E Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0340] This is a less efficient mode for vehicle heating between -15° C. and 15° C. while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 15F—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE

[0341] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 17 and is operated with R1234yf and with RE1B6. The use of Refrigerant RE1B6 (R444A) is reported in Table E15F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00183 TABLE E15F Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0342] This is a less efficient mode for vehicle heating between -15° C. and 15° C. while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 15G—Vehicle Heating with Ambient Between -15° C. and 5° C. With Battery Charging-Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE

[0343] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 17 and is operated with R1234yf and with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E15G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00184 TABLE E15G Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0344] This is a less efficient mode for vehicle heating between -15° C. and 15° C. while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 15H—Vehicle Heating with Ambient Between -15°C . and 5°C . With Battery Charging-Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE

[0345] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 17 and is operated with R1234yf and with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E15H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00185 TABLE E15H Evaporative Heat Source 14: Motor, Inverter, Battery and PTC at OHE Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0346] This is a less efficient mode for vehicle heating between -15°C . and 15°C . while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 16A—Vehicle Heating with Ambient Between -15°C . and 15°C . With Component Warm Up-Evaporative Heat Source 15: Battery and PTC at OHE Using R454C

[0347] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 18 and is operated with R1234yf and with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E16A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00186 TABLE E16A Evaporative Heat Source 15: Battery and PTC at OHE Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0348] This is a less efficient mode for vehicle heating between -15°C . and 15°C . while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 16B—Vehicle Heating with Ambient Between -15°C . and 15°C . With Component Warm Up-Evaporative Heat Source 15: Battery and PTC at OHE Using R455A

[0349] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 18 and is operated with R1234yf and with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E16B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00187 TABLE E16B Evaporative Heat Source 15: Battery and PTC at OHE Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0350] This is a less efficient mode for vehicle heating between -15°C . and 15°C . while the motor

and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 16C—Vehicle Heating with Ambient Between -15° C. and 15° C. With Component Warm Up-Evaporative Heat Source 15: Battery and PTC at OHE Using R474A

[0351] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 18 and is operated with R1234yf and with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E16C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00188 TABLE E16C Evaporative Heat Source 15: Battery and PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0352] This is a less efficient mode for vehicle heating between -15° C. and 15° C. while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 16D—Vehicle Heating with Ambient Between -15° C. and 15° C. With Component Warm Up-Evaporative Heat Source 15: Battery and PTC at OHE Using R474B

[0353] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 18 and is operated with R1234yf and with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E16D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00189 TABLE E16D Evaporative Heat Source 15: Battery and PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0354] This is a less efficient mode for vehicle heating between -15° C. and 15° C. while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 16E—Vehicle Heating with Ambient Between -15° C. and 15° C. With Component Warm Up-Evaporative Heat Source 15: Battery and PTC at OHE Using R516A

[0355] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 18 and is operated with R1234yf and with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E16E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00190 TABLE E16E Evaporative Heat Source 15: Battery and PTC at OHE Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0356] This is a less efficient mode for vehicle heating between -15°C . and 15°C . while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 16F—Vehicle Heating with Ambient Between -15°C . and 15°C . With Component Warm Up-Evaporative Heat Source 15: Battery and PTC at OHE Using R444A

[0357] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 18 and is operated with R1234yf and with RE1B6. The use of Refrigerant RE1B2 (R444A) is reported in Table E16F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00191 TABLE E16F Evaporative Heat Source 15: Battery and PTC at OHE Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No** -5°C . **OK OK OK Start Yes Cold Cold Cold Comfort No** -25°C . Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35°C . Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0358] This is a less efficient mode for vehicle heating between -15°C . and 15°C . while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 16G—Vehicle Heating with Ambient Between -15°C . and 15°C . With Component Warm Up-Evaporative Heat Source 15: Battery and PTC at OHE

[0359] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 18 and is operated with R1234yf and with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E16G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00192 TABLE E16G Evaporative Heat Source 15: Battery and PTC at OHE Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No** -5°C . **OK OK OK Start Yes Cold Cold Cold Comfort No** -25°C . Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35°C . Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0360] This is a less efficient mode for vehicle heating between -15°C . and 15°C . while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 16H—Vehicle Heating with Ambient Between -15°C . and 15°C . With Component Warm Up-Evaporative Heat Source 15: Battery and PTC at OHE Using R457D

[0361] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 18 and is operated with R1234yf and with RE1B8. The use of Refrigerant RE1B4 (R457D) is reported in Table E16H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00193 TABLE E16H Evaporative Heat Source 15: Battery and PTC at OHE Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No** -5°C . **OK OK OK Start Yes Cold Cold Cold Comfort No** -25°C . Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35°C . Cool Cool Cool Start Yes Cold Cold Cold Comfort No

Cold Cold Comfort No -35 Cold Cold Cold Start Yes Cold Cold Cold Comfort No
[0362] This is a less efficient mode for vehicle heating between -15° C. and 15° C. while the motor and inverter and battery are warming up (the PTC energy can be used). This can also be used to de-ice the OHE.

Example 17A—Vehicle Heating with Ambient Between -15° C. and 5° C.—Evaporative Heat Source 16: Radiator (Air) at Chiller Using R454C

[0363] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 19 and is operated with R1234yf and with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E17A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00194 TABLE E17A Evaporative Heat Source 16: Radiator (Air) at Chiller Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0364] This is also an efficient mode for vehicle heating between -5° C. and 15° C. Energy from the air can be used at the chiller while not affecting the motor and Inverter or the battery.

Example 17B—Vehicle Heating with Ambient Between -15° C. and 5° C.—Evaporative Heat Source 16: Radiator (Air) at Chiller Using R455A

[0365] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 19 and is operated with R1234yf and with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E17B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00195 TABLE E17B Evaporative Heat Source 16: Radiator (Air) at Chiller Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0366] This is also an efficient mode for vehicle heating between -5° C. and 15° C. Energy from the air can be used at the chiller while not affecting the motor and Inverter or the battery.

Example 17C—Vehicle Heating with Ambient Between -15° C. and 5° C.—Evaporative Heat Source 16: Radiator (Air) at Chiller Using R 474A

[0367] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 19 and is operated with R1234yf and with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E17C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00196 TABLE E17C Evaporative Heat Source 16: Radiator (Air) at Chiller Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0368] This is also an efficient mode for vehicle heating between -5°C . and 15°C . Energy from the air can be used at the chiller while not affecting the motor and Inverter or the battery.

Example 17D—Vehicle Heating with Ambient Between -15°C . and 5°C .—Evaporative Heat Source 16: Radiator (Air) at Chiller Using R474B

[0369] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 19 and is operated with R1234yf and with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E17D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00197 TABLE E17D Evaporative Heat Source 16: Radiator (Air) at Chiller Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No** **OK Start Yes Cool Cool Cool Comfort No** -5°C . **OK OK OK Start Yes Cold Cold Cold Comfort No** -15°C . **OK OK OK Start Yes Cold Cold Cold Comfort No** -25°C . Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35°C . Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0370] This is also an efficient mode for vehicle heating between -5°C . and 15°C . Energy from the air can be used at the chiller while not affecting the motor and Inverter or the battery.

Example 17E—Vehicle Heating with Ambient Between -15°C . and 5°C .—Evaporative Heat Source 16: Radiator (Air) at Chiller Using R 416A

[0371] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 19 and is operated with R1234yf and with RE1B5. The use of Refrigerant RE1B5 (R416A) is reported in Table E17E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00198 TABLE E17E Evaporative Heat Source 16: Radiator (Air) at Chiller Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No** **OK Start Yes Cool Cool Cool Comfort No** -5°C . **OK OK OK Start Yes Cold Cold Cold Comfort No** -15°C . **OK OK OK Start Yes Cold Cold Cold Comfort No** -25°C . Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35°C . Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0372] This is also an efficient mode for vehicle heating between -5°C . and 15°C . Energy from the air can be used at the chiller while not affecting the motor and Inverter or the battery.

Example 17F—Vehicle Heating with Ambient Between -15°C . and 5°C .—Evaporative Heat Source 16: Radiator (Air) at Chiller Using R444A

[0373] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 19 and is operated with R1234yf and with RE1B6. The use of Refrigerant RE1B6 (R444A) is reported in Table E17E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00199 TABLE E17E Evaporative Heat Source 16: Radiator (Air) at Chiller Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No** **OK Start Yes Cool Cool Cool Comfort No** -5°C . **OK OK OK Start Yes Cold Cold Cold Comfort No** -15°C . **OK OK OK Start Yes Cold Cold Cold Comfort No** -25°C . Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35°C . Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0374] This is also an efficient mode for vehicle heating between -5°C . and 15°C . Energy from the air can be used at the chiller while not affecting the motor and Inverter or the battery.

Example 17G—Vehicle Heating with Ambient Between -15°C. and 5°C. —Evaporative Heat

Source 16: Radiator (Air) at Chiller Using R457C

[0375] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 19 and is operated with R1234yf and with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E17G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00200 TABLE E17G Evaporative Heat Source 16: Radiator (Air) at Chiller Ambient, Motor & $^{\circ}\text{C.}$ Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No** -5°C. **OK OK OK Start Yes Cold Cold Cold Comfort No** -15°C. **OK OK OK Start Yes Cold Cold Cold Comfort No** -25°C. Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35°C. Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0376] This is also an efficient mode for vehicle heating between -5°C. and 15°C. Energy from the air can be used at the chiller while not affecting the motor and Inverter or the battery.

Example 17H—Vehicle Heating with Ambient Between -15°C. and 5°C. —Evaporative Heat Source 16: Radiator (Air) at Chiller

[0377] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 19 and is operated with R1234yf and with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E17H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00201 TABLE E17H Evaporative Heat Source 16: Radiator (Air) at Chiller Ambient, Motor & $^{\circ}\text{C.}$ Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No** -5°C. **OK OK OK Start Yes Cold Cold Cold Comfort No** -15°C. **OK OK OK Start Yes Cold Cold Cold Comfort No** -25°C. Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35°C. Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0378] This is also an efficient mode for vehicle heating between -5°C. and 15°C. Energy from the air can be used at the chiller while not affecting the motor and Inverter or the battery.

Example 18A—Vehicle Heating with Ambient Between -15°C. and 5°C. After Component Warm-Up-Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller

[0379] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 20 and is operated with R1234yf and with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E18A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00202 TABLE E18A Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller Ambient, Motor & $^{\circ}\text{C.}$ Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No** -5°C. **OK OK OK Start Yes Cold Cold Cold Comfort No** -15°C. **OK OK OK Start Yes Cold Cold Cold Comfort No** -25°C. Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35°C. Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0380] This is also an efficient mode for vehicle heating between -15°C. and 15°C. after the motor and inverter and battery are warmed up and need (or can tolerate) some cooling. It can also be used to cool the motor and inverter and battery in warm weather.

Example 18B—Vehicle Heating with Ambient Between -15°C . and 5°C . After Component Warm-Up-Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller

[0381] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 20 and is operated with R1234yf and with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E18B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00203 TABLE E18B Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0382] This is also an efficient mode for vehicle heating between -15°C . and 15°C . after the motor and inverter and battery are warmed up and need (or can tolerate) some cooling. It can also be used to cool the motor and inverter and battery in warm weather.

Example 18C—Vehicle Heating with Ambient Between -15°C . and 5°C . After Component Warm-Up-Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller

[0383] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 20 and is operated with R1234yf and with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E18C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00204 TABLE E18C Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0384] This is also an efficient mode for vehicle heating between -15°C . and 15°C . after the motor and inverter and battery are warmed up and need (or can tolerate) some cooling. It can also be used to cool the motor and inverter and battery in warm weather.

Example 18D—Vehicle Heating with Ambient Between -15°C . and 5°C . After Component Warm-Up-Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller

[0385] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 20 and is operated with R1234yf and with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E18D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00205 TABLE E18D Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

Comfort No

[0386] This is also an efficient mode for vehicle heating between -15°C . and 15°C . after the motor and inverter and battery are warmed up and need (or can tolerate) some cooling. It can also be used to cool the motor and inverter and battery in warm weather.

Example 18E—Vehicle Heating with Ambient Between -15°C . and 5°C . After Component Warm-Up-Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller

[0387] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 20 and is operated with R1234yf and with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E18E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00206 TABLE E18E Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0388] This is also an efficient mode for vehicle heating between -15°C . and 15°C . after the motor and inverter and battery are warmed up and need (or can tolerate) some cooling. It can also be used to cool the motor and inverter and battery in warm weather.

Example 18F—Vehicle Heating with Ambient Between -15°C . and 5°C . After Component Warm-Up-Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller

[0389] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 20 and is operated with R1234yf and with RE1B6. The use of Refrigerant RE1B6 (R444A) is reported in Table E18F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00207 TABLE E18F Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0390] This is also an efficient mode for vehicle heating between -15°C . and 15°C . after the motor and inverter and battery are warmed up and need (or can tolerate) some cooling. It can also be used to cool the motor and inverter and battery in warm weather.

Example 18G—Vehicle Heating with Ambient Between -15°C . and 5°C . After Component Warm-Up-Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller

[0391] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 20 and is operated with R1234yf and with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E18G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00208 TABLE E18G Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller Ambient, Motor & $^{\circ}\text{C}$. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot

Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0392] This is also an efficient mode for vehicle heating between -15° C. and 15° C. after the motor and inverter and battery are warmed up and need (or can tolerate) some cooling. It can also be used to cool the motor and inverter and battery in warm weather.

Example 18H—Vehicle Heating with Ambient Between -15° C. and 5° C. After Component Warm-Up-Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller

[0393] In this example, a system of the present invention is configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 20 and is operated with R1234yf and with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E18D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00209 TABLE E18H Evaporative Heat Source 17: Battery, Motor and Inverter at Chiller Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0394] This is also an efficient mode for vehicle heating between -15° C. and 15° C. after the motor and inverter and battery are warmed up and need (or can tolerate) some cooling. It can also be used to cool the motor and inverter and battery in warm weather.

Example 19A1-De-humidification-Evaporative Heat Source 18: Dehumidification at Evaporator

[0395] In this example, a system of the present invention configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21A. This system is operated with R1234yf and with RE1B1 and produces the results as reported in the following Table E19A11. The use of Refrigerant RE1B1 (R454C) is reported in Table E19A12 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00210 TABLE E19A11 Temperature and pressure data for Ambient Temperature of 25° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 25.0 17.04 NA 35.0 9.0 NA 5.4 9.0 RE1B1 25.0 17.04 NA 35.0 15.3 NA 7.8 15.3 (R454C)

TABLE-US-00211 TABLE E19A12 Evaporative Heat Source 18: Dehumidification at Evaporator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 **Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0396] This is the normal mode for dehumidification where the air is cooled (below the dew point to remove moisture) and then reheated to achieve a more comfortable temperature for passengers. In very warm weather there would be less or no reheat but in milder conditions dehumidification is necessary.

Example 19A2-De-humidification-Evaporative Heat Source 18: Dehumidification at Evaporator

[0397] In this example, a system of the present invention configured for operation to heat cabin air

during periods of normal ambient temperatures is illustrated in FIG. 21A. This system is operated with R1234yf and with RE1B2 and produces the results as reported in the following Table E19A21. The use of Refrigerant RE1B2 (R455A) is reported in Table E19A22 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00212 TABLE E19A21 Temperature and pressure data for Ambient Temperature of 25° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 25.0 17.04 NA 35.0 9.0 NA 5.4 9.0 RE1B2 25.0 17.04 NA 35.0 17.6 NA 8.3 17.6 (R455A)

TABLE-US-00213 TABLE E19A22 Evaporative Heat Source 18: Dehumidification at Evaporator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 **Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0398] Example 19A3-De-humidification-Evaporative Heat Source 18: Dehumidification at Evaporator

[0399] In this example, a system of the present invention configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21A. This system is operated with R1234yf and with RE1B3 and produces the results as reported in the following Table E19A31. The use of Refrigerant RE1B3 (R474A) is reported in Table E19A32 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00214 TABLE E19A31 Temperature and pressure data for Ambient Temperature of 25° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 25.0 17.04 NA 35.0 9.0 NA 5.4 9.0 RE1B3 25.0 17.04 NA 35.0 13.7 NA 7.3 13.7 (R474A)

TABLE-US-00215 TABLE E19A32 Evaporative Heat Source 18: Dehumidification at Evaporator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 **Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0400] This is the normal mode for dehumidification where the air is cooled (below the dew point to remove moisture) and then reheated to achieve a more comfortable temperature for passengers. In very warm weather there would be less or no reheat but in milder conditions dehumidification is necessary.

Example 19A4-De-humidification-Evaporative Heat Source 18: Dehumidification at Evaporator

[0401] In this example, a system of the present invention configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21A. This system is operated with R1234yf and with RE1B4 and produces the results as reported in the following Table E19A41. The use of Refrigerant RE1B4 (R474B) is reported in Table E19A42 below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00216 TABLE E19A41 Temperature and pressure data for Ambient Temperature of 25° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 25.0 17.04 NA 35.0 9.0 NA 5.4 9.0 RE1B4 25.0 17.04 NA 35.0 14.9 NA 8.0 14.9 (R474B)

TABLE-US-00217 TABLE E19A42 Evaporative Heat Source 18: Dehumidification at Evaporator

Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 **Hot Hot Hot Start Yes OK**
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot
Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool
Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start
Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool
Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold
Comfort No

[0402] This is the normal mode for dehumidification where the air is cooled (below the dew point to remove moisture) and then reheated to achieve a more comfortable temperature for passengers. In very warm weather there would be less or no reheat but in milder conditions dehumidification is necessary.

Example 19E1-De-humidification-Evaporative Heat Source 18: Dehumidification at Evaporator

[0403] In this example, a system of the present invention configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21A. This system is operated with R1234yf and with RE1B5 and produces the results as reported in the following Table E19E1A. The use of Refrigerant RE1B5 (R516A) is reported in Table E19E1B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00218 TABLE E19E1A Temperature and pressure data for Ambient Temperature of 25° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 25.0 17.04 NA 35.0 9.0 NA 5.4 9.0 RE1B5 25.0 17.04 NA 35.0 9.2 NA 5.5 9.2 (R516A)

TABLE-US-00219 TABLE E19E1B Evaporative Heat Source 18: Dehumidification at Evaporator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 **Hot Hot Hot Start Yes OK**
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot
Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool
Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start
Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool
Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold
Comfort No

[0404] This is the normal mode for dehumidification where the air is cooled (below the dew point to remove moisture) and then reheated to achieve a more comfortable temperature for passengers. In very warm weather there would be less or no reheat but in milder conditions dehumidification is necessary.

Example 19F1-De-humidification-Evaporative Heat Source 18: Dehumidification at Evaporator

[0405] In this example, a system of the present invention configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21A. This system is operated with R1234yf and with RE1B6 and produces the results as reported in the following Table E19F1A. The use of Refrigerant RE1B6 (R444A) is reported in Table E19F1A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00220 TABLE E19F1A Temperature and pressure data for Ambient Temperature of 25° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 25.0 17.04 NA 35.0 9.0 NA 5.4 9.0 RE1B6 25.0 17.04 NA 35.0 10.6 NA 4.9 10.6 (R444A)

TABLE-US-00221 TABLE E19F1A Evaporative Heat Source 18: Dehumidification at Evaporator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 **Hot Hot Hot Start Yes OK**
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot
Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool
Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start
Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool
Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold

Comfort No

[0406] Example 19G1-De-humidification-Evaporative Heat Source 18: Dehumidification at Evaporator

[0407] In this example, a system of the present invention configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21A. This system is operated with R1234yf and with RE1B7 and produces the results as reported in the following Table E19G1A. The use of Refrigerant RE1B7 (R457C) is reported in Table E19G1B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00222 TABLE E19G1A Temperature and pressure data for Ambient Temperature of 25° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 25.0 17.04 NA 35.0 9.0 NA 5.4 9.0 RE1B3 25.0 17.04 NA 35.0 11.4 NA 6.2 11.4 (R457C)

TABLE-US-00223 TABLE E19G1A Evaporative Heat Source 18: Dehumidification at Evaporator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 **Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0408] This is the normal mode for dehumidification where the air is cooled (below the dew point to remove moisture) and then reheated to achieve a more comfortable temperature for passengers. In very warm weather there would be less or no reheat but in milder conditions dehumidification is necessary.

Example 19H1-De-humidification-Evaporative Heat Source 18: Dehumidification at Evaporator

[0409] In this example, a system of the present invention configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21A. This system is operated with R1234yf and with RE1B8 and produces the results as reported in the following Table E19H1A. The use of Refrigerant RE1B8 (R457D) is reported in Table E19H1A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00224 TABLE E19H1A Temperature and pressure data for Ambient Temperature of 25° C. T4 T5 T6 T7 p1 p2 p3 p4 Refrigerant [C.] [C.] [C.] [C.] [bar] [bar] [bar] [bar] R-1234yf 25.0 17.04 NA 35.0 9.0 NA 5.4 9.0 RE1B8 25.0 17.04 NA 35.0 10.5 NA 5.9 10.5 (R457D)

TABLE-US-00225 TABLE E19H1A Evaporative Heat Source 18: Dehumidification at Evaporator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 **Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0410] This is the normal mode for dehumidification where the air is cooled (below the dew point to remove moisture) and then reheated to achieve a more comfortable temperature for passengers. In very warm weather there would be less or no reheat but in milder conditions dehumidification is necessary.

Example 19B1-De-humidification-Using Inside Condenser for Dehumidification Reheat

[0411] In this example, a system of the present invention is configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21B and is operated with RE1B1 (R454C).

Example 19B2-De-humidification-Using Inside Condenser for Dehumidification Reheat
[0412] In this example, a system of the present invention is configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21B and is operated with RE1B2 (R455A).

Example 19B3-De-humidification-Using Inside Condenser for Dehumidification Reheat
[0413] In this example, a system of the present invention is configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21B and is operated with RE1B3 (R474A).

Example 19B4-De-humidification-Using Inside Condenser for Dehumidification Reheat
[0414] In this example, a system of the present invention is configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21B and is operated with RE1B4 (R474B).

Example 19B5-De-humidification-Using Inside Condenser for Dehumidification Reheat
[0415] In this example, a system of the present invention is configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21B and is operated with RE1B5 (R516A).

Example 19B6-De-humidification-Using Inside Condenser for Dehumidification Reheat
[0416] In this example, a system of the present invention is configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21B and is operated with RE1B6 (R444A).

Example 19B7-De-humidification-Using Inside Condenser for Dehumidification Reheat
[0417] In this example, a system of the present invention is configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21B and is operated with RE1B7 (R457C).

Example 19B8-De-humidification-Using Inside Condenser for Dehumidification Reheat
[0418] In this example, a system of the present invention is configured for operation to heat cabin air during periods of normal ambient temperatures is illustrated in FIG. 21B and is operated with RE1B8 (R457D).

Examples 20-27

[0419] In addition to the evaporative heat source flexibility provided by the present invention, several energy saving, warming and cooling configurations are advantageously provided.
[0420] The configurations shown on the following page show energy saving opportunities for many conditions. [0421] Warming the battery with self-heating [0422] Warming the battery with PTC heating [0423] Warming the motor and inverter with self-heating [0424] Warming the motor and inverter with PTC [0425] Warming the battery, motor and inverter with self-heating [0426] Warming the battery, motor and inverter with PTC [0427] Cooling the motor and inverter at the radiator Cooling the motor and inverter @ the radiator (Battery @ chiller) [0428] Cooling the battery at the radiator [0429] Cooling the motor, inverter and battery at the radiator

Example 20A—Battery Warming—Warming the Battery (Self-Heating)

[0430] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures while also warming the battery is illustrated in FIG. 22 and is operated with R1234yf and with RE1B1.

[0431] The use of Refrigerant RE1B1 (R454C) is reported in Table E20A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00226 TABLE E20A Warming the Battery (Self-Heating) Ambient, Motor & ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start**
Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15

OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0432] Circulating coolant to the battery helps the battery warm up more consistently either when charging or while driving.

Example 20B—Battery Warming—Warming the Battery (Self-Heating)

[0433] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures while also warming the battery is illustrated in FIG. 22 and is operated with R1234yf and with RE1B2.

[0434] The use of Refrigerant RE1B2 (R455A) is reported in Table E20B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00227 TABLE E20B Warming the Battery (Self-Heating) Ambient, Motor & ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start
Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15
OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0435] Circulating coolant to the battery helps the battery warm up more consistently either when charging or while driving.

Example 20C—Battery Warming—Warming the Battery (Self-Heating)

[0436] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures while also warming the battery is illustrated in FIG. 22 and is operated with R1234yf and with RE1B3.

[0437] The use of Refrigerant RE1B3 (R474A) is reported in Table E20C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00228 TABLE E20C Warming the Battery (Self-Heating) Ambient, Motor & ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start
Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15
OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0438] Circulating coolant to the battery helps the battery warm up more consistently either when charging or while driving.

Example 20D—Battery Warming—Warming the Battery (Self-Heating)

[0439] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures while also warming the battery is illustrated in FIG. 22 and is operated with R1234yf and with RE1B4.

[0440] The use of Refrigerant RE1B4 (R474B) is reported in Table E20D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00229 TABLE E20D Warming the Battery (Self-Heating) Ambient, Motor & ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start
Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15
OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0441] Circulating coolant to the battery helps the battery warm up more consistently either when charging or while driving.

Example 20E—Battery Warming—Warming the Battery (Self-Heating)

[0442] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures while also warming the battery is illustrated in FIG. 22 and is operated with R1234yf and with RE1B5.

[0443] The use of Refrigerant RE1B5 (R516A) is reported in Table E20E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00230 TABLE E20E Warming the Battery (Self-Heating) Ambient, Motor & ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start
Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15
OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0444] Circulating coolant to the battery helps the battery warm up more consistently either when charging or while driving.

Example 20F—Battery Warming—Warming the Battery (Self-Heating)

[0445] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures while also warming the battery is illustrated in FIG. 22 and is operated with R1234yf and with RE1B6.

[0446] The use of Refrigerant RE1B6 (R444A) is reported in Table E20F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00231 TABLE E20F Warming the Battery (Self-Heating) Ambient, Motor & ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start
Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15
OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0447] Circulating coolant to the battery helps the battery warm up more consistently either when charging or while driving.

Example 20G—Battery Warming—Warming the Battery (Self-Heating)

[0448] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures while also warming the battery is illustrated in FIG. 22 and is operated with R1234yf and with RE1B7.

[0449] The use of Refrigerant RE1B7 (R457C) is reported in Table E20G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00232 TABLE E20G Warming the Battery (Self-Heating) Ambient, Motor & ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start
Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15
OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0450] Circulating coolant to the battery helps the battery warm up more consistently either when charging or while driving.

Example 20H—Battery Warming—Warming the Battery (Self-Heating)

[0451] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures while also warming the battery is illustrated in FIG. 22 and is operated with R1234yf and with RE1B8.

[0452] The use of Refrigerant RE1B8 (R457D) is reported in Table E20H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00233 TABLE E20H Warming the Battery (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0453] Circulating coolant to the battery helps the battery warm up more consistently either when charging or while driving.

Example 21A—Batter Heating—Very Cold Ambient—Warming the Battery with PTC

[0454] In this example, a system of the present invention configured for operation to heat cabin air during periods of very low ambient temperatures is illustrated in FIG. 23 and is operated with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E21A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00234 TABLE E21A Warming the Battery with PTC Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0455] In cool to very cold weather the battery can be heated by circulating coolant heated by the PTC. This could be necessary before charging the battery in very cold weather. This could also improve charging time (decrease) in more mild but cool conditions. This could also be used to warm up the battery at the beginning of the drive cycle.

Example 21B—Batter Heating—Very Cold Ambient—Warming the Battery with PTC

[0456] In this example, a system of the present invention configured for operation to heat cabin air during periods of very low ambient temperatures is illustrated in FIG. 23 and is operated with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E21B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00235 TABLE E21B Warming the Battery with PTC Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0457] In cool to very cold weather the battery can be heated by circulating coolant heated by the PTC. This could be necessary before charging the battery in very cold weather. This could also improve charging time (decrease) in more mild but cool conditions. This could also be used to warm up the battery at the beginning of the drive cycle.

Example 21C—Batter Heating—Very Cold Ambient—Warming the Battery with PTC
[0458] In this example, a system of the present invention configured for operation to heat cabin air during periods of very low ambient temperatures is illustrated in FIG. 23 and is operated with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E21C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00236 TABLE E21C Warming the Battery with PTC Ambient, Motor & ° C. Cabin
Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot
Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes**
Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK
OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold
Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0459] In cool to very cold weather the battery can be heated by circulating coolant heated by the PTC. This could be necessary before charging the battery in very cold weather. This could also improve charging time (decrease) in more mild but cool conditions. This could also be used to warm up the battery at the beginning of the drive cycle.

Example 21D—Batter Heating—Very Cold Ambient—Warming the Battery with PTC
[0460] In this example, a system of the present invention configured for operation to heat cabin air during periods of very low ambient temperatures is illustrated in FIG. 23 and is operated with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E21D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00237 TABLE E21D Warming the Battery with PTC Ambient, Motor & ° C. Cabin
Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot
Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes**
Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK
OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold
Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0461] In cool to very cold weather the battery can be heated by circulating coolant heated by the PTC. This could be necessary before charging the battery in very cold weather. This could also improve charging time (decrease) in more mild but cool conditions. This could also be used to warm up the battery at the beginning of the drive cycle.

Example 21E—Battery Heating—Very Cold Ambient—Warming the Battery with PTC
[0462] In this example, a system of the present invention configured for operation to heat cabin air during periods of very low ambient temperatures is illustrated in FIG. 23 and is operated with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E2E1E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00238 TABLE E21E Warming the Battery with PTC Ambient, Motor & ° C. Cabin
Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot
Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes**
Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK
OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold
Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0463] In cool to very cold weather the battery can be heated by circulating coolant heated by the PTC. This could be necessary before charging the battery in very cold weather. This could also improve charging time (decrease) in more mild but cool conditions. This could also be used to

warm up the battery at the beginning of the drive cycle.

Example 21F—Battery Heating—Very Cold Ambient—Warming the Battery with PTC

[0464] In this example, a system of the present invention configured for operation to heat cabin air during periods of very low ambient temperatures is illustrated in FIG. 23 and is operated with RE1B6. The use of Refrigerant RE1B6 (R444A) is reported in Table E21F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00239 TABLE E21F Warming the Battery with PTC Ambient, Motor & ° C. Cabin
Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot
Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes
Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK
OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold
Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0465] In cool to very cold weather the battery can be heated by circulating coolant heated by the PTC. This could be necessary before charging the battery in very cold weather. This could also improve charging time (decrease) in more mild but cool conditions. This could also be used to warm up the battery at the beginning of the drive cycle.

Example 21G—Battery Heating—Very Cold Ambient—Warming the Battery with PTC

[0466] In this example, a system of the present invention configured for operation to heat cabin air during periods of very low ambient temperatures is illustrated in FIG. 23 and is operated with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E21G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00240 TABLE E21G Warming the Battery with PTC Ambient, Motor & ° C. Cabin
Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot
Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes
Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK
OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold
Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0467] In cool to very cold weather the battery can be heated by circulating coolant heated by the PTC. This could be necessary before charging the battery in very cold weather. This could also improve charging time (decrease) in more mild but cool conditions. This could also be used to warm up the battery at the beginning of the drive cycle.

Example 21H—Battery Heating—Very Cold Ambient—Warming the Battery with PTC

[0468] In this example, a system of the present invention configured for operation to heat cabin air during periods of very low ambient temperatures is illustrated in FIG. 23 and is operated with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E21H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00241 TABLE E21H Warming the Battery with PTC Ambient, Motor & ° C. Cabin
Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot
Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm
Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes
Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK
OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold
Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0469] In cool to very cold weather the battery can be heated by circulating coolant heated by the PTC. This could be necessary before charging the battery in very cold weather. This could also

improve charging time (decrease) in more mild but cool conditions. This could also be used to warm up the battery at the beginning of the drive cycle.

Example 22A—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating) [0470] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 24 and is operated with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E22A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00242 TABLE E22A Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0471] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. Coolant can be circulated without removing heat.

Example 22B—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating) [0472] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 24 and is operated with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E22B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00243 TABLE E22B Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0473] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. Coolant can be circulated without removing heat.

Example 22C—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating) [0474] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 24 and is operated with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E22C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00244 TABLE E22C Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0475] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. Coolant can be circulated without removing heat.

Example 22C—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating) [0476] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 24 and is operated with RE1B4.

The use of Refrigerant RE1B4 (R474B) is reported in Table E22D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00245 TABLE E22D Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cold Cold Cold Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0477] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. Coolant can be circulated without removing heat.

Example 22E—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating)

[0478] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 24 and is operated with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E22E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00246 TABLE E22E Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0479] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. Coolant can be circulated without removing heat.

Example 22F—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating)

[0480] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 24 and is operated with RE1B6. The use of Refrigerant RE1B6 (R444A) is reported in Table E22F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00247 TABLE E22F Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0481] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. Coolant can be circulated without removing heat.

Example 22G—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating)

[0482] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 24 and is operated with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E22G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00248 TABLE E22G Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm

Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK**
OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort
No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold
Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0483] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. Coolant can be circulated without removing heat.

Example 22H—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating)

[0484] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 24 and is operated with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E22H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00249 TABLE E22H Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK**
OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort
No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold
Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0485] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. Coolant can be circulated without removing heat.

Example 23A—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating)

[0486] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 25 and is operated with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E23A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00250 TABLE E23A Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK**
OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort
No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold
Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0487] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. PTC heat can be used to speed up the warmup. This may improve vehicle efficiency after charging and just prior to drive start.

Example 23B—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating)

[0488] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 25 and is operated with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E23B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00251 TABLE E23B Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK**
OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort
No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold

Cold Cold Comfort No -35 **Cool Cold Cool Start Yes Cold Cold Cold Comfort No**
[0489] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. PTC heat can be used to speed up the warmup. This may improve vehicle efficiency after charging and just prior to drive start.

Example 23C—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating)
[0490] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 25 and is operated with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E23C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00252 TABLE E23C Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0491] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. PTC heat can be used to speed up the warmup. This may improve vehicle efficiency after charging and just prior to drive start.

Example 23D—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating)
[0492] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 25 and is operated with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E23D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00253 TABLE E23D Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0493] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. PTC heat can be used to speed up the warmup. This may improve vehicle efficiency after charging and just prior to drive start.

Example 23E—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating)
[0494] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 25 and is operated with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E23E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00254 TABLE E23E Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0495] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important

to achieve the best efficiency. PTC heat can be used to speed up the warmup.

[0496] This may improve vehicle efficiency after charging and just prior to drive start. Example 23F—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating) In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 25 and is operated with RE1B6. The use of Refrigerant RE1B6 (R444A) is reported in Table E23F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00255 TABLE E23F Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0497] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. PTC heat can be used to speed up the warmup. This may improve vehicle efficiency after charging and just prior to drive start.

Example 23G—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating) [0498] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 25 and is operated with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E23G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00256 TABLE E23G Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0499] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. PTC heat can be used to speed up the warmup. This may improve vehicle efficiency after charging and just prior to drive start.

Example 23H—Motor and Inverter Self Heating—Warming the Motor and Inverter (Self-Heating) [0500] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 25 and is operated with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E23H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00257 TABLE E23H Warming the Motor and Inverter (Self-Heating) Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 **OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0501] Allowing the motor and Inverter in cold to cool weather to uniformly self-heat is important to achieve the best efficiency. PTC heat can be used to speed up the warmup. This may improve vehicle efficiency after charging and just prior to drive start.

Example 24A—Battery, Motor and Inverter Heating—Warming the Battery, Motor and Inverter with Self Heating

[0502] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 26 and is operated with RE1B1. [0503] The use of Refrigerant RE1B1 (R454C) is reported in Table E24A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00258 TABLE E24A Warming the Battery, Motor and Inverter with Self Heating
Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 **OK** Warm Warm **Start Yes Cool Cool Cool** Comfort
No 5 OK OK OK Start Yes Cool Cool Cool Comfort **No -5 OK OK OK Start Yes Cold Cold**
Cold Comfort **No -15 OK OK OK Start Yes Cold Cold Cold** Comfort **No -25 Cool Cool Cool**
Start Yes Cold Cold Cold Comfort **No -35 Cool Cool Cool Start Yes Cold Cold Cold** Comfort
No

[0504] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. This would likely be used during charging until cooling is necessary for the battery. It could also be used during driving to warm the battery until the desired temperature is reached when the battery would be removed from the loop.

Example 24B—Battery, Motor and Inverter Heating—Warming the Battery, Motor and Inverter with Self Heating

[0505] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 26 and is operated with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E24B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00259 TABLE E24B Warming the Battery, Motor and Inverter with Self Heating
Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 **OK** Warm Warm **Start Yes Cool Cool Cool** Comfort
No 5 OK OK OK Start Yes Cool Cool Cool Comfort **No -5 OK OK OK Start Yes Cold Cold**
Cold Comfort **No -15 OK OK OK Start Yes Cold Cold Cold** Comfort **No -25 Cool Cool Cool**
Start Yes Cold Cold Cold Comfort **No -35 Cool Cool Cool Start Yes Cold Cold Cold** Comfort
No

[0506] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. This would likely be used during charging until cooling is necessary for the battery. It could also be used during driving to warm the battery until the desired temperature is reached when the battery would be removed from the loop.

Example 24C—Battery, Motor and Inverter Heating—Warming the Battery, Motor and Inverter with Self Heating

[0507] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 26 and is operated with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E24C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00260 TABLE E24C Warming the Battery, Motor and Inverter with Self Heating
Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 **OK** Warm Warm **Start Yes Cool Cool Cool** Comfort
No 5 OK OK OK Start Yes Cool Cool Cool Comfort **No -5 OK OK OK Start Yes Cold Cold**

Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0508] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. This would likely be used during charging until cooling is necessary for the battery. It could also be used during driving to warm the battery until the desired temperature is reached when the battery would be removed from the loop.

Example 24D—Battery, Motor and Inverter Heating—Warming the Battery, Motor and Inverter with Self Heating

[0509] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 26 and is operated with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E24D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00261 TABLE E24D Warming the Battery, Motor and Inverter with Self Heating
Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort**
No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold
Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort
No

[0510] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. This would likely be used during charging until cooling is necessary for the battery. It could also be used during driving to warm the battery until the desired temperature is reached when the battery would be removed from the loop.

Example 24E—Battery, Motor and Inverter Heating—Warming the Battery, Motor and Inverter with Self Heating

[0511] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 26 and is operated with RE1B5.

[0512] The use of Refrigerant RE1B5 (R516A) is reported in Table E24E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00262 TABLE E24E Warming the Battery, Motor and Inverter with Self Heating
Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK
Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot
Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort**
No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold
Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool
Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort
No

[0513] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. This would likely be used during charging until cooling is necessary for the battery. It could also be used during driving to warm the battery until the desired temperature is reached when the battery would be removed from the loop.

Example 24F—Battery, Motor and Inverter Heating—Warming the Battery, Motor and Inverter with Self Heating

[0514] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 26 and is operated with RE1B6.

The use of Refrigerant RE1B6 (R444A) is reported in Table E24F below by using bold type as an

indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00263 TABLE E24F Warming the Battery, Motor and Inverter with Self Heating Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK** Warm Warm **Start Yes Cool Cool Cool** Comfort No 5 **OK OK OK Start Yes Cool Cool Cool** Comfort No -5 **OK OK OK Start Yes Cold Cold Cold** Comfort No -15 **OK OK OK Start Yes Cold Cold Cold** Comfort No -25 **Cool Cool Cool Start Yes Cold Cold Cold** Comfort No -35 **Cool Cool Cool Start Yes Cold Cold Cold** Comfort No

[0515] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. This would likely be used during charging until cooling is necessary for the battery. It could also be used during driving to warm the battery until the desired temperature is reached when the battery would be removed from the loop.

Example 24G—Battery, Motor and Inverter Heating—Warming the Battery, Motor and Inverter with Self Heating

[0516] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 26 and is operated with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E24G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00264 TABLE E24G Warming the Battery, Motor and Inverter with Self Heating Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK** Warm Warm **Start Yes Cool Cool Cool** Comfort No 5 **OK OK OK Start Yes Cool Cool Cool** Comfort No -5 **OK OK OK Start Yes Cold Cold Cold** Comfort No -15 **OK OK OK Start Yes Cold Cold Cold** Comfort No -25 **Cool Cool Cool Start Yes Cold Cold Cold** Comfort No -35 **Cool Cool Cool Start Yes Cold Cold Cold** Comfort No

[0517] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. This would likely be used during charging until cooling is necessary for the battery. It could also be used during driving to warm the battery until the desired temperature is reached when the battery would be removed from the loop.

Example 24H—Battery, Motor and Inverter Heating—Warming the Battery, Motor and Inverter with Self Heating

[0518] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 26 and is operated with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E24H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00265 TABLE E24H Warming the Battery, Motor and Inverter with Self Heating Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK** Warm Warm **Start Yes Cool Cool Cool** Comfort No 5 **OK OK OK Start Yes Cool Cool Cool** Comfort No -5 **OK OK OK Start Yes Cold Cold Cold** Comfort No -15 **OK OK OK Start Yes Cold Cold Cold** Comfort No -25 **Cool Cool Cool Start Yes Cold Cold Cold** Comfort No -35 **Cool Cool Cool Start Yes Cold Cold Cold** Comfort No

[0519] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. This would likely be used during charging until cooling is necessary for the

battery. It could also be used during driving to warm the battery until the desired temperature is reached when the battery would be removed from the loop.

Example 25A—Battery, Motor and Inverter Self Heating—Warming the Battery, Motor and Inverter with PTC

[0520] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 27 and is operated with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E25A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00266 TABLE E25A Warming the Battery, Motor and Inverter with PTC Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK** Warm Warm **Start Yes Cool Cool Cool** Comfort **No 5 OK OK OK Start Yes Cool Cool Cool** Comfort **No -5 OK OK OK Start Yes Cold Cold Cold** Comfort **No -15 OK OK OK Start Yes Cold Cold Cold** Comfort **No -25 Cool Cool Cool Start Yes Cold Cold Cold** Comfort **No -35 Cool Cool Cool Start Yes Cold Cold Cold** Comfort **No**

[0521] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. PTC heat could also be used to augment the warming. This would be appropriate while or prior to charging.

Example 25B—Battery, Motor and Inverter Self Heating—Warming the Battery, Motor and Inverter with PTC

[0522] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 27 and is operated with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E25B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00267 TABLE E25B Warming the Battery, Motor and Inverter with PTC Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK** Warm Warm **Start Yes Cool Cool Cool** Comfort **No 5 OK OK OK Start Yes Cool Cool Cool** Comfort **No -5 OK OK OK Start Yes Cold Cold Cold** Comfort **No -15 OK OK OK Start Yes Cold Cold Cold** Comfort **No -25 Cool Cool Cool Start Yes Cold Cold Cold** Comfort **No -35 Cool Cool Cool Start Yes Cold Cold Cold** Comfort **No**

[0523] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. PTC heat could also be used to augment the warming. This would be appropriate while or prior to charging.

Example 25C—Battery, Motor and Inverter Self Heating—Warming the Battery, Motor and Inverter with PTC

[0524] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 27 and is operated with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E25C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00268 TABLE E25C Warming the Battery, Motor and Inverter with PTC Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK** Warm Warm **Start Yes Cool Cool Cool** Comfort **No 5 OK OK OK Start Yes Cool Cool Cool** Comfort **No -5 OK OK OK Start Yes Cold Cold Cold** Comfort **No -15 OK OK OK Start Yes Cold Cold Cold** Comfort **No -25 Cool Cool Cool Start Yes Cold Cold Cold** Comfort **No -35 Cool Cool Cool Start Yes Cold Cold Cold** Comfort **No**

[0525] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. PTC heat could also be used to augment the warming. This would be appropriate while or prior to charging.

Example 25D—Battery, Motor and Inverter Self Heating—Warming the Battery, Motor and Inverter with PTC

[0526] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 27 and is operated with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E25D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00269 TABLE E25D Warming the Battery, Motor and Inverter with PTC Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK** Warm Warm **Start Yes Cool Cool Cool** Comfort **No 5 OK OK OK Start Yes Cool** Cool **Cool** Comfort **No -5 OK OK OK Start Yes Cold Cold Cold** Comfort **No -15 OK OK OK Start Yes Cold Cold Cold** Comfort **No -25 Cool Cool Cool Start Yes Cold Cold Cold** Comfort **No -35 Cool Cool Cool Start Yes Cold Cold Cold** Comfort **No**

[0527] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. PTC heat could also be used to augment the warming. This would be appropriate while or prior to charging.

Example 25E—Battery, Motor and Inverter Self Heating—Warming the Battery, Motor and Inverter with PTC

[0528] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 27 and is operated with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E25E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00270 TABLE E25RE Warming the Battery, Motor and Inverter with PTC Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK** Warm Warm **Start Yes Cool Cool Cool** Comfort **No 5 OK OK OK Start Yes Cool** Cool **Cool** Comfort **No -5 OK OK OK Start Yes Cold Cold Cold** Comfort **No -15 OK OK OK Start Yes Cold Cold Cold** Comfort **No -25 Cool Cool Cool Start Yes Cold Cold Cold** Comfort **No -35 Cool Cool Cool Start Yes Cold Cold Cold** Comfort **No**

[0529] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. PTC heat could also be used to augment the warming. This would be appropriate while or prior to charging.

Example 25F—Battery, Motor and Inverter Self Heating—Warming the Battery, Motor and Inverter with PTC

[0530] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 27 and is operated with RE1B6. The use of Refrigerant RE1B6 (R444A) is reported in Table E25F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00271 TABLE E25F Warming the Battery, Motor and Inverter with PTC Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK** Warm Warm **Start Yes Cool Cool Cool** Comfort **No 5 OK OK OK Start Yes Cool** Cool **Cool** Comfort **No -5 OK OK OK Start Yes Cold Cold Cold** Comfort **No -15 OK OK OK Start Yes Cold Cold Cold** Comfort **No -25 Cool Cool Cool Start Yes Cold**

Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0531] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. PTC heat could also be used to augment the warming. This would be appropriate while or prior to charging.

Example 25G—Battery, Motor and Inverter Self Heating—Warming the Battery, Motor and Inverter with PTC

[0532] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 27 and is operated with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E25G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00272 TABLE E25G Warming the Battery, Motor and Inverter with PTC Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0533] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. PTC heat could also be used to augment the warming. This would be appropriate while or prior to charging.

Example 25H—Battery, Motor and Inverter Self Heating—Warming the Battery, Motor and Inverter with PTC

[0534] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 27 and is operated with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E25H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00273 TABLE E25H Warming the Battery, Motor and Inverter with PTC Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot Hot Hot Start Yes OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm Warm Comfort No 15 **OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No**

[0535] Circulating coolant through the motor/inverter and battery would be an efficient way to warm both devices. PTC heat could also be used to augment the warming. This would be appropriate while or prior to charging.

Example 26A—Motor and Inverter Cooling—Cooling the Motor and Inverter at the Radiator

[0536] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 28 and is operated with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E26A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00274 TABLE E26A Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No 35 Hot Hot Hot** Start Yes **OK Warm Warm Comfort No 25 Hot Hot Hot** Start Yes **OK Warm Warm Comfort No 15 OK Warm Warm** Start Yes **Cool Cool Cool Comfort No 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes**

Cold Cold Cold Comfort No -35 Cold Cold Cold Start Yes Cold Cold Cold Comfort No
[0537] In many ambient driving conditions, it should be very efficient to cool the motor and inverter at the radiator as necessary without increasing the load on the AC system (at the chiller) which takes additional energy that is likely being used to keep the vehicle cool.

Example 26B—Motor and Inverter Cooling—Cooling the Motor and Inverter at the Radiator
[0538] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 28 and is operated with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E26B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00275 TABLE E26B Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0539] In many ambient driving conditions, it should be very efficient to cool the motor and inverter at the radiator as necessary without increasing the load on the AC system (at the chiller) which takes additional energy that is likely being used to keep the vehicle cool.

Example 26C—Motor and Inverter Cooling—Cooling the Motor and Inverter at the Radiator
[0540] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 28 and is operated with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E26C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00276 TABLE E26C Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0541] In many ambient driving conditions, it should be very efficient to cool the motor and inverter at the radiator as necessary without increasing the load on the AC system (at the chiller) which takes additional energy that is likely being used to keep the vehicle cool.

Example 26D—Motor and Inverter Cooling—Cooling the Motor and Inverter at the Radiator
[0542] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 28 and is operated with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E26D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00277 TABLE E26D Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0543] In many ambient driving conditions, it should be very efficient to cool the motor and

inverter at the radiator as necessary without increasing the load on the AC system (at the chiller) which takes additional energy that is likely being used to keep the vehicle cool.

Example 26E—Motor and Inverter Cooling—Cooling the Motor and Inverter at the Radiator [0544] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 28 and is operated with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E26E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00278 TABLE E26E Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0545] In many ambient driving conditions, it should be very efficient to cool the motor and inverter at the radiator as necessary without increasing the load on the AC system (at the chiller) which takes additional energy that is likely being used to keep the vehicle cool.

Example 26F—Motor and Inverter Cooling—Cooling the Motor and Inverter at the Radiator [0546] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 28 and is operated with RE1B6. The use of Refrigerant RE1B6 (R444A) is reported in Table E26F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00279 TABLE E26F Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0547] In many ambient driving conditions, it should be very efficient to cool the motor and inverter at the radiator as necessary without increasing the load on the AC system (at the chiller) which takes additional energy that is likely being used to keep the vehicle cool.

Example 26G—Motor and Inverter Cooling—Cooling the Motor and Inverter at the Radiator [0548] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 28 and is operated with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E26G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00280 TABLE E26G Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0549] In many ambient driving conditions, it should be very efficient to cool the motor and inverter at the radiator as necessary without increasing the load on the AC system (at the chiller) which takes additional energy that is likely being used to keep the vehicle cool.

Example 26H—Motor and Inverter Cooling—Cooling the Motor and Inverter at the Radiator
[0550] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 28 and is operated with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E26H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00281 TABLE E26H Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0551] In many ambient driving conditions, it should be very efficient to cool the motor and inverter at the radiator as necessary without increasing the load on the AC system (at the chiller) which takes additional energy that is likely being used to keep the vehicle cool.

Example 27A—Motor and Inverter Self Heating

[0552] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 29A and is operated with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E27A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00282 TABLE E27A Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0553] This mode (cooling the motor and inverter at the radiator) could also be used while cooling the battery at the Chiller. This would be the predominant component cooling mode in warm to hot weather. The Evaporator is used to cool the vehicle. In such a mode where the OHE is used, enhanced performance can be expected for such systems in this and/or similar configurations in which the OHE is used.

[0554] The COP and heat capacity of the system operating in three modes according to this example is approximated by the results shown in FIG. 29B.

Example 27B—Motor and Inverter Self Heating

[0555] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 29A and is operated with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E27B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00283 TABLE E27B Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0556] This mode (cooling the motor and inverter at the radiator) could also be used while cooling

the battery at the Chiller. This would be the predominant component cooling mode in warm to hot weather. The Evaporator is used to cool the vehicle. In such a mode where the OHE is used, enhanced performance can be expected for such systems in this and/or similar configurations in which the OHE is used.

[0557] The COP and heat capacity of the system operating in three modes according to this example is approximated by the results shown in FIG. 29B.

Example 27C—Motor and Inverter Self Heating

[0558] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 29A and is operated with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E27C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00284 TABLE E27C Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0559] This mode (cooling the motor and inverter at the radiator) could also be used while cooling the battery at the Chiller. This would be the predominant component cooling mode in warm to hot weather. The Evaporator is used to cool the vehicle. In such a mode where the OHE is used, enhanced performance can be expected for such systems in this and/or similar configurations in which the OHE is used.

[0560] The COP and heat capacity of the system operating in three modes according to this example is approximated by the results shown in FIG. 29B.

Example 27D—Motor and Inverter Self Heating

[0561] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 29A and is operated with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E27D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00285 TABLE E27D Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0562] This mode (cooling the motor and inverter at the radiator) could also be used while cooling the battery at the Chiller. This would be the predominant component cooling mode in warm to hot weather. The Evaporator is used to cool the vehicle. In such a mode where the OHE is used, enhanced performance can be expected for such systems in this and/or similar configurations in which the OHE is used.

[0563] The COP and heat capacity of the system operating in three modes according to this example is approximated by the results shown in FIG. 29B.

Example 27E—Motor and Inverter Self Heating

[0564] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 29A and is operated with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E27E below by using bold type as an

indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00286 TABLE E27E Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0565] This mode (cooling the motor and inverter at the radiator) could also be used while cooling the battery at the Chiller. This would be the predominant component cooling mode in warm to hot weather. The Evaporator is used to cool the vehicle. In such a mode where the OHE is used, enhanced performance can be expected for such systems in this and/or similar configurations in which the OHE is used.

Example 27F—Motor and Inverter Self Heating

[0566] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 29A and is operated with RE1B6. The use of Refrigerant RE1B6 (R444A) is reported in Table E27F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00287 TABLE E27F Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0567] This mode (cooling the motor and inverter at the radiator) could also be used while cooling the battery at the Chiller. This would be the predominant component cooling mode in warm to hot weather. The Evaporator is used to cool the vehicle. In such a mode where the OHE is used, enhanced performance can be expected for such systems in this and/or similar configurations in which the OHE is used.

Example 27G—Motor and Inverter Self Heating

[0568] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 29A and is operated with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E27G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00288 TABLE E27G Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0569] This mode (cooling the motor and inverter at the radiator) could also be used while cooling the battery at the Chiller. This would be the predominant component cooling mode in warm to hot weather. The Evaporator is used to cool the vehicle. In such a mode where the OHE is used, enhanced performance can be expected for such systems in this and/or similar configurations in which the OHE is used.

Example 27H—Motor and Inverter Self Heating

[0570] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 29A and is operated with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E27H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00289 TABLE E27H Cooling the Motor and Inverter at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start Yes **Cool Cool Cool Comfort No** 5 OK OK OK Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0571] This mode (cooling the motor and inverter at the radiator) could also be used while cooling the battery at the Chiller. This would be the predominant component cooling mode in warm to hot weather. The Evaporator is used to cool the vehicle. In such a mode where the OHE is used, enhanced performance can be expected for such systems in this and/or similar configurations in which the OHE is used.

Example 28A—Battery Cooling—Cooling of the Battery at the Radiator

[0572] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 30 and is operated with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E28A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00290 TABLE E28A Cooling of the Battery at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 **Hot Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start **Yes Cool Cool Cool Comfort No** 5 **OK OK OK Start Yes Cool Cool Cool Comfort No** -5 **OK OK OK Start Yes Cold Cold Cold Comfort No** -15 **OK OK OK Start Yes Cold Cold Cold Comfort No** -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0573] In many ambient cases the battery could also be cooled at the radiator to reduce energy usage (over the chiller). In warm to mild conditions this will be used during charging to remove heat from the battery.

Example 28B—Battery Cooling—Cooling of the Battery at the Radiator

[0574] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 30 and is operated with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E28B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00291 TABLE E28B Cooling of the Battery at the Radiator Ambient, Motor & ° C. Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No 35 Hot **Hot Hot** Start Yes **OK Warm Warm Comfort No** 25 **Hot Hot Hot** Start Yes **OK Warm Warm Comfort No** 15 **OK Warm Warm** Start **Yes Cool Cool Cool Comfort No** 5 **OK OK OK Start Yes Cool Cool Cool Comfort No** -5 **OK OK OK Start Yes Cold Cold Cold Comfort No** -15 **OK OK OK Start Yes Cold Cold Cold Comfort No** -25 Cool Cool Cool Start Yes Cold Cold Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0575] In many ambient cases the battery could also be cooled at the radiator to reduce energy usage (over the chiller). In warm to mild conditions this will be used during charging to remove heat from the battery.

Example 28C—Battery Cooling—Cooling of the Battery at the Radiator

[0576] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 30 and is operated with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E28C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00292 TABLE E28C Cooling of the Battery at the Radiator Ambient, Motor & ° C.

Cabin	Battery	Inverter	Driving	Charging	45	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No				
35	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	25	Hot	Hot	Hot	Start	Yes	OK	Warm	
	Warm	Comfort	No	15	OK	Warm	Warm	Start	Yes	Cool	Cool	Cool	Comfort	No	5	OK	OK	OK	
	Start	Yes	Cool	Cool	Cool	Comfort	No	-5	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	
	-15	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-25	Cool	Cool	Cool	Start	Yes	Cold	Cold
	Cold	Comfort	No	-35	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No					

[0577] In many ambient cases the battery could also be cooled at the radiator to reduce energy usage (over the chiller). In warm to mild conditions this will be used during charging to remove heat from the battery.

Example 28D—Battery Cooling—Cooling of the Battery at the Radiator

[0578] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 30 and is operated with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E28D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00293 TABLE E28D Cooling of the Battery at the Radiator Ambient, Motor & ° C.

Cabin	Battery	Inverter	Driving	Charging	45	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No				
35	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	25	Hot	Hot	Hot	Start	Yes	OK	Warm	
	Warm	Comfort	No	15	OK	Warm	Warm	Start	Yes	Cool	Cool	Cool	Comfort	No	5	OK	OK	OK	
	Start	Yes	Cool	Cool	Cool	Comfort	No	-5	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	
	-15	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-25	Cool	Cool	Cool	Start	Yes	Cold	Cold
	Cold	Comfort	No	-35	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No					

[0579] In many ambient cases the battery could also be cooled at the radiator to reduce energy usage (over the chiller). In warm to mild conditions this will be used during charging to remove heat from the battery.

Example 28E—Battery Cooling—Cooling of the Battery at the Radiator

[0580] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 30 and is operated with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E28E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00294 TABLE E28E Cooling of the Battery at the Radiator Ambient, Motor & ° C.

Cabin	Battery	Inverter	Driving	Charging	45	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No				
35	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	25	Hot	Hot	Hot	Start	Yes	OK	Warm	
	Warm	Comfort	No	15	OK	Warm	Warm	Start	Yes	Cool	Cool	Cool	Comfort	No	5	OK	OK	OK	
	Start	Yes	Cool	Cool	Cool	Comfort	No	-5	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	
	-15	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-25	Cool	Cool	Cool	Start	Yes	Cold	Cold
	Cold	Comfort	No	-35	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No					

[0581] In many ambient cases the battery could also be cooled at the radiator to reduce energy usage (over the chiller). In warm to mild conditions this will be used during charging to remove heat from the battery.

Example 28F—Battery Cooling—Cooling of the Battery at the Radiator

[0582] In this example, a system of the present invention configured for operation to heat cabin air

during periods of low ambient temperatures is illustrated in FIG. 30 and is operated with RE1B6. The use of Refrigerant RE1B6 (R444A) is reported in Table E28F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00295 TABLE E28F Cooling of the Battery at the Radiator Ambient, Motor & ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot **Hot** Hot Start Yes **OK Warm Warm Comfort No** 25 **Hot Hot** Hot Start Yes **OK Warm**
Warm Comfort No 15 **OK Warm Warm** Start **Yes Cool Cool Cool Comfort No** 5 **OK OK OK**
Start Yes Cool Cool Cool Comfort No -5 **OK OK OK Start Yes** Cold Cold **Cold** Comfort No
-15 **OK OK OK Start Yes** Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0583] In many ambient cases the battery could also be cooled at the radiator to reduce energy usage (over the chiller). In warm to mild conditions this will be used during charging to remove heat from the battery.

Example 28G—Battery Cooling—Cooling of the Battery at the Radiator

[0584] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 30 and is operated with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E28G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00296 TABLE E28G Cooling of the Battery at the Radiator Ambient, Motor & ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot **Hot** Hot Start Yes **OK Warm Warm Comfort No** 25 **Hot Hot** Hot Start Yes **OK Warm**
Warm Comfort No 15 **OK Warm Warm** Start **Yes Cool Cool Cool Comfort No** 5 **OK OK OK**
Start Yes Cool Cool Cool Comfort No -5 **OK OK OK Start Yes** Cold Cold **Cold** Comfort No
-15 **OK OK OK Start Yes** Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0585] In many ambient cases the battery could also be cooled at the radiator to reduce energy usage (over the chiller). In warm to mild conditions this will be used during charging to remove heat from the battery.

Example 28H—Battery Cooling—Cooling of the Battery at the Radiator

[0586] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 30 and is operated with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E28H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00297 TABLE E28H Cooling of the Battery at the Radiator Ambient, Motor & ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot **Hot** Hot Start Yes **OK Warm Warm Comfort No** 25 **Hot Hot** Hot Start Yes **OK Warm**
Warm Comfort No 15 **OK Warm Warm** Start **Yes Cool Cool Cool Comfort No** 5 **OK OK OK**
Start Yes Cool Cool Cool Comfort No -5 **OK OK OK Start Yes** Cold Cold **Cold** Comfort No
-15 **OK OK OK Start Yes** Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0587] In many ambient cases the battery could also be cooled at the radiator to reduce energy usage (over the chiller). In warm to mild conditions this will be used during charging to remove heat from the battery.

Example 29A—Battery, Motor and Inverter Self Heating-Cooling of the Battery at the Radiator

[0588] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 31 and is operated with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E29A below by using bold type as an

indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00298 TABLE E29A Cooling of the Battery at the Radiator Ambient, Motor & ° C.

Cabin	Battery	Inverter	Driving	Charging	45	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No																																																																								
35	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	25	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	15	OK	Warm	Warm	Start	Yes	Cool	Cool	Cool	Comfort	No	5	OK	OK	OK	Start	Yes	Cool	Cool	Cool	Comfort	No	-5	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-15	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-25	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No	-35	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No

[0589] In many conditions, the motor, inverter and battery can be cooled at the radiator reducing energy consumption at the chiller. In cool to warm conditions this might be the predominant method of cooling the components.

Example 29B—Battery, Motor and Inverter Self Heating-Cooling of the Battery at the Radiator

[0590] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 31 and is operated with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E29B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00299 TABLE E29B Cooling of the Battery at the Radiator Motor & Ambient, ° C.

Cabin	Battery	Inverter	Driving	Charging	45	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No																																																																								
35	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	25	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	15	OK	Warm	Warm	Start	Yes	Cool	Cool	Cool	Comfort	No	5	OK	OK	OK	Start	Yes	Cool	Cool	Cool	Comfort	No	-5	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-15	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-25	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No	-35	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No

[0591] In many conditions, the motor, inverter and battery can be cooled at the radiator reducing energy consumption at the chiller. In cool to warm conditions this might be the predominant method of cooling the components.

Example 29C—Battery, Motor and Inverter Self Heating-Cooling of the Battery at the Radiator

[0592] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 31 and is operated with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E29C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00300 TABLE E29C Cooling of the Battery at the Radiator Motor & Ambient, ° C.

Cabin	Battery	Inverter	Driving	Charging	45	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No																																																																								
35	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	25	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	15	OK	Warm	Warm	Start	Yes	Cool	Cool	Cool	Comfort	No	5	OK	OK	OK	Start	Yes	Cool	Cool	Cool	Comfort	No	-5	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-15	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-25	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No	-35	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No

[0593] In many conditions, the motor, inverter and battery can be cooled at the radiator reducing energy consumption at the chiller. In cool to warm conditions this might be the predominant method of cooling the components.

Example 29D—Battery, Motor and Inverter Self Heating-Cooling of the Battery at the Radiator

[0594] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 31 and is operated with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E29D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00301 TABLE E29D Cooling of the Battery at the Radiator Motor & Ambient, ° C.
 Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
 35 Hot **Hot Hot Start** Yes **OK Warm Warm Comfort No** 25 **Hot Hot Hot Start** Yes **OK Warm**
Warm Comfort No 15 **OK Warm Warm Start** Yes **Cool Cool Cool Comfort No** 5 **OK OK OK**
Start Yes **Cool Cool Cool** Comfort No -5 **OK OK OK Start** Yes Cold Cold **Cold** Comfort No
 -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
 Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0595] In many conditions, the motor, inverter and battery can be cooled at the radiator reducing energy consumption at the chiller. In cool to warm conditions this might be the predominant method of cooling the components.

Example 29E—Battery, Motor and Inverter Self Heating-Cooling of the Battery at the Radiator
 [0596] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 31 and is operated with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E29E below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00302 TABLE E29E Cooling of the Battery at the Radiator Motor & Ambient, ° C.
 Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
 35 Hot **Hot Hot Start** Yes **OK Warm Warm Comfort No** 25 **Hot Hot Hot Start** Yes **OK Warm**
Warm Comfort No 15 **OK Warm Warm Start** Yes **Cool Cool Cool Comfort No** 5 **OK OK OK**
Start Yes **Cool Cool Cool** Comfort No -5 **OK OK OK Start** Yes Cold Cold **Cold** Comfort No
 -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
 Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0597] In many conditions, the motor, inverter and battery can be cooled at the radiator reducing energy consumption at the chiller. In cool to warm conditions this might be the predominant method of cooling the components.

Example 29F—Battery, Motor and Inverter Self Heating-Cooling of the Battery at the Radiator
 [0598] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 31 and is operated with RE1B6. The use of Refrigerant RE1B6 (R444A) is reported in Table E29F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00303 TABLE E29F Cooling of the Battery at the Radiator Motor & Ambient, ° C.
 Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
 35 Hot **Hot Hot Start** Yes **OK Warm Warm Comfort No** 25 **Hot Hot Hot Start** Yes **OK Warm**
Warm Comfort No 15 **OK Warm Warm Start** Yes **Cool Cool Cool Comfort No** 5 **OK OK OK**
Start Yes **Cool Cool Cool** Comfort No -5 **OK OK OK Start** Yes Cold Cold **Cold** Comfort No
 -15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
 Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0599] In many conditions, the motor, inverter and battery can be cooled at the radiator reducing energy consumption at the chiller. In cool to warm conditions this might be the predominant method of cooling the components.

Example 29G—Battery, Motor and Inverter Self Heating-Cooling of the Battery at the Radiator
 [0600] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 31 and is operated with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E29G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00304 TABLE E29G Cooling of the Battery at the Radiator Motor & Ambient, ° C.
 Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No

35 Hot Hot Start Yes OK OK Warm Warm Comfort No 25 Hot Hot Hot Start Yes OK Warm
Warm Comfort No 15 OK Warm Warm Start Yes Cool Cool Cool Comfort No 5 OK OK OK
Start Yes Cool Cool Cool Comfort No -5 OK OK OK Start Yes Cold Cold Cold Comfort No
-15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0601] In many conditions, the motor, inverter and battery can be cooled at the radiator reducing energy consumption at the chiller. In cool to warm conditions this might be the predominant method of cooling the components.

Example 29H—Battery, Motor and Inverter Self Heating-Cooling of the Battery at the Radiator

[0602] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures is illustrated in FIG. 31 and is operated with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E29H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00305 TABLE E29H Cooling of the Battery at the Radiator Motor & Ambient, ° C.

Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot **Hot Hot Start** Yes **OK Warm Warm Comfort No 25 Hot Hot Hot Start** Yes **OK Warm
Warm Comfort No 15 OK Warm Warm Start** Yes **Cool Cool Cool Comfort No 5 OK OK OK
Start** Yes **Cool Cool Cool** Comfort No -5 **OK OK OK Start** Yes Cold Cold **Cold** Comfort No
-15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0603] In many conditions, the motor, inverter and battery can be cooled at the radiator reducing energy consumption at the chiller. In cool to warm conditions this might be the predominant method of cooling the components.

Example 30A—Embodiments—Enhanced Efficiency without Compressor—Cooling of the Battery at the Radiator

[0604] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 31 and is operated with RE1B1. The use of Refrigerant RE1B1 (R454C) is reported in Table E30A below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00306 TABLE E30A Cooling of the Battery at the Radiator Motor & Ambient, ° C.

Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot **Hot Hot Start** Yes **OK Warm Warm Comfort No 25 Hot Hot Hot Start** Yes **OK Warm
Warm Comfort No 15 OK Warm Warm Start** Yes **Cool Cool Cool Comfort No 5 OK OK OK
Start** Yes **Cool Cool Cool** Comfort No -5 **OK OK OK Start** Yes Cold Cold **Cold** Comfort No
-15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0605] However, in this example, the present system provides operation as a heat pump in an EV to achieve efficiency improvement by rejecting heat without running the compressor and using radiator heat dissipation as opposed to using the chiller, as illustrated by the performance data for this Example 30 as shown in FIGS. 31A and 31B.

Example 30B—Embodiments—Enhanced Efficiency without Compressor—Cooling of the Battery at the Radiator

[0606] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 31C and in FIG. 3D is provided. In accordance with FIG. 3D, heat removal occurs at the radiator, that is, the relative positions of the of the second cooling pump and the radiator and the motor/inverter are configured to achieve overall improvement in system efficiency and at the same time providing excellent and preferred control of battery and motor/inverter temperatures. In particular, a second coolant pump is located

down stream of the battery and upstream of the motor/inverter and directs the coolant through the radiator and back to the first liquid pump. In preferred embodiments, with piping and valving preferably arranged to allow the options of allowing at least a portion and possibly all (in which case the second liquid pump would not need to be in operation) of the coolant to be diverted around the motor inverter and directly to the radiator. In preferred embodiments, the coolant in such an arrangement is able to cool the battery to within a temperature range of about 25° C.-35° C. and to cool motor/inverter temperatures to within about 40° C. to about 65° C. For operation in which ambient air is at about 15° C., the coolant (preferably water/glycol) would exit the radiator and return to the first liquid pump (with the chiller and the PTC not operating as heat transfer devices in this mode) at a temperature of about 20° C. An exemplary version of this configuration is illustrated in FIG. 31C, showing the use of hot radiator air at the OHE and/or the IHE and/or the inner condenser in order to provide necessary heating to the cabin air.

[0607] In this example, the system is operated with RE1B2. The use of Refrigerant RE1B2 (R455A) is reported in Table E30B below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00307 TABLE E30B Cooling of the Battery at the Radiator Motor & Ambient, ° C.																					
Cabin	Battery	Inverter	Driving	Charging	45	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No						
35	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	25	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No
15	OK	Warm	Warm	Start	Yes	Cool	Cool	Cool	Comfort	No	5	OK	OK	OK	Start	Yes	Cool	Cool	Cool	Comfort	No
-5	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-15	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No
-25	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No	-35	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No

However, in this example, the present system provides operation as a heat pump in an EV to achieve efficiency improvement by rejecting heat without running the compressor and using radiator heat dissipation as opposed to using the chiller, as illustrated by the performance data for this Example 30 as shown in FIGS. 31A and 31B.

Example 30C—Enhanced Efficiency without Compressor—Cooling of the Battery at the Radiator [0608] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 31 and FIG. 3D1, and in particular showing heat removal at the radiator being in accordance with preferred embodiments.

That is, the relative positions of the of the second cooling pump and the radiator and the motor/inverter achieve overall improvement in system efficiency and at the same time providing excellent and preferred control of battery and motor/inverter temperatures. In particular, a second coolant pump is located downstream of the battery and upstream of the motor/inverter and directs the coolant through the radiator and back to the first liquid pump. In preferred embodiments, with piping and valving preferably arranged to allow the options of allowing at least a portion and possibly all (in which case the second liquid pump would not need to be in operation) of the coolant to be diverted around the motor inverter and directly to the radiator. In preferred embodiments, the coolant in such an arrangement is able to cool the battery to within a temperature range of about 25° C.-35° C. and to cool motor/inverter temperatures to within about 40° C. to about 65° C. For operation in which ambient air is at about 15° C., the coolant (preferably water/glycol) would exit the radiator and return to the first liquid pump (with the chiller and the PTC not operating as heat transfer devices in this mode) at a temperature of about 20° C. An exemplary version of this configuration is illustrated in FIG. 31C, with the additional optional enhancement of using hot radiator air at the OHE and/or the IHE and/or the inner condenser in order to provide necessary heating to the cabin air. and is operated with RE1B3. The use of Refrigerant RE1B3 (R474A) is reported in Table E30C below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00308 TABLE E30C Cooling of the Battery at the Radiator Motor & Ambient, ° C.

Cabin	Battery	Inverter	Driving	Charging	45	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No
35	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	25	Hot	Hot	Hot	Start
	Yes	OK	Warm	Warm	Comfort	No	15	OK	Warm	Warm	Start	Yes	Cool	Cool	Cool
	Comfort	No	5	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-5	OK
	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-25	Cool	Cool	Cool	Start
	Yes	Cold	Cold	Comfort	No	-35	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort
	No	-35	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No			

[0609] However, in this example, the present system provides operation as a heat pump in an EV to achieve efficiency improvement by rejecting heat without running the compressor and using radiator heat dissipation as opposed to using the chiller, as illustrated by the performance data for this Example 30 as shown in FIGS. 32A and 32B.

Example 30D—Embodiments—Enhanced Efficiency without Compressor—Cooling of the Battery at the Radiator

[0610] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 31C, with heat removal at the radiator being adjusted in accordance with the configuration illustrated in FIG. 3D1. In particular, the relative positions of the of the second cooling pump and the radiator and the motor/inverter are changed to achieve overall improvement in system efficiency and at the same time providing excellent and preferred control of battery and motor/inverter temperatures. In particular, a second coolant pump is located downstream of the battery and upstream of the motor/inverter and directing the coolant through the radiator and back to the first liquid pump. In preferred embodiments, with piping and valving is preferably arranged to allow the options of allowing at least a portion and possibly all (in which case the second liquid pump would not need to be in operation) of the coolant to be diverted around the motor inverter and directly to the radiator. In preferred embodiments, the coolant in such an arrangement is able to cool the battery to within a temperature range of about 25° C.-35° C. and to cool motor/inverter temperatures to within about 40° C. to about 65° C. For operation in which ambient air is at about 15° C., the coolant (preferably water/glycol) would exit the radiator and return to the first liquid pump (with the chiller and the PTC not operating as heat transfer devices in this mode) at a temperature of about 20° C. An exemplary version of this configuration is illustrated in FIG. 31C, with the additional optional enhancement of using hot radiator air at the OHE and/or the IHE and/or the inner condenser in order to provide necessary heating to the cabin air. In this example the system is operated with RE1B4. The use of Refrigerant RE1B4 (R474B) is reported in Table E30D below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00309 TABLE E30D Cooling of the Battery at the Radiator Motor & Ambient, ° C.

Cabin	Battery	Inverter	Driving	Charging	45	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No
35	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	25	Hot	Hot	Hot	Start
	Yes	OK	Warm	Warm	Comfort	No	15	OK	Warm	Warm	Start	Yes	Cool	Cool	Cool
	Comfort	No	5	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-5	OK
	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-25	Cool	Cool	Cool	Start
	Yes	Cold	Cold	Comfort	No	-35	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort
	No	-35	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No			

[0611] However, in this example, the present system provides operation as a heat pump in an EV to achieve efficiency improvement by rejecting heat without running the compressor and using radiator heat dissipation as opposed to using the chiller, as illustrated by the performance data for this Example 30 as shown in FIGS. 31A and 31B.

Example 30E—Enhanced Efficiency without Compressor—Cooling of the Battery at the Radiator

[0612] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 31C and FIG. 3D1 and is operated with RE1B5. The use of Refrigerant RE1B5 (R516A) is reported in Table E30E below by

using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00310 TABLE E30E Cooling of the Battery at the Radiator Motor & Ambient, ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot **Hot Hot Start** Yes **OK Warm Warm Comfort No** 25 **Hot Hot Hot Start** Yes **OK Warm**
Warm Comfort No 15 **OK Warm Warm Start** Yes **Cool Cool Cool Comfort No** 5 **OK OK OK**
Start Yes **Cool Cool Cool** Comfort No -5 **OK OK OK Start** Yes Cold Cold **Cold** Comfort No
-15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0613] However, in this example, the present system provides operation as a heat pump in an EV to achieve efficiency improvement by rejecting heat without running the compressor and using radiator heat dissipation as opposed to using the chiller, as illustrated by the performance data for this Example 30A as shown in FIGS. 31A and 31B.

Example 30F—Enhanced Efficiency without Compressor—Cooling of the Battery at the Radiator
[0614] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 31C and FIG. 3D1 and is operated with RE1B6. The use of Refrigerant RE1B2 (R444A) is reported in Table E30F below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00311 TABLE E30F Cooling of the Battery at the Radiator Motor & Ambient, ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot **Hot Hot Start** Yes **OK Warm Warm Comfort No** 25 **Hot Hot Hot Start** Yes **OK Warm**
Warm Comfort No 15 **OK Warm Warm Start** Yes **Cool Cool Cool Comfort No** 5 **OK OK OK**
Start Yes **Cool Cool Cool** Comfort No -5 **OK OK OK Start** Yes Cold Cold **Cold** Comfort No
-15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0615] However, in this example, the present system provides operation as a heat pump in an EV to achieve efficiency improvement by rejecting heat without running the compressor and using radiator heat dissipation as opposed to using the chiller, as illustrated by the performance data for this Example 30B as shown in FIGS. 31A and 31B.

Example 30G—Enhanced Efficiency without Compressor—Cooling of the Battery at the Radiator
[0616] In this example, a system of the present invention configured for operation to heat cabin air during periods of low ambient temperatures as illustrated in FIG. 31C and FIG. 3D1 and is operated with RE1B7. The use of Refrigerant RE1B7 (R457C) is reported in Table E30G below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00312 TABLE E30G Cooling of the Battery at the Radiator Motor & Ambient, ° C.
Cabin Battery Inverter Driving Charging 45 Hot Hot Hot Start Yes OK Warm Warm Comfort No
35 Hot **Hot Hot Start** Yes **OK Warm Warm Comfort No** 25 **Hot Hot Hot Start** Yes **OK Warm**
Warm Comfort No 15 **OK Warm Warm Start** Yes **Cool Cool Cool Comfort No** 5 **OK OK OK**
Start Yes **Cool Cool Cool** Comfort No -5 **OK OK OK Start** Yes Cold Cold **Cold** Comfort No
-15 OK OK OK Start Yes Cold Cold Cold Comfort No -25 Cool Cool Cool Start Yes Cold Cold
Cold Comfort No -35 Cool Cool Cool Start Yes Cold Cold Cold Comfort No

[0617] However, in this example, the present system provides operation as a heat pump in an EV to achieve efficiency improvement by rejecting heat without running the compressor and using radiator heat dissipation as opposed to using the chiller, as illustrated by the performance data for this Example 30 as shown in FIGS. 31A and 31B.

Example 30H—Embodiments—Enhanced Efficiency without Compressor—Cooling of the Battery at the Radiator

[0618] In this example, a system of the present invention configured for operation to heat cabin air

during periods of low ambient temperatures as illustrated in FIG. 31C and FIG. 3D1 and is operated with RE1B8. The use of Refrigerant RE1B8 (R457D) is reported in Table E30H below by using bold type as an indicator for acceptable operation at the indicated locations and for indicated ambient conditions when using the identified evaporative heat source(s).

TABLE-US-00313 TABLE E30H Cooling of the Battery at the Radiator Motor & Ambient, ° C.

Cabin	Battery	Inverter	Driving	Charging	45	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No																																																																								
35	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	25	Hot	Hot	Hot	Start	Yes	OK	Warm	Warm	Comfort	No	15	OK	Warm	Warm	Start	Yes	Cool	Cool	Cool	Comfort	No	5	OK	OK	OK	Start	Yes	Cool	Cool	Cool	Comfort	No	-5	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-15	OK	OK	OK	Start	Yes	Cold	Cold	Cold	Comfort	No	-25	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No	-35	Cool	Cool	Cool	Start	Yes	Cold	Cold	Cold	Comfort	No

[0619] However, in this example, the present system provides operation as a heat pump in an EV to achieve efficiency improvement by rejecting heat without running the compressor and using radiator heat dissipation as opposed to using the chiller, as illustrated by the performance data for this Example 30 as shown in FIGS. 31A and 31B.

Claims

1. A heat transfer system to alternatively and/or simultaneously provide heating and/or cooling in a mobile vehicle that includes an electrical power source requiring heating and/or cooling during charging and/or operation and that includes a cabin that requires heat input during low temperature ambient conditions, said system comprising: a) a vapor compression refrigeration circuit located in said mobile vehicle comprising: (i) a heat transfer composition comprising a first refrigerant, (ii) a compressor for compressing said first refrigerant in the vapor state from a first pressure to a higher second pressure, (iii) an inner condenser for selectively condensing during low temperature ambient conditions at least a portion of said first refrigerant vapor from said compressor by rejecting heat to said cabin, (iv) an outside heat exchanger located downstream of said inner condenser to selectively either (1) condense during low temperature ambient conditions at least a portion of said higher pressure refrigerant vapor not condensed in said inner condenser by rejecting heat, directly or indirectly, to ambient air and/or to a circulating coolant or (2) evaporate during high temperature ambient conditions low pressure refrigerant liquid from said inner condenser vapor; (v) a first OCE connected between said inner condenser and said outside heat exchanger for selectively (1) providing in an expansion mode a flow of reduced pressure liquid refrigerant from said inner condenser to said outside heat exchanger; (2) allowing in an open mode said condensed high pressure refrigerant from said condenser to pass to said outside condenser without pressure drop to said outside heat exchanger; or (3) preventing in a closed mode the flow of refrigerant from said inner condenser to said outside heat exchanger; (vi) an inside heat exchanger fluidly connectable to said refrigerant downstream of said inner condenser for selectively providing heating to a flow of cabin air; (vii) a chiller fluidly connectable to said refrigerant downstream of said inner condenser for selectively heating a flow of liquid coolant; (viii) a bypass channel system connected upstream of said first OCE and downstream of said outside heat exchanger for selectively routing said refrigerant from said inner condenser and/or from said outside heat exchanger (1) around said first expansion device and to either (A) a second OCE fluidly connected to said inside heat exchanger for selectively (a) providing in an expansion mode a flow of reduced pressure liquid refrigerant from said inner condenser to said inside heat exchanger; (b) allowing in an open mode said condensed high pressure refrigerant from said condenser or from said outside heat exchanger to pass without pressure reduction to said inside heat exchanger; or (c) preventing in a closed mode the flow of refrigerant to said inside heat exchanger; and/or (B) an expansion device fluidly connected to said chiller for selectively (a) providing in an expansion mode a flow of reduced pressure liquid refrigerant to said chiller; or (b) preventing in a closed mode the flow of

refrigerant to said chiller; or (2) through said first OCE operating in the expansion mode through said outside heat exchanger; and b) a heat exchange network interconnected with said vapor compression refrigeration circuit to selectively; (i) deliver, directly or indirectly, at said outside heat exchanger and/or at said chiller evaporative heat from one or more of ambient air and/or heat associated with the generation or use of electrical power within the vehicle and/or at said inside heat exchanger either directly or indirectly from (1) ambient air and/or (2) said electrical power source located in said vehicle.

2. The system of claim 1 wherein said first refrigerant comprises 2,3,3,3-tetrafluoropropene (R1234yf).

3. The system of claim 2 wherein said first refrigerant further comprises from about 20% to about 35% by weight of a second component selected from R32, R134a, R1132E, CO.sub.2 and combinations of two or more of these.

4. The system of claim 1 wherein said first refrigerant comprises trans-1,3,3,3-tetrafluoropropene (R1234ze(E)).

5. The system of claim 1 wherein said first refrigerant comprises from about 65% to about 80% by weight of 2,3,3,3-tetrafluoropropene (R1234yf) and from about 20% to about 35% by weight of a second component selected from R32, R1132E and combinations of these.

6. The system of claim 1 further comprising a refrigerant accumulator, wherein said compressor is being connected upstream to said refrigerant accumulator.

7. The system of claim 1 wherein said refrigerant comprises R-1234yf and R32.

8. The system of claim 1 wherein said refrigerant comprises from about 75% to about 89% by weight of R-1234yf and about 21% by weight of R32.

9. The system of claim 8 wherein said refrigerant further comprises CO.sub.2.

10. The system of claim 1 wherein said refrigerant comprises R-1234yf and R1132(E).

11. The system of claim 1 wherein said refrigerant comprises about 23% by weight of R1132(E).

12. The system of claim 1 wherein the heat exchange network comprises a coolant circuit that comprises a coolant that absorbs waste heat from an electrical power source located in said vehicle during low temperature ambient conditions and rejects heat to said refrigerant in said chiller.

13. A mobile vehicle having one or more electric traction motors and one or more batteries and/or capacitors providing electric current to said one or more traction motors comprising a heat transfer system carried by said mobile vehicle to alternatively and/or simultaneously provide heating and cooling in said mobile vehicle, wherein heating and/or cooling is provided to said one or more batteries and/or capacitors during charging and/or operation and that includes a cabin that requires heat input during low temperature ambient conditions, wherein said heat transfer system comprises: a) a vapor compression refrigeration circuit located in said mobile vehicle comprising: (i) a first refrigerant comprising from about 65% to about 80% by weight of 2,3,3,3-tetrafluoropropene (R1234yf) and from about 20% to about 35% by weight of a second component selected from R32, R1132E and combinations of these, (ii) a compressor for compressing said first refrigerant in the vapor state from a first pressure to a higher second pressure, said compressor being connected upstream to a refrigerant accumulator, (iii) an inner condenser for selectively condensing during low temperature ambient conditions at least a portion of said first refrigerant vapor from said compressor by rejecting heat to said cabin, (iv) an outside heat exchanger located downstream of said inner condenser to selectively either (1) condense during low temperature ambient conditions at least a portion of said higher pressure refrigerant vapor not condensed in said inner condenser by rejecting heat, directly or indirectly, to ambient air and/or to a circulating coolant or (2) evaporate during high temperature ambient conditions low pressure refrigerant liquid from said inner condenser vapor; (v) a first open/closed/expansion device connected between said inner condenser and said outside heat exchanger for selectively (1) providing in an expansion mode a flow of reduced pressure liquid refrigerant from said inner condenser to said outside heat exchanger; (2) allowing in an open mode said condensed high pressure refrigerant from said condenser to pass to

said outside condenser without pressure drop to said outside heat exchanger; or (3) preventing in a closed mode the flow of refrigerant from said inner condenser to said outside heat exchanger; (vi) an inside heat exchanger fluidly connectable to said refrigerant downstream of said inner condenser for selectively providing heating to a flow of cabin air; (vii) a chiller fluidly connectable to said refrigerant downstream of said inner condenser for selectively heating a flow of liquid coolant; (viii) a bypass channel system connected upstream of said first open/closed/expansion device and downstream of said outside heat exchanger for selectively routing said refrigerant from said inner condenser and/or from said outside heat exchanger (1) around said first expansion device and to either (A) a second open/closed/expansion device fluidly connected to said inside heat exchanger for selectively (a) providing in an expansion mode a flow of reduced pressure liquid refrigerant from said inner condenser to said inside heat exchanger; (b) allowing in an open mode said condensed high pressure refrigerant from said condenser or from said outside heat exchanger to pass without pressure reduction to said inside heat exchanger; or (c) preventing in a closed mode the flow of refrigerant to said inside heat exchanger; and/or (B) an expansion device fluidly connected to said chiller for selectively (a) providing in an expansion mode a flow of reduced pressure liquid refrigerant to said chiller; or (b) preventing in a closed mode the flow of refrigerant to said chiller; or (2) through said first open/closed/expansion device operating in the expansion mode through said outside heat exchanger to said accumulator; and b) a heat exchange network interconnected with said vapor compression refrigeration circuit to selectively; (i) deliver, directly or indirectly, at said outside heat exchanger and/or at said chiller evaporative heat from one or more of ambient air and/or heat associated with the generation or use of electrical power within the vehicle and/or at said inside heat exchanger either directly or indirectly from (1) ambient air and/or (2) said electrical power source located in said vehicle.

14. The system of claim 13 wherein said refrigerant comprises R-1234yf and R32.

15. The system of claim 13 wherein said refrigerant comprises from about 75% to about 89% by weight of R-1234yf and about 21% by weight of R32.

16. The system of claim 15 wherein said refrigerant further comprises CO₂, vehicle.

17. The system of claim 13 wherein said refrigerant comprises R-1234yf and R1132(E).

18. The system of claim 13 wherein said refrigerant comprises about 23% by weight of R1132(E).

19. The system of claim 13 wherein said refrigerant comprises R-1234ze(E).

20. The system of claim 13 wherein the heat exchange network comprises a coolant circuit that comprises a coolant that absorbs waste heat from an electrical power source located in said vehicle during low temperature ambient conditions and rejects heat to said refrigerant in said chiller.
